ENGINEERING PHYSICS

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1 LASERS AND FIBRE OPTICS

OBJECTIVES:

- To explain basic interactions of radiation with matter.
- To comprehend the basic principle of lasing system and their applications.
- To understand the basic working principle of optical fibre and their applications.
- To classify different types of optical fibres and explain their merits & demerits.

1.1 INTRODUCTION

1.1.1 Characteristics of laser

Laser light is highly monochromatic, highly coherent, highly directional and can be sharply focused. Each of these characteristics that are not normally found in ordinary light makes laser a unique and the most powerful tool. Lasers find a wide variety of applications in the field of scientific research, engineering and medicine.

1.1.2 Interaction of radiation with matter

There are three possible ways through which interaction of radiation and matter can take place, absorption of radiation is one means, and the emission of radiation, as suggested by Einstein in 1917, can occur in two ways namely the spontaneous emission, and the stimulated emission.

All the three are described as follows by considering an atom as a system.

Absorption: Absorption of a photon of frequency f takes place when the energy difference $E_2 - E_1$ of the allowed energy states of the atomic system equals the energy hf of the photon. Then the photon disappears and the atomic system moves to upper energy state E_2.

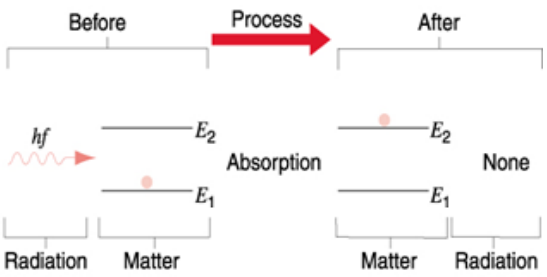


Figure 1.1.1: Absorption

Spontaneous Emission: Spontaneous emission is the emission of a photon when a system transits from a higher energy state to a lower energy state without the aid of any external agency. The average life time of the atomic system in the excited state is of the order of 10^{-8} s. After the life time of the atomic system in the excited state, it comes back to the state of lower energy on its own accord by emitting a photon of energy $hf = E_2 - E_1$.

In an ordinary light source, the radiation of light from different atoms is not coherent. The radiations are emitted in different directions in random manner. Such type of emission of radiation is called spontaneous emission.

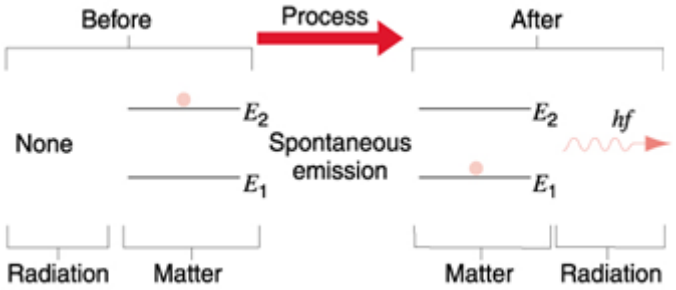


Figure 1.1.2: Spontaneous Emission

Stimulated Emission: When a photon (called stimulating photon) of suitable frequency interacts with an excited atomic system, the latter comes down to ground state before its life time. Such an emission of radiation is called stimulated emission. In stimulated emission, both the stimulating photon and the stimulated photon are of same frequency, same phase and are in same state of polarization. These photons are emitted in the same direction. In other words, these two photons are coherent. This phenomenon is responsible for laser action.

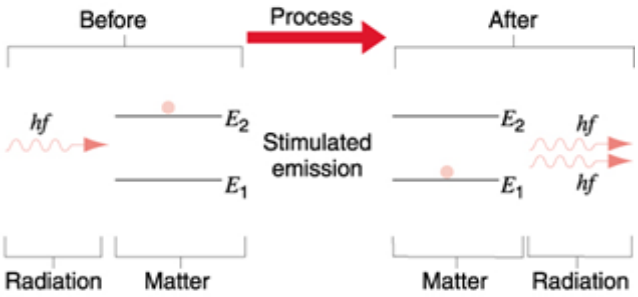


Figure 1.3: Stimulated Emission

1.1.3 Population inversion

From Boltzmann statistics, the ratio between population of atoms in two energy states E_1 and E_2 at equilibrium temperature T is,

$$\frac{N_2}{N_1} = e^{-(E_2-E_1)/kT} \quad \textbf{1.1}$$

where k is Boltzmann constant, N_1 is density of atoms with energy E_1 & N_2 is density of atoms with energy E_2. Under normal condition, where populations are determined only by the action of thermal agitation, population of the atoms in upper energy state is less than that in lower energy state (i.e. $N_2 < N_1$, Figure 1.4 a).

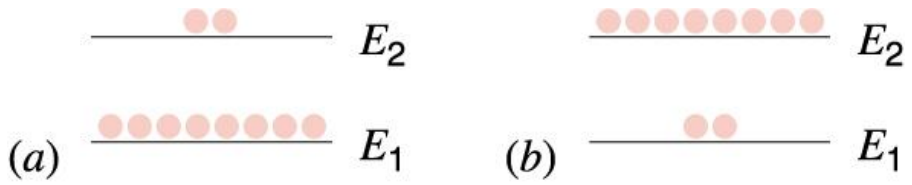


Figure 1.4: (a) Normal thermal equilibrium distribution of atomic systems (b) An inverted population.

For the stimulated emission rate to exceed the absorption rate it is necessary to have higher population of upper energy state than that of lower energy state. This condition is called population inversion ($N_2 > N_1$, Figure 1.4b). This is a non-equilibrium condition and is facilitated by the presence of energy states called '**metastable states**' where the average life time of the atom is 10^{-3} s which is much longer than that of the ordinary excited state ($\approx 10^{-8}$s).

1.2 EINSTEIN'S COEFFICIENTS

Einstein coefficients give the probability associated with the absorption and emission processes. Consider two energy states E_1 and E_2 of a system of atoms ($E_2 > E_1$). Let N_1 and N_2 be the density of atoms with energies E_1 and E_2 respectively, per unit volume of the system. Let radiations with a continuous spectrum of frequencies be incident upon the system. Let there be a radiation of frequency f such that $f = (E_2 - E_1)/h$. Also let $I_f\, df$ be the energy incident/unit volume of the system considering only those radiations whose frequencies lie in the range f and $f + df$. Then, I_f represents the energy density of frequency f.

Let us consider the absorption, and the two emission processes case by case.

(1) In the case of **absorption,** an atom in the level E_1 can go to the level E_2, when it absorbs a radiation of frequency f such that $f = (E_2 - E_1)/h$ (Fig. 1.1).

The number of such absorptions per unit time, per unit volume is called rate of absorption. The rate of absorption depends upon,

(a) the number density of lower energy state, i.e., N_1, and, (b) the energy density i.e., I_f.

$\therefore$ Rate of absorption $\propto \boldsymbol{N_1 I_f}$

Rate of absorption $= \boldsymbol{B_{12}\, N_1\, I_f}$ **1.2**

where, B_{12} is the constant of proportionality called Einstein coefficient of induced absorption.

(2) In the case of **spontaneous emission**, an atom in the higher energy level E_2 undergoes transition to the lower energy level E_1, voluntarily by emitting a photon (Fig. 1.2).

Since it is a voluntary transition, it is independent of the energy density of any frequency in the incident radiation. The number of such spontaneous emissions per unit time, per unit volume is called rate of spontaneous emission which is proportional to only the number density in the higher energy state, i.e., N_2.

$\therefore$ Rate of spontaneous emission $= \boldsymbol{A_{21}\, N_2}$ **1.3**

where, A_{21} is the constant of proportionality called the Einstein coefficient of spontaneous emission.

(3) In case of **stimulated emission**, the system requires an external photon of appropriate frequency $= [(E_2 - E_1)/h]$, to stimulate the atom for the corresponding downward transition, and thereby cause emission of stimulated photons (Fig. 1.3), the energy density I_f, has a role to play in this case.

The number of stimulated emissions per unit time per unit volume, called rate of stimulated emission, is proportional to, a) the number density of the higher energy state, i.e., N_2 and, b) the energy density, i.e., I_f.

$\therefore$ Rate of stimulated emission $\propto \boldsymbol{N_2 I_f}$

Rate of stimulated emission $= \boldsymbol{B_{21}\, N_2 I_f}$

1.4

where, B_{21} is the constant of proportionality called the Einstein coefficient of simulated emission.

In thermal equilibrium, the total energy of the system remains unchanged in spite of the interaction that is taking place between itself and the incident radiation. Under such a condition, the number of photons absorbed by the system per second must be equal to the number of photons it emits per second by both the stimulated and the spontaneous emission processes.

∴ At thermal equilibrium,

Rate of absorption = Rate of spontaneous emission + Rate of stimulated emission,

∴ From Eqs (1.1), (1.2) & (1.3), we have $\boldsymbol{B_{12}\, N_1\, I_f = A_{21}\, N_2 + B_{21}\, N_2\, I_f}$ **1.5**

Or $$I_f(B_{12}\, N_1 - B_{21}\, N_2) = A_{21}\, N_2$$

Or, $$I_f = \frac{A_{21}\, N_2}{(B_{12}\, N_1 - B_{21}\, N_2)}$$

By rearranging the above equation, we get,

$$\boldsymbol{I_f = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{B_{12}\, N_1}{B_{21} N_2} - 1}\right]}$$ **1.6**

But, by Boltzmann's law, we have,

$$\boldsymbol{\frac{N_2}{N_1} = e^{-(E_2 - E_1)/kT} = e^{-\frac{hf}{kT}} \qquad \therefore \frac{N_1}{N_2} = e^{\frac{hf}{kT}}}$$ **1.7**

Equation (1.6) becomes,

$$\boldsymbol{I_f = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{B_{12}}{B_{21}} e^{\frac{hf}{kT}} - 1}\right]}$$ **1.8**

According to Planck's law, the equation for I_f, is,

$$\boldsymbol{I_f = \frac{8\pi h f^3}{c^3} \left[\frac{1}{e^{\frac{hf}{kT}} - 1}\right]}$$ **1.9**

Now, comparing the equations (1.8) and (1.9), term by term on the basis of positional identity, we have,

$\frac{A_{21}}{B_{21}} = \frac{8\pi hf^3}{c^3}$ and, $\frac{B_{12}}{B_{21}} = 1$ or $B_{12} = B_{21}$

Hence the probability of induced absorption is equal to the probability of stimulated emission.

Hence equation (1.8) can be rearranged as $\frac{A_{21}}{B_{21}} = I_f\left(e^{\frac{hf}{kT}} - 1\right)$ **1.10**

Because of the above identity, the subscripts could be dropped, and A_{21} & B_{21} can be represented simply as A and B and Eq (1.8) can be rewritten.

∴ At thermal equilibrium the equation for energy density is, $I_f = \frac{A}{B\left[e^{\frac{hf}{kT}} - 1\right]}$ **1.11**

1.2.1 Interpretation of Einstein's Coefficients

Dependence of nature of emission on frequency:

We have, $\frac{A_{21}}{B_{21}} = \frac{8\pi hf^3}{c^3}$ $implies$ $\frac{A_{21}}{B_{21}} \propto f^3$

Since $f = (\Delta E/h)$, the above proportionality implies that for large f values.

For higher ΔE values, the transitions in the system are such that, the probability of spontaneous emission supersedes that for stimulated emissions by a large margin.

Case (i) $hf >> kT$:

When the frequency of radiation is high, $hf >> kT$, due to which $e^{\frac{hf}{kT}} \gg 1$

Hence in Equation (1.10), $\frac{A_{21}}{B_{21}} \gg 1$, i.e.. $A_{21} \gg B_{21}$

Hence the spontaneous emissions will be much larger in number than the stimulated emissions.

Case (ii) ≈ kT :

For $hf \approx kT$, $e^{\frac{hf}{kT}}$ will be low and comparable to 1.

∴ A_{21} & B_{21} become comparable, which means, the process of stimulated emissions become significant.

Case (iii) << kT :

For $hf \ll kT$, $\left(e^{\frac{hf}{kT}} - 1\right) << 1$, and, $\frac{A_{21}}{B_{21}} \ll 1$, Or, $B_{21} \gg A_{21}$

Hence for lower frequencies, stimulated emissions dominate the emission process. This is what we observe at room temperature in the atomic transitions which generate microwaves.

Non-Equilibrium conditions leading to amplification (Laser action):

We have from equation (1.4),

$$\frac{\text{Rate of emission}}{\text{Rate of absorption}} = \frac{A_{21}\,N_2 + B_{21}\,N_2 I_f}{B_{12}\,N_1\,I_f} = \frac{N_2}{N_1}\left[\frac{A_{21} + B_{21} I_f}{B_{12}\;I_f}\right]$$

Since in the Einstein's theory we have $B_{12} = B_{21}$,

$$\frac{\textbf{Rate of emission}}{\textbf{Rate of absorption}} = \frac{N_2}{N_1}\left[\frac{A_{21}}{B_{21} I_f} + \mathbf{1}\right] \qquad \textbf{1.12}$$

Now let us again get back to Eq (1.10). If for the system, the energy levels E_1 & E_2, involved in the transitions are such that $\Delta E << kT$, i.e., $hf << kT$, then as per Eq (1.10).

$$\frac{A_{21}}{B_{21} I_f} \ll 1$$

∴ Eq (1.12) can be expressed as, $$\frac{\textbf{Rate of emission}}{\textbf{Rate of absorption}} = \frac{N_2}{N_1} \qquad \textbf{1.13}$$

Under normal conditions in all the systems, N_2 will always be lesser than N_1.

Now let us consider a system which can support the energy level population status with $N_2 > N_1$, between certain pair of energy levels i.e., the system in which population inversion could be achieved. By some suitable means of energy supply when the population is inverted between the pair of energy levels in the system, the system will then be under non-equilibrium conditions since the Boltzmann factor conditions are violated.

As per Eq (1.10), under non-equilibrium conditions, rate of emission exceeds the rate of absorption. Further if the photons emitted in a particular direction are returned into the system by reflecting them back and forth, then the rate of stimulated emission soon exceeds the absorption rate. If right conditions are provided, all the stimulated emissions could be arranged to be identical in respect of wavelength, phase & direction to the one starting stimulating radiation. In other words, we can say that, the single starting stimulating radiation itself has been multiplied or, amplified. But the amplification takes place at the cost of downward transitions which will soon reverse the condition, $N_2 > N_1$. In order to sustain the status of

population inversion, energy is provided in the right manner to keep lifting atoms to higher energy level continuously so long as amplification is required to be maintained.

This forms the basis for functioning of every laser.

1.3 CONSTRUCTION OF LASER SYSTEM

1.3.1 Essential components of a laser

The laser device basically consists of three elements (Fig 1.5).

- Pumping system
- Lasing medium/ active medium
- A resonant cavity or an optical cavity

Lasing medium has atomic systems (active centers), with special energy levels which are suitable for laser action. This medium may be a gas, or a liquid, or a crystal or a semiconductor. The atomic systems in this may have energy levels including a ground state (E_1), an excited state (E_3) and a metastable state (E_2).

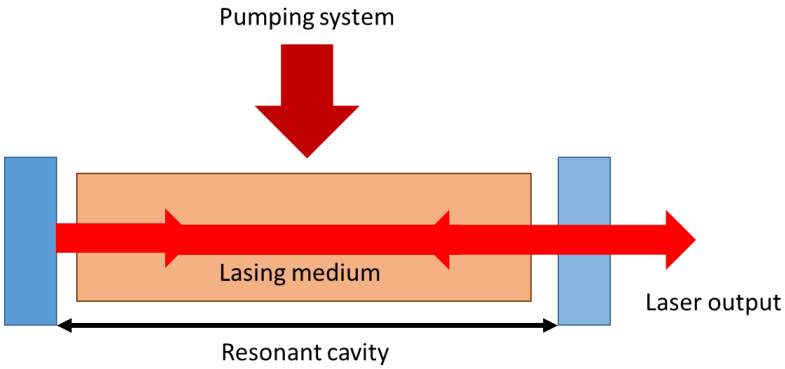


Figure 1.5: Block diagram of laser system

1.3.2 He-Ne laser

He-Ne Laser has a glass discharge tube filled with He (80%) and Ne (20%) at low pressure. Helium gas is the "pumping" medium and Neon gas is the "lasing" medium (Figure 1.6). The simplified energy level diagram (Figure 1.7) shows four levels: E_o, E_1, E_2 and E_3. Electrons and ions in the electrical gas discharge occasionally collide with He-atoms, raising them to level E_3 (a metastable state). During collisions between He- and Ne- atoms, the excitation energy of He-atom is transferred to Ne-atom (level E_2), selectively populating E_2 due to resonant energy transfer. Thus, population inversion occurs between levels E_2 and E_1. This population inversion is maintained because (i) the

metastability of level E_3 ensures a ready supply of Ne-atoms in level E_2 and (ii) level E_1 decays rapidly to E_o. Stimulated emission from level E_2 to level E_1 predominates, and red laser light is generated. The mirror M_1 is fully reflective and the mirror M_2 is partially reflective to allow the laser beam to come out. The Brewster's windows W 's are at polarizing angles to the mirrors, to reduce the reflection loss.

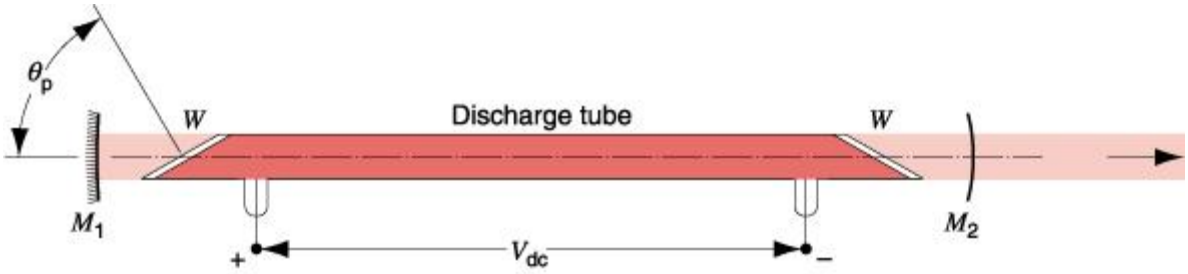


Figure 1.6 Basic elements of He-Ne gas laser

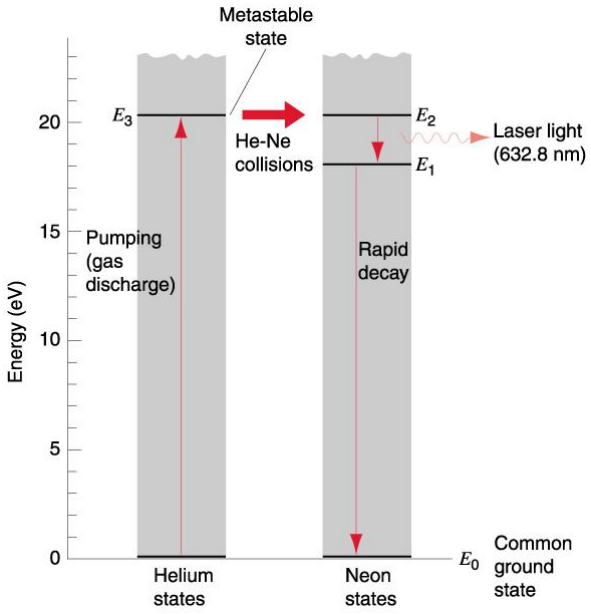


Figure 1.7: Atomic levels involved in the operation of He-Ne gas laser

1.3.3 Ruby laser

In ruby laser (Figure 1.8), the lasing medium is a ruby rod which has chemical composition of Al_2O_3 doped with Cr_2O_3. Cr^{3+} ions are the active centers, which has approximately similar energy level structure as shown in Figure 1.9. The resonant cavity is a pair of parallel mirrors to reflect the radiation back into the lasing medium. Pumping is a process of exciting more number of atoms in the ground state to higher energy states, which is required for attaining the population inversion. The atoms in the state E_3 may

come down to state E_1 by spontaneous emission or they may come down to metastable state (E_2) by collision. The atoms in the state E_2 come down to state E_1 by stimulated emission. These radiations may be reflected due to mirror action of the end faces. When population inversion takes place at E_2, a stray photon of right energy stimulates chain reaction, accumulates more photons, all coherent. The reflecting ends turn the coherent beam back into active region so that the regenerative process continues and part of the light beam comes out from the partial mirror as a laser pulse. The output is an intense beam of coherent light.

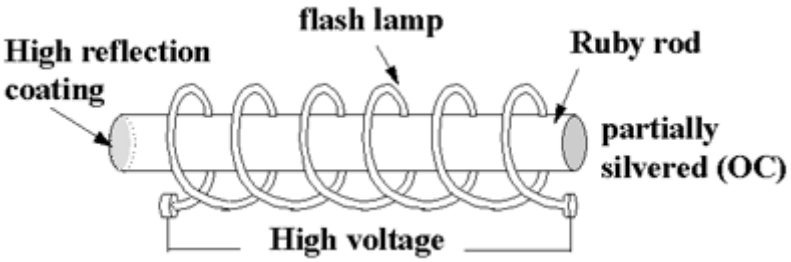


Fig. 1.8 Construction of Ruby laser

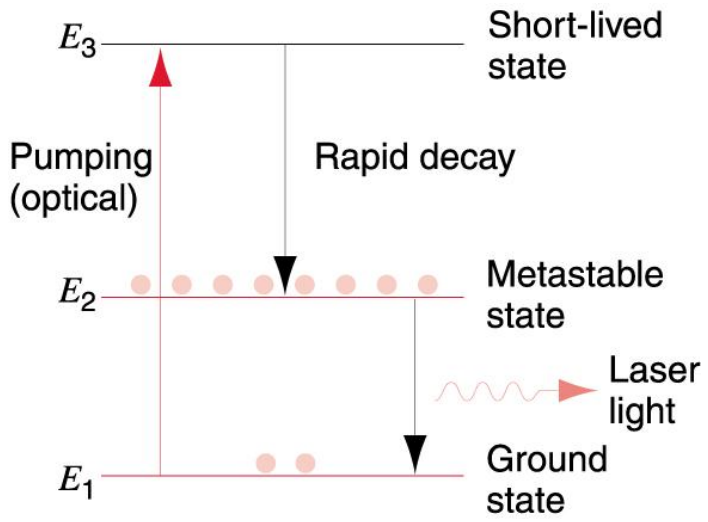


Figure 1.9: The basic three-level scheme for laser operation

1.3.4 Semiconductor laser

The semiconductor laser was first constructed by Hall et al. in 1962. The active medium in a semiconductor laser is a p-n junction diode and hence the semiconductor laser is also called diode laser.

Principle: It is based on the principle of the emission of the recombination energy in form of light in certain semiconducting materials such as GaAs, GaAlAs, etc. By heavily

doping the p-n junction diode and by applying a high current density, the stimulated emission of light is produced and hence the laser operation is made possible in the semiconductor laser.

Construction: The p-n junction of a semiconductor laser is prepared using n-GaAs and p-GaAs. The height, width and breadth of this laser are 1 mm x 1 mm x 1 mm respectively. The upper and bottom surface of the p-n junction are used for ohmic contact, the front and back surfaces are used as resonators. For resonators, the end faces are made perfectly parallel to each other and highly polished. One end is made fully reflecting, whereas the other end is made partially reflecting. The remaining two surfaces are made rough, so as to prevent the leakage of light through these faces. The diode is forward biased, with a voltage nearly equal to the band gap voltage of the material. The p-type and n-type layers are heavily doped.

Working: The heavily doping of the p-type and n-type regions produces population inversion in the depletion region of the diode. The diode is forward biased with a voltage equal to the $V= E_g/e$ energy gap voltage, and hence the recombination of the holes and electrons takes place. In order to produce and maintain the stimulated emission continuously, a high current density nearly in the order of 20 kA/cm^2 is applied. Due to the high current density, more and more number of electrons are injected into the n-region and more number of holes are injected into the p-region and hence the continuous emission of light is made possible. Since, more number of charge carriers are injected, the laser is also called injection laser. The output laser beam emerges from the p-n junction of the partially reflecting surface as shown in Fig.1.10. The output wavelength is given by

$$\lambda = \frac{hc}{E_g}$$

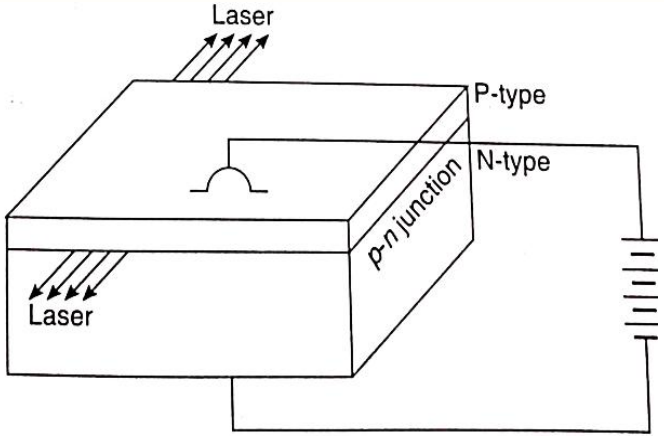


Figure 1.10: Semiconductor laser

1.4 APPLICATIONS OF LASER

Laser is used in various scientific, engineering and medical applications. It is used in investigating the basic laws of interaction of atoms and molecules with electromagnetic wave of high intensity. Laser is widely used in engineering applications like optical communication, micro welding and sealing etc. In medical field, laser is used in bloodless and painless surgery especially in treating the retinal detachment. Also used as a tool in treating dental decay, tooth extraction, cosmetic surgery.

1.4.1 Bar code scanner

Laser technology plays a crucial role in reading barcodes, primarily due to its precision and directionality. When a laser scanner directs a narrow beam of light onto a barcode, the light interacts with the pattern of black and white stripes. The barcode, typically in the form of a Universal Product Code (UPC), consists of vertical stripes that represent data. Each digit is encoded using a pattern of seven blocks, which are either black or white, to denote numbers from 0 to 9 as shown in the Figure 1.11.

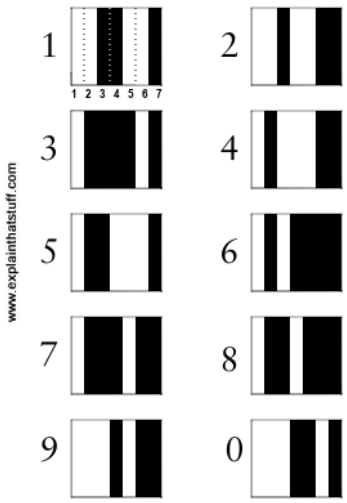


Figure 1.11: Universal Product Code (UPC)

The directionality of the laser beam ensures it can consistently scan the barcode from various angles, enabling reliable reading even in different orientations. The laser beam's narrow focus allows it to distinguish fine details, making it effective in reading densely packed barcodes. The physics of laser interaction with barcodes involves the reflection of light from the barcode's surface. Black stripes absorb more light, reflecting less, while white spaces reflect more light back to the scanner. This variation in reflected light intensity is captured by a photodiode or a charge-coupled device (CCD) within the scanner. The device converts these light variations into electrical signals, which the scanner's software decodes into the barcode's

data, enabling quick and accurate data processing. The primary advantage of laser scanners is their ability to read barcodes quickly and accurately from a distance, making them indispensable in retail, warehousing, and industrial applications.

1.4.2 Laser printer

A laser printer operates based on the principle of electrophotography (xerography), which involves the combined action of electrostatics, photoconductivity, and laser optics to produce printed images. The central component of the printer is a photoconductive drum, usually coated with semiconductor material such as selenium or organic photoconductors. In darkness, this material behaves as an insulator, but when exposed to light it becomes conductive. This property allows a laser beam to selectively remove charge from the drum surface and create an electrostatic image corresponding to the document to be printed. Fine powdered ink called toner, which carries an electric charge, is then attracted to these charged regions and subsequently transferred onto paper. Finally, heat and pressure permanently fix the toner onto the paper, producing the printed output. Steps involved in laser printing:

• Charging the drum: A primary charging roller applies a uniform electrostatic charge (usually negative) over the entire surface of the photoconductive drum.

• Laser exposure (writing the image): A laser beam, guided by a rotating mirror system, scans across the drum surface. Wherever the laser strikes, the photoconductive surface becomes conductive and the charge is reduced, forming an electrostatic latent image of the document.

• Developing the image: Negatively charged toner particles from the toner cartridge are brought near the drum. These particles are attracted to the regions where the charge has been altered by the laser, making the image visible on the drum.

• Transfer to paper: The paper is given a positive charge by a transfer roller. Due to electrostatic attraction, the toner particles move from the drum surface to the paper.

• Fusing: The paper passes through a fuser unit, which consists of heated rollers. The heat melts the toner particles and pressure presses them into the paper fibers, producing a permanent print.

• Cleaning and resetting: Any remaining toner on the drum is removed by a cleaning blade, and the drum is discharged to prepare for the next printing cycle.

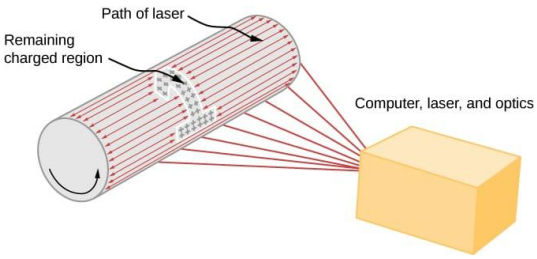


Figure 1.12: Laser Printer

1.4.3 Laser cooling

Principle of Laser cooling (Figure 1.13) is the use of dissipative light forces for reducing the random motion and thus the temperature of small particles, typically atoms or ions. Depending on the mechanism used, the temperature achieved can be in the milli-kelvin, micro-kelvin, or even nano-kelvin regime. If an atom is traveling toward a laser beam and absorbs a photon from the laser, it will be slowed by the fact that the photon has momentum, $p = E/c = h/\lambda$. It would take a large number of such absorptions to cool the atoms to near 0 K. The following are the two types of laser cooling: i. Doppler Cooling and ii. Sisyphus Cooling.

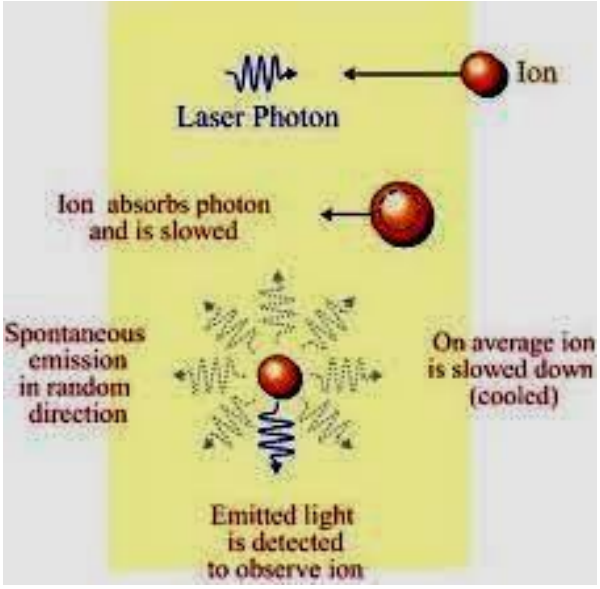


Figure 1.13: Laser Cooling

1.5 OPTICAL FIBRES

Optical fibres are thin, flexible strands of transparent dielectric material such as glass or plastic. They are basically used to guide infrared & visible light waves through curved paths.

1.5.1 Construction of optical fibre

It consists of a central cylindrical core made of pure glass or plastic of refractive index n_1 surrounded by a cladding made of similar material but of lower refractive index n_2 ($n_2 < n_1$).

But there is a material continuity from core to cladding. The cladding is enclosed in a polyurethane jacket that protects the fibre from external damaging factors such as abrasion, crushing & chemical reactions (Figure 1.14). Many such protected fibres are grouped to form a cable. The diameter of the core varies between 10 to 200 μm that of the cladding varies between 50 to 250 μm.

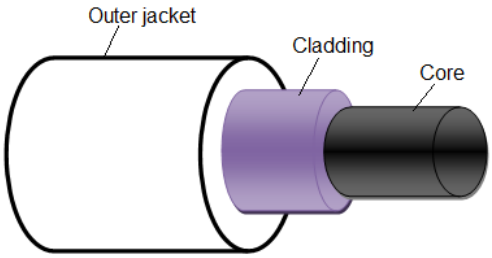


Figure 1.14: Optical fibre – sectional view

1.6 PRINCIPLE OF OPTICAL FIBRES

Optical fibres work on the principle of total internal reflection of light. When a beam of light traveling in an optically *denser* medium falls on interface separating denser medium from relatively *less dense* medium, if the angle of incidence is greater than particular angle called critical angle (θ_C) for the pair of media, the light undergoes total internal reflection (Figure 1.15). Total internal reflection is the most superior type of reflection. Reflection is total in the sense that almost the entire energy is returned to the first medium through reflection without any loss of energy. Due to this the optical fibres are able to sustain light signal transmission over very long distances despite infinite number of reflections.

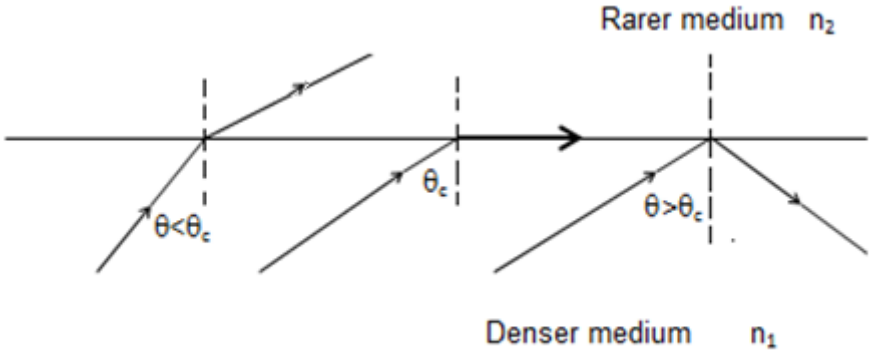


Figure 1.15: Total internal reflection

1.6.1 Acceptance angle, acceptance cone and numerical aperture

Consider an optical fibre with refractive index of the material of the core n_1 and cladding n_2 placed in a surrounding medium of refractive index n_0 (Figure 1.16). Let a ray AO of light enter the core of the fibre at an angle θ_0. Let this ray after refraction through an angle θ_1 at O strikes the interface between the core and the cladding at the critical angle such that the refracted ray grazes the interface.

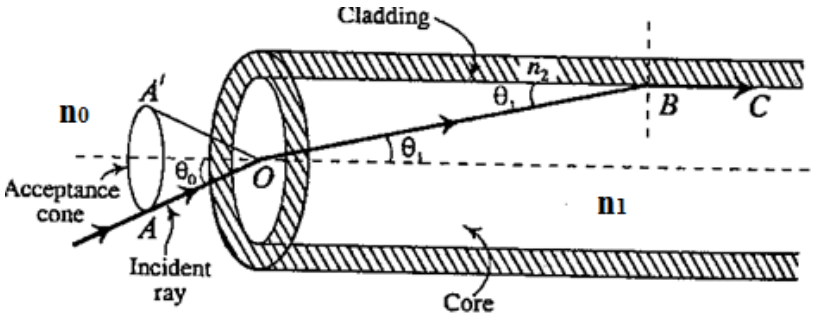


Figure 1.16: Wave propagation through fibre

Applying Snell's law of refraction at O, we have,

$$\frac{\sin\theta_0}{\sin\theta_1} = \frac{n_1}{n_0}$$

$$\therefore \sin\theta_0 = \frac{n_1}{n_0}\sin\theta_1 \qquad \textbf{1.14}$$

Similarly, applying Snell's law at B,

$$\frac{\sin(90^\circ - \theta_1)}{\sin 90^\circ} = \frac{n_2}{n_1} \qquad \textbf{or} \qquad \cos\theta_1 = \frac{n_2}{n_1}$$

$$\sin\theta_1 = \sqrt{1 - \frac{n_2^2}{n_1^2}} \qquad \textbf{1.15}$$

Substituting Eq. (1.15) in Eq. (1.14) and simplifying,

$$\sin\theta_0 = \frac{1}{n_0}\sqrt{n_1^2 - n_2^2} \qquad \textbf{1.16}$$

θ_0 is called the **acceptance angle** or **half angle of the acceptance cone**. The acceptance angle is generally about 5° for a single mode fibre & 10° to 15° for multimode fibres. The term $n_0 \sin\theta_0$ is called **numerical aperture (*NA*)**, which indicates the light gathering power of the optical fibre. It is evident that any ray that enters the fibre at an angle less than θ_0, strikes the

core-cladding interface at angle greater than the critical angle and undergoes total internal reflection each time it strikes the interface. The optical fibre sustains the light signal transmission over a long distance.

Fractional refractive index change (Δ): It is the ratio of the difference in the refractive indices $(n_1 - n_2)$ between the core & the cladding to the refractive index n_1 of the core.

$$\Delta = \frac{(n_1 - n_2)}{n_1} \qquad \textbf{1.17}$$

Since $n_1 > n_2$, Δ is always positive.

Relation between NA & Δ :

From Eq. 1.16, assuming $n_0 = 1$, $NA = \sqrt{n_1^2 - n_2^2} = \sqrt{(n_1 - n_2)(n_1 + n_2)}$

From Eq. 1.17, $(n_1 - n_2) = n_1 \Delta$ and since $n_1 \approx n_2$, we can approximate $(n_1 + n_2) \approx 2\, n_1$.

Therefore, $NA = \sqrt{(n_1 \Delta)(2n_1)} = n_1 \sqrt{2\Delta}$ **1.18**

The light accepting capacity of a fibre can be increased by making Δ large. But there are practical limitations to achieve this. Also a very large Δ may cause signal distortion.

Skip distance (L_s): Skip distance is the distance between two successive reflections of the ray of light which propagates through the optical fibre. Consider a portion of the optical fibre through which a light signal is transmitted (Figure 1.17).

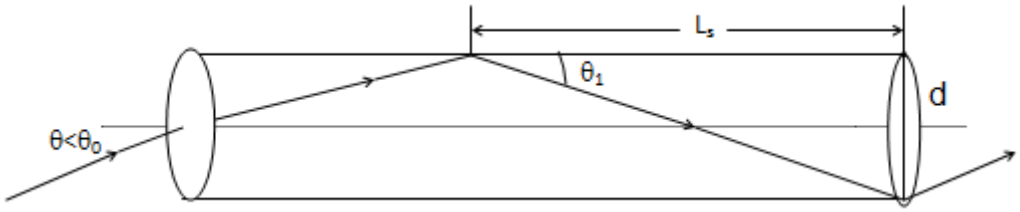


Figure 1.17: Skip distance

From the Figure 1.17,

$$L_s = d \cot\theta_1 = d\sqrt{\operatorname{cosec}^2\theta_1 - 1}$$

$$L_s = d\sqrt{\frac{n_1^2}{n_0^2 \sin^2\theta} - 1} \quad (\because \sin\theta_1 = \frac{n_0}{n_1}\sin\theta) \qquad \textbf{1.19}$$

1.7 TYPES OF OPTICAL FIBRES

Based on their refractive index profile and geometry, optical fibres may be broadly classified into

a) single mode step index optical fibres
b) multi-mode step index optical fibres
c) multi-mode graded index optical fibres

Number of modes of transmission through an optical fibre: Depending on the launch angle into the fibre, there can be hundreds of ray paths or modes by which energy can propagate down the core. The ray paths corresponding to the same propagating wave front is called a mode. An optical fibre permits a discrete number of modes to propagate through it. Not all the rays that enter the acceptance cone sustain propagation. Only those modes that satisfy the coherent phase condition are successfully propagated. The rays belonging to the same propagating wave-front must remain in step despite the phase changes that occur on reflection and traversing different optical paths.

The number of modes supported for propagation through an optical fibre is determined by a parameter called normalized frequency (V) given by

$V = \frac{\pi d}{\lambda}\sqrt{n_1^2 - n_2^2}$ where d is the diameter of the core and λ is the wavelength of the light propagated. If $V >> 1$, then the number of successfully propagated modes is $\frac{V^2}{2}$.

1.7.1 Single mode step index optical fibre

A single mode optical fibre consists of a core having a uniform refractive index n_1 that abruptly decreases at the core-cladding interface to a lower value n_2, the refractive index of the cladding. The diameter of the core is narrow (5-10μm) generally a few times the wavelength of the light propagating through it. Only rays nearly parallel to the fibre axis will travel through. It supports a single mode propagation because of its narrow core (Figure 1.18).

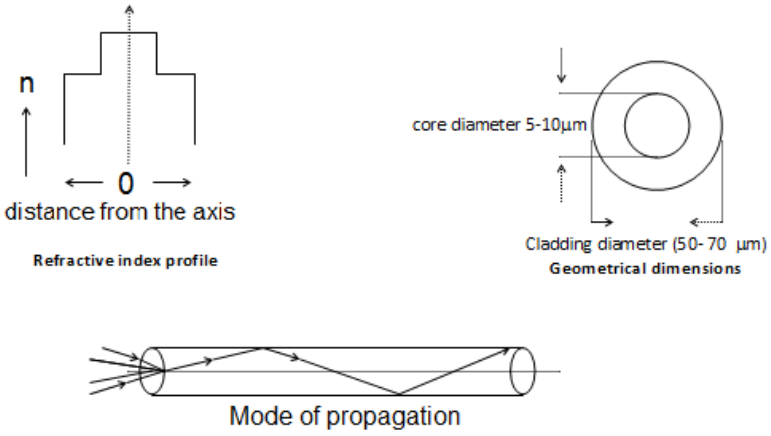


Figure 1.18: Single mode step-index fibre

1.7.2 Step-Index Multimode fibre

In this case also the refractive index profile is similar to step index fibre i.e., fibre consists of a core having a uniform refractive index n_1 that abruptly decreases at the core-cladding interface to a lower value n_2 the refractive index of the cladding. But the diameter of the core is large (50-200 µm). The comparatively large central core makes it rugged and easily infused with light, as well as easily terminated and coupled. It also supports a large number of modes for propagation (Figure 1.19).

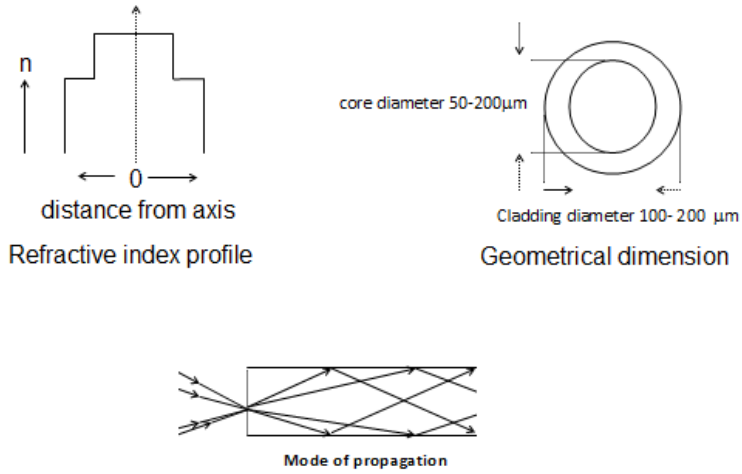


Figure 1.19: Multi mode step-index fibre

1.7.3 Graded-Index Multimode fibre (GRIN)

It consists of a core whose refractive index decreases gradually from its axis radially outward and becomes equal to the refractive index of the cladding at the core-cladding interface. The refractive index of the cladding remains uniform. Dimensions of the core and cladding are

similar to that of step index multimode fibres. It supports a large number of modes for propagation because of its large core diameter (Figure 1.20).

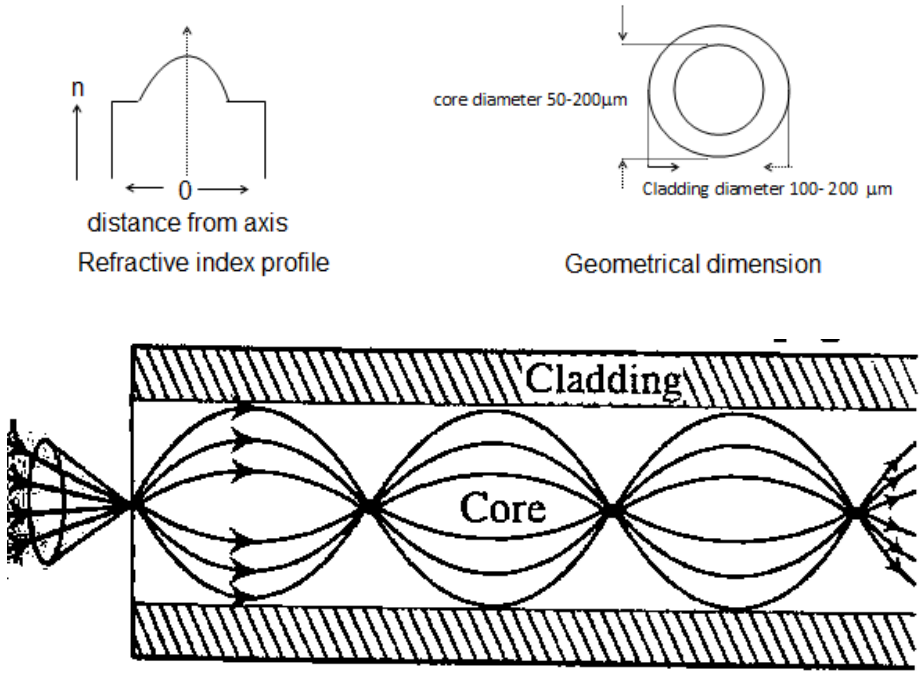


Figure 1.20: Multi mode graded index fibre

1.8 ATTENUATION IN OPTICAL FIBRES

Attenuation is the loss of power of the light signal that occurs during its propagation through the optical fibre. The main sources of attenuation are

1. absorption
2. scattering
3. bending losses

1.8.1 Absorption

Absorption of light during propagation occurs due to the impurities present in the fibre material and also due to the intrinsic nature of the material itself.

The impurities generally present are

a) Transition metals such as iron, chromium, cobalt, copper etc.
b) The hydroxy ions (OH^-) that enter into the fibre material at the time of fabrication.

The photons absorbed by the impurities may be lost as heat or may be reemitted as light energy of different wavelengths and different phase from the one that is propagated. Hence it results in a loss. Intrinsic absorption occurs by the pure material itself even if the material is free from

impurities and inhomogeneities. Intrinsic absorption though quite less compared to the loss due to the impurities, it cannot be eliminated.

1.8.2 Scattering

Glass is a heterogeneous mixture of oxides of silicon, phosphorus, germanium etc. Structural inhomogeneities in the core index will set in the fibre material during solidification of glass from its molten state. It will also result in a fluctuation of the molecular density. These inhomogeneities act as scattering centers. Since their dimensions are smaller than λ the wavelength of the light propagated through the fibre, the energy loss that occurs due to such scattering resemble Rayleigh scattering that is proportional to $\frac{1}{\lambda^4}$. The losses due to these scattering cannot be eliminated by any process. There are other structural inhomogeneities & defects that set in during fabrication of the fibre that contribute to the loss due to scattering. Their sources are trapped gas bubbles, unreacted starting materials etc. However, these can be reduced to a great extent by improved methods of manufacturing.

1.8.3 Bending losses

a) Due to dimensional irregularities and imperfections in the fibres (that are called microscopic bends) (Figure 1.21) the light may not sustain total internal reflection. The energy will escape from the core.

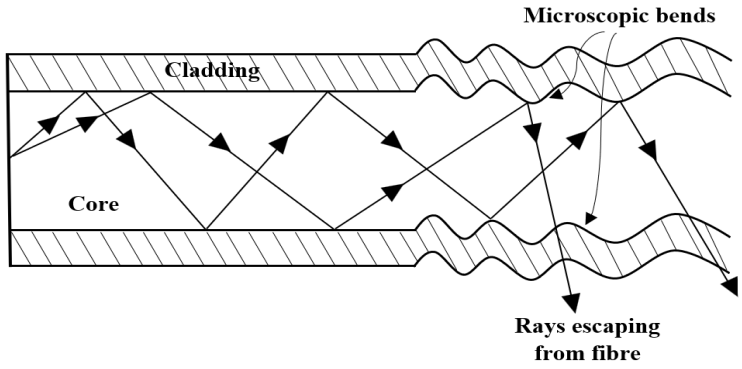


Figure 1.21: Signal attenuation in optical fibre due to microscopic bending

b) Macroscopic bends occur during wrapping the fibre on a spool or negotiating a curve during cable laying. Fibres can withstand bends of curvature up to about 10cm without significant loss. For higher curvature (smaller radius of curvature) than this, the loss increases exponentially.

Amplification is therefore needed in communication applications at regular intervals in order to compensate for the losses that occur despite all precautions. An optical repeater is used to boost the signal.

1.9 DISTORTION IN OPTICAL FIBRES

A light pulse launched into a fibre decreases in amplitude, as it travels along the fibre, due to losses in the fibre. In addition, it also spreads during its travel. The pulse received at the output is wider than input pulse, as shown in Figure 1.22. Such distortion arises due to dispersion effects.

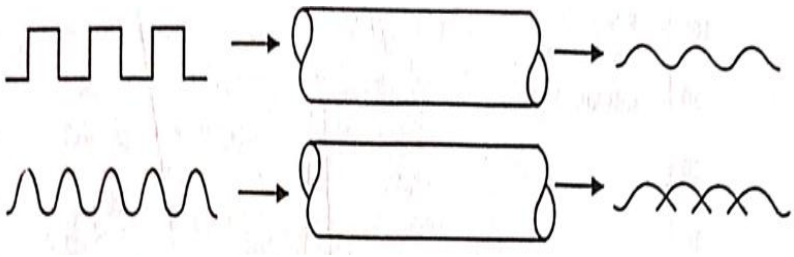


Figure 1.22: Broadening of the signal due to distortion

There are three mechanisms which contribute to the distortion (signal spreading) of the light pulse in a fibre.

1.9.1 Material dispersion

The material dispersion is due to the dependence of the refractive indices of glass and consequently the group velocity of the optical signal. A LED emits a broad spectrum of light. Whenever a LED is used as a source, the broad spectrum contains a number of wavelengths. These different wavelengths of light are travelling through optical fibre at different speeds because the refractive indices of the glass vary with wavelengths. The short wavelength waves travel slower than long wavelength waves in a material. Hence, narrow pulses of light tend to broaden as they travel down the optical fibre. This is known as **material dispersion** (Fig. 1.23 (a)). Material dispersion is present in both single mode and multimode fibres. The material dispersion is given by the equation

$$D_m = \frac{\lambda\,(\Delta\lambda)}{c}\,L\,\frac{d^2n}{d\lambda^2} \qquad \textbf{1.20}$$

where λ is the peak wavelength, $\Delta\lambda$ is the spectral width, L is the length of the core, and n is the refractive index of the core.

To cite an example, an LED operating at 820 nm and having a spectral width of 38 nm results in a dispersion of about 3 ns/km in a certain fibre. In the same fiber, dispersion can be reduced to 0.3 ns/km using a laser diode operating at 1140 nm and having a 3 nm spectral width. Thus, material dispersion can be reduced using a more and more monochromatic source.

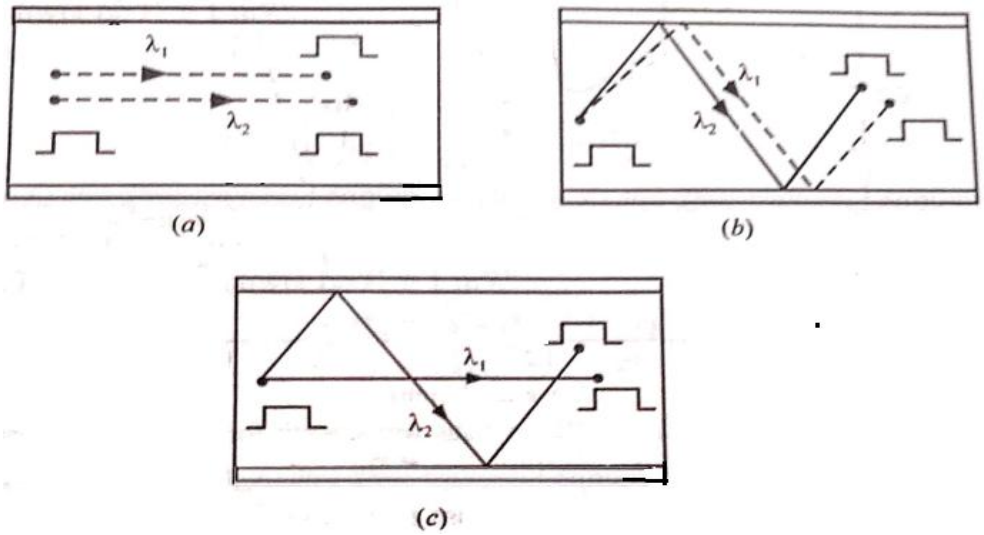


Figure 1.23: (a) Material dispersion (b) Waveguide dispersion (c) Intermodal dispersion

1.9.2 Waveguide dispersion

Waveguide dispersion arises from the guiding properties of the fibre. Pulses at different wavelengths, though propagating in the same mode, travel at slightly different angles. This is known as **waveguide dispersion** (Fig. 1.23 (b)). It exists in both single mode and multimode fibres.

1.9.3 Intermodal dispersion

A ray of light launched into a fibre follows different zigzag paths. When numerous modes are propagating in a fibre, they travel with different net velocities with respect to fibre axis. Parts of the wave arrive at the output before other parts leading to a spread of the input pulse. This is known as **intermodal dispersion** (Fig. 1.23 (c)). It does not depend on the spectral width of the source. A light pulse from an ideal monochromatic source ($\Delta\lambda = 0$) would still exhibit spreading. In single- mode fibres, since there in only one mode, there is no intermodal dispersion.

1.10 APPLICATIONS OF OPTICAL FIBRES

Optical communication is communication at a distance using light to carry information. An optical communication system uses a transmitter, which encodes a message into an optical

signal, a channel (optical fibre), which carries the signal to its destination, and a receiver, which reproduces the message from the received optical signal (Fig.1.24). Optical communication has several advantages – can carry large data in digital form, interference and noise free.

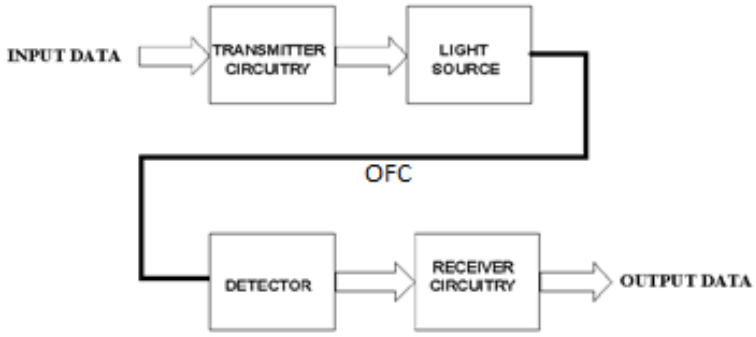


Figure 1.24: Simplified block diagram of optical communication system

Optical fibres are also used in sensors, flexible fibrescope (endoscope) and other industrial applications.

1.11 QUESTIONS

1.1 LASER

1 Mention the characteristics of a laser beam.

2 Explain the following terms with reference to lasers:

(a) spontaneous emission

(b) stimulated emission (c) metastable state

(d) population inversion (e) pumping

(f) active medium (g) resonant cavity.

3 Explain the principle of a laser.

4 Explain construction and operation of ruby laser with necessary diagrams.

5 Explain construction and operation of He-Ne laser with necessary diagrams.

6 Write a note on semiconductor lasers.

7 Mention any four applications of laser.

8 Explain LASER Cooling and its application

1.2 OPTICAL FIBRES

11 What is a critical angle? Define total internal reflection

12 With necessary diagram, derive an expression for angle of acceptance and numerical aperture.

13 Define fractional refractive index change Δ and get the relation between Δ & numerical aperture NA.

14 What is skip distance? With neat diagram, derive an expression for it.

15 What are different types of optical fibres? Briefly explain them with diagrams.

16 What are the different types of attenuations possible in optical fibres? Explain them. Explain different types of distortion in optical fibres.

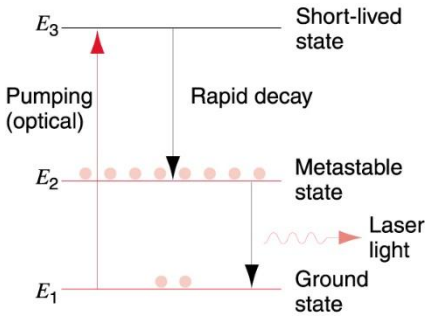
17 Write a note on Fiber optic networking

18 Explain Fiber optic communication

1.12 PROBLEMS

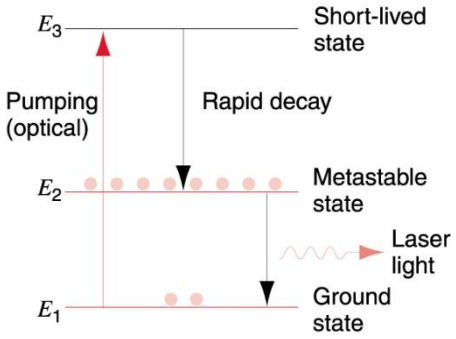
1.1 LASER

1 **A three level laser of the type shown in figure, emits laser light at a wavelength 550 nm, near the centre of the visible band. If the optical mechanism is shut off, what will be the ratio of the population of the upper level E_2 to that of the lower level E_1 at 300 K ? At what temperature for the condition of (a) would the ratio of populations be half ?**



Ans: (a) 1.77×10^{-38} (b) 37800 K

2 A pulsed ruby laser has a ruby rod (Al_2O_3 doped with Cr_2O_3) as an active medium, which is 6 cm long and 1 cm in diameter. There is one chromium ion (active centre, with energy levels of the type shown in the figure) for every 3500 aluminium ions. The ruby laser light has a wavelength of 694.4 nm. Suppose that all the chromium ions are in metastable state (E_2) and none are in ground state (E_1). How much energy is there in a single laser pulse if all these ions come down to ground state in a single stimulated emission chain reaction episode ? Density of Al_2O_3 is 3700 kg/m^3. Molar mass of Al_2O_3 is 0.102 kg/mol.



Ans: Total energy per pulse= 16.9 J

3 **A ruby laser emits light at a wavelength of 694.4 nm. If a laser pulse is emitted for 12.0 ps and the energy release per pulse is 150 mJ, (a) what is the length of the pulse, and (b) how many photons are there in each pulse ?**

(a) 3.6×10^{-3} m (b) 5.25×10^{17}

4 It is entirely possible that techniques for modulating the frequency or amplitude of a laser beam will be developed so that such a beam can serve as a carrier for television signals, much as microwave beams do now. Assume also that laser systems will be available whose wavelengths can be precisely tuned to anywhere in the visible range (400 nm to 700 nm). If a television channel occupies a bandwidth of 10 MHz, how many channels could be accommodated with this laser technology ? Comment on the intrinsic superiority of visible light to microwaves as carriers of information. Ans: 3.27×10^{7}

5 A He-Ne laser emits light at a wavelength of 632.8 nm and has an output power of 2.3 mW. How many photons are emitted each minute by this laser when operating ? Ans: 4.4×10^{17} /min

6 An atom has two energy levels with a transition wavelength of 582 nm. At 300 K, 4.0×10^{20} atoms are in the lower state. (a) How many occupy the upper state under conditions of thermal equilibrium ? (b) Suppose, instead, that 7.0×10^{20} atoms are pumped into upper state, with 4.0×10^{20} atoms in the lower state. How much energy could be released in a single laser pulse ?

Ans: (a) $6.6 \times 10^{-16} \approx 0$ (b) 240 J

7 **Calculate the ratio of, (i) Einstein's coefficients (ii) rate of stimulated to spontaneous emissions, for a system at 300 K in which radiations of wavelength 1.39 μm are emitted.** [Ans: The ratio of $[A_{21}/B_{12}]$ is 6.2×10^{-15}, and ratio of rate of stimulated to spontaneous emissions 10^{-15}]

1.2 OPTICAL FIBRE

8 **A step index optical fibre 63.5 μm in core-diameter has a core of refractive index 1.53 and a cladding of index 1.39. Determine (a) the numerical aperture for the fibre, (b) the critical angle for core-cladding interface, (c) the acceptance cone half-angle (the maximum entrance angle) (d) the number of reflections in 1.0 m length of the fibre for a ray at the maximum entrance angle, (e) the number of**

reflections in 1.0 m length of the fibre for a ray at half the maximum entrance angle.

Ans: a) 0.639 b) 65.29° (c) 39.71° (d) 7.25×10^3 (e) 358×10^3

9 A glass optical fibre of refractive index 1.450 is to be clad with another to ensure total internal reflection that will contain light traveling within 5° of the fibre-axis. What maximum index of refraction is allowed for the cladding? Ans: 1.444

10 The numerical aperture of an optical fibre is 0.2 when surrounded by air. Determine the refractive index of its core. The refractive index of the cladding is 1.59. Also find the acceptance cone half-angle when the fibre in water. Refractive index of water is 1.33.

Ans: 1.6, 8.65°

11 The angle of acceptance of an optical fibre is 30° when kept in air. Find the angle of acceptance when it is in a medium of refractive index 1.33. Ans: 22°

2 QUANTUM PHYSICS

2.1 BLACKBODY RADIATION AND PLANCK'S HYPOTHESIS

A black body is an object that absorbs all incident radiation. A small hole cut into a cavity is the most popular and realistic example. None of the incident radiation escapes. The radiation is absorbed in the walls of the cavity. This causes a heating of the cavity walls. The oscillators in the cavity walls vibrate and re-radiate at wavelengths corresponding to the temperature of the cavity, thereby producing standing waves. Some of the energy from these standing waves can leave through the opening. The electromagnetic radiation emitted by the black body is called **black-body radiation.**

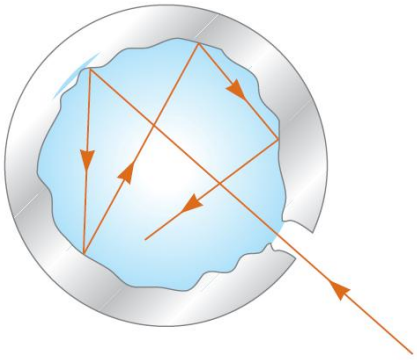


Figure 2.1: A physical model of a blackbody.

The black body is an ideal absorber of incident radiation.

- A black-body reaches thermal equilibrium with the surroundings when the incident radiation power is balanced by the power re-radiated.
- The emitted "thermal" radiation from a black body characterizes the equilibrium temperature of the black-body.
- The nature of radiation from a blackbody does not depend on the material of which the walls are made.

2.1.1 Basic laws of radiation

(1) All objects emit radiant energy.

(2) Hotter objects emit more energy (per unit area) than colder objects. The total power of the emitted radiation is proportional to the fourth power of temperature. This is called **Stefan's Law** and is given by,

$$P = \sigma A e T^4 \qquad \textbf{2-1}$$

where P is power radiated from the surface of the object (W), T is equilibrium surface temperature (K), σ is Stefan-Boltzmann constant (= 5.670 x 10^{-8} W/m^2K^4), A is surface area of the object (m^2) and e is emissivity of the surface (e =1 for a perfect blackbody).

(3) The peak of the wavelength distribution shifts to shorter wavelengths as the black body temperature increases. This is **Wien's Displacement Law** and is given by,

$$\lambda_m T = \text{constant} = 2.898 \times 10^{-3}\ \text{m.K}, \text{ or } \lambda_m \propto T^{-1} \qquad \textbf{2-2}$$

where λ_m is the wavelength corresponding to peak intensity and T is equilibrium temperature of the blackbody.

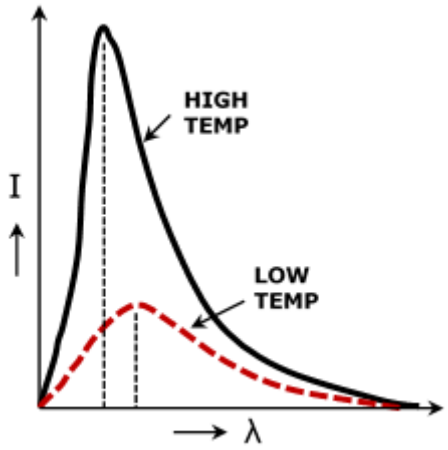


Figure 2.2: Intensity of blackbody radiation versus wavelength at two temperatures.

(4) **Rayleigh-Jeans Law**: This law tries to explain the distribution of energy from a black body. The intensity or power per unit area $I(\lambda,T)d\lambda$, emitted in the wavelength interval λ to $\lambda+d\lambda$ from a blackbody is given by,

$$I(\lambda,T) = \frac{2\pi c k_B T}{\lambda^4} \qquad \textbf{2-3}$$

k_B is Boltzmann's constant, c is speed of light in vacuum, T is equilibrium blackbody temperature. It agrees with experimental measurements only for long wavelengths. It predicts an energy output that diverges towards infinity as wavelengths become smaller and is known as the **ultraviolet catastrophe**.

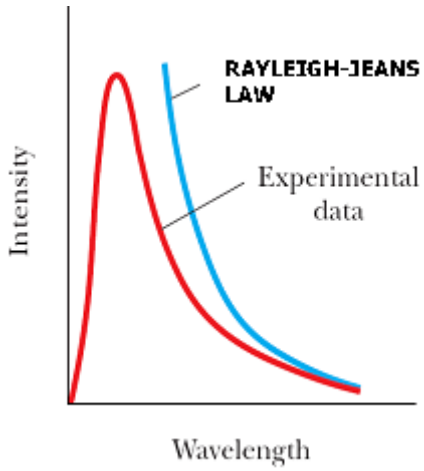


Figure 2.3: Comparison of experimental results and the curve predicted by the Rayleigh–Jeans law for the distribution of blackbody radiation

5) **Planck's Law**:

Max Planck developed a theory of blackbody radiation that leads to an equation for $I(\lambda,T)$ that is in complete agreement with experimental results. To derive the law, Planck made two assumptions concerning the nature of the oscillators in the cavity walls:

(i) The energy of an oscillator is quantized hence it can have only certain discrete values:

$$E_n = n\,h\,f \qquad \textbf{2-4}$$

where ***n*** is a positive integer called a **quantum number**, ***f*** is the **frequency** of cavity oscillators, and ***h*** is a constant called **Planck's constant**. Each discrete energy value corresponds to a different quantum state, represented by the quantum number *n*.

(ii) The oscillators emit or absorb energy only when making a transition from one quantum state to another. Difference in energy will be integral multiples of *hf*.

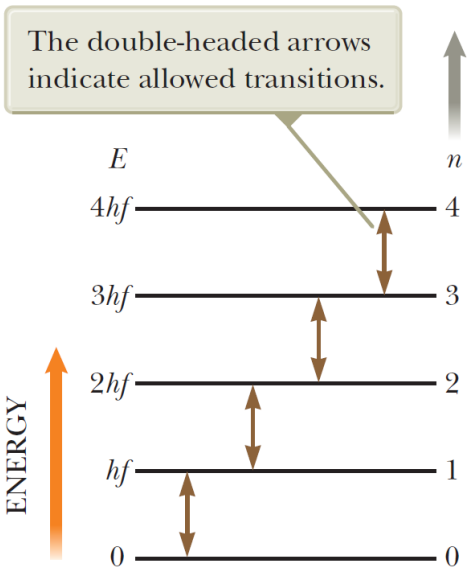


Figure 2.4: Allowed energy levels for an oscillator with frequency f

Planck's law explains the distribution of energy from a black body which is given by,

$$I(\lambda,T) = \frac{2\pi hc^2}{\lambda^5}\frac{1}{e^{\frac{hc}{\lambda k_B T}}-1} \qquad \textbf{2-5}$$

where $I(\lambda,T)\,d\lambda$ is the intensity or power per unit area emitted in the wavelength interval $d\lambda$ from a blackbody, h is Planck's constant, k_B is Boltzmann's constant, c is speed of light in vacuum and T is equilibrium temperature of blackbody .

The Planck's Law gives a distribution that peaks at a certain wavelength, the peak shifts to shorter wavelengths for higher temperatures, and the area under the curve grows rapidly with increasing temperature. This law is in agreement with the experimental data.

The results of Planck's law:

- The denominator $[exp(hc/\lambda kT)]$ tends to infinity faster than the numerator (λ^{-5}), thus resolving the ultraviolet catastrophe and hence arriving at experimental observation:
  $I(\lambda, T) \to 0$ as $\lambda \to 0$.
- For very large λ, $exp\left(\frac{hc}{\lambda kT}\right) - 1 \cong \frac{hc}{\lambda kT} \Rightarrow I(\lambda,T) \to 2\pi c\lambda^{-4} kT$

  i.e. $I(\lambda, T) \to 0$ as $\lambda \to \infty$.

From a fit between Planck's law and experimental data, Planck's constant was derived to be $\boldsymbol{h = 6.626 \times 10^{-34}}$ **J-s.**

2.2 PHOTOELECTRIC EFFECT

Ejection of electrons from the surface of certain metals when it is irradiated by an electromagnetic radiation of suitable frequency is known as photoelectric effect.

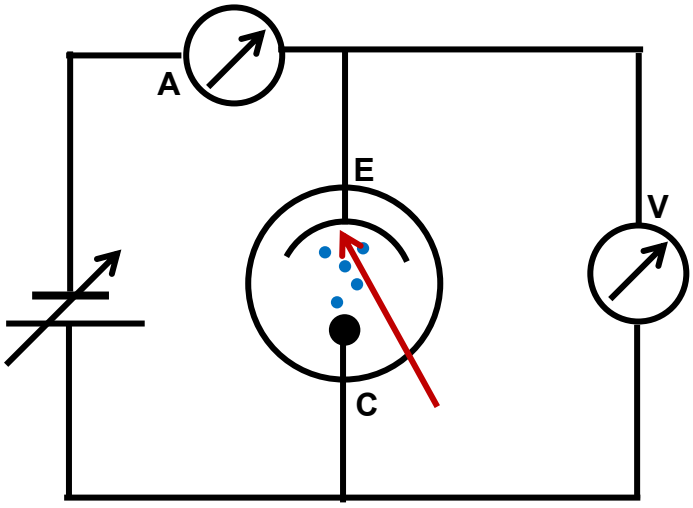


Figure 2.5: (a) Apparatus (b) circuit for studying Photoelectric Effect (T – Evacuated glass/ quartz tube, E – Emitter Plate / Photosensitive material / Cathode, C – Collector Plate / Anode, V – Voltmeter, A - Ammeter)

2.2.1 Experimental observations

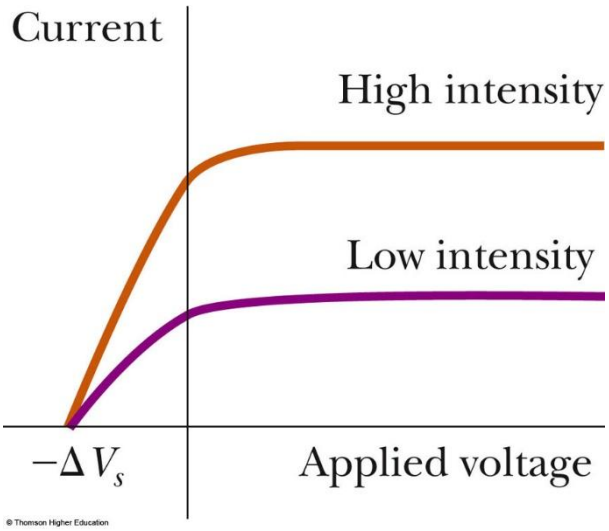


Figure 2.6: Photoelectric current versus applied potential difference for two light intensities

1. When plate *E* is illuminated by light of suitable frequency, electrons are emitted from *E* and a current is detected in *A* (Figure 2.5).
2. Photocurrent produced vs potential difference graph shows that kinetic energy of the most energetic photoelectrons is,

   $$\mathbf{K_{max} = e\,\Delta V_s} \qquad \textbf{2-6}$$

   where ΔV_s is stopping potential
3. Kinetic energy of the most energetic photoelectrons is independent of light intensity.
4. Electrons are emitted from the surface of the emitter almost instantaneously
5. No electrons are emitted if the incident light frequency falls below a cutoff frequency.
6. Kinetic energy of the most energetic photoelectrons increases with increasing light frequency.

2.2.2 Classical predictions

1. If light is really a wave, it was thought that if one shine of light of any fixed wavelength, at sufficient intensity on the emitter surface, electrons should absorb energy continuously from the **em** waves and electrons should be ejected.
2. As the intensity of light is increased (made it brighter and hence classically, a more energetic wave), kinetic energy of the emitted electrons should increase.
3. Measurable / larger time interval between incidence of light and ejection of photoelectrons.
4. Ejection of photoelectron should not depend on light frequency
5. In short experimental results contradict classical predictions.
6. Photoelectron kinetic energy should not depend upon the frequency of the incident light.

2.2.3 Einstein's interpretation of electromagnetic radiation

1. Electromagnetic waves carry discrete energy packets (light quanta called photons now).
2. The energy E, per packet depends on frequency f: E = hf.
3. More intense light corresponds to more photons, not higher energy photons.
4. Each photon of energy E moves in vacuum at the speed of light: $c = 3 \times 10^8$ m/s and each photon carries a momentum, p = E/c.

Einstein's theory of photoelectric effect:

A photon of the incident light gives all its energy hf to a single electron (absorption of energy by the electrons is not a continuous process as envisioned in the wave model) and the kinetic energy of the most energetic photoelectron,

$$\mathbf{K_{max} = hf - \phi \ (Einstein's\ photoelectric\ equation)} \qquad \textbf{2-7}$$

ϕ is called the work function of the metal. It is the minimum energy with which an electron is bound in the metal.

All the observed features of photoelectric effect could be explained by Einstein's photoelectric equation:

1. Equation shows that K_{max} depends only on frequency of the incident light.
2. Almost instantaneous emission of photoelectrons due to one -to –one interaction between photons and electrons.
3. Ejection of electrons depends on light frequency since photons should have energy greater than the work function ϕ in order to eject an electron.
4. The cutoff frequency f_c is related to ϕ by $f_c = \phi/h$. If the incident frequency f is less than f_c, there is no emission of photoelectrons.

The graph of kinetic energy of the most energetic photoelectron K_{max} vs frequency f is a straight line, according to Einstein's equation.

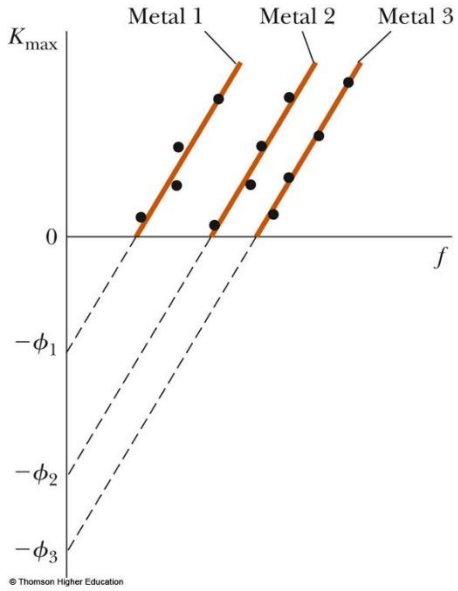


Figure 2.7: A representative plot of K_{max} versus frequency of incident light for three different metals

2.3 COMPTON EFFECT

When X-rays are scattered by free/nearly free electrons, they suffer a change in their wavelength which depends on the scattering angle. This scattering phenomenon is known as Compton Effect.

2.3.1 Classical Predictions

Oscillating **electromagnetic waves** (classically, X-rays are em waves) incident on electrons should have two effects: i) oscillating electromagnetic field causes oscillations in electrons. Each electron first absorbs radiation as a moving particle and then re-radiates in all directions as a moving particle and thereby exhibiting two Doppler shifts in the frequency of radiation. ii) radiation pressure should cause the electrons to accelerate in the direction of propagation of the waves. Because different electrons will move at different speeds after the interaction, depending on the amount of energy absorbed from electromagnetic waves, the scattered waves at a given angle will have all frequencies (Doppler- shifted values).

Compton's experiment and observation: Compton measured the intensity of scattered X-rays from a solid target (graphite) as a function of wavelength for different angles. The experimental setup is shown in Figure 2.8. Contrary to the classical prediction, only one frequency for scattered radiation was seen at a given angle. This is shown in the Figure 2.9.

The graphs for three nonzero angles show two peaks, one at λ_o and the other at $\lambda' > \lambda_o$. The shifted peak at λ' is caused by the scattering of X-rays from free electrons. Shift in wavelength was predicted by Compton to depend on scattering angle as

$$\lambda' - \lambda = \frac{h}{m c}(1 - \cos\theta) \qquad \text{2-8}$$

where m is the mass of the electron, c is velocity of light, h is Planck's constant.

This is known as Compton shift equation, and the factor $\frac{h}{m c}$ is called the Compton wavelength and $\frac{h}{m c}$ = 2.43 pm for electron.

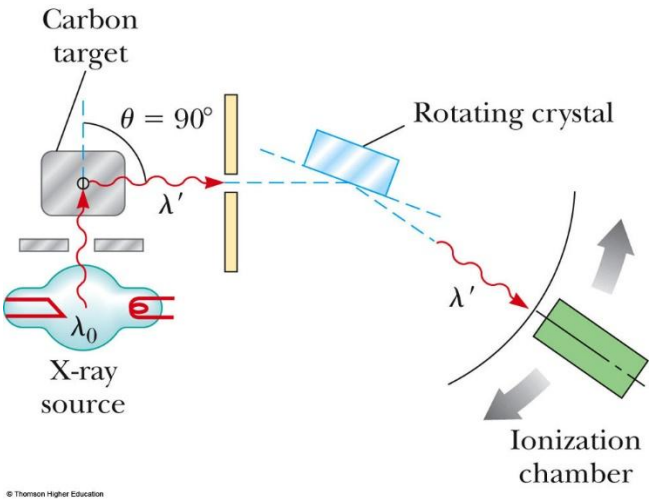


Figure 2.8 Schematic diagram of Compton's apparatus. The wavelength is measured with a rotating crystal spectrometer for various scattering angles θ.

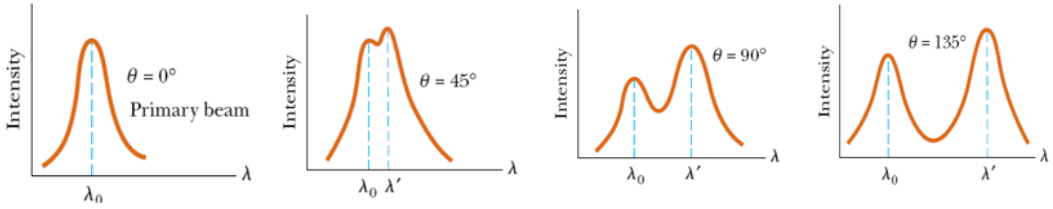


Figure 2.9: Scattered x-ray intensity versus wavelength for Compton scattering at θ = 0°, 45°, 90°, and 135° showing single frequency at a given angle

2.3.2 Derivation of the Compton shift equation

Compton could explain the experimental result by treating the X-rays not as waves but rather as point like particles (photons) having energy $E = hf_o = hc/\lambda_o$, momentum $p = hf/c = h/\lambda$ and zero rest energy. Photons collide elastically with free electrons initially at rest and moving relativistically after collision.

Let λ_o , $p_o = h/\lambda_o$ and $E_o = hc/\lambda_o$ be the wavelength, momentum and energy of the incident photon respectively. λ', $p' = h/\lambda'$ and $E' = hc/\lambda'$ be the corresponding quantities for the scattered photon.

We know that, for the electron, the total relativistic energy $E = \sqrt{p^2c^2 + m^2c^4}$

Kinetic energy $K = E - m c^2$

And momentum $p = \gamma mv$. where $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$

v and m are the speed and mass of the electron respectively.

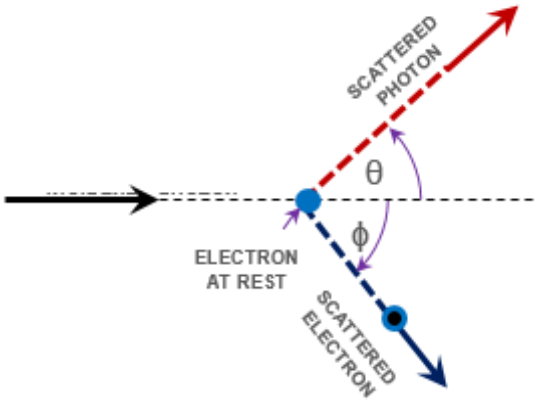


Figure 2.10: Quantum model for X-ray scattering from an electron

In the scattering process, the **total energy and total linear momentum of the system must be conserved.**

For conservation of energy we must have, $E_o = E' + K$

ie, $E_o = E' + (E - m c^2)$

Or $E_o - E' + m c^2 = E = \sqrt{p^2c^2 + m^2c^4}$

Squaring both the sides, $(E_o - E')^2 + 2(E_o - E')\, mc^2 + m^2c^4 = p^2c^2 + m^2c^4$

For conservation of momentum, x-component: $p_o = p' \cos\theta + p \cos\phi$

y-component: $0 = p' \sin\theta - p \sin\phi$

Rewriting these two equations

$$p_o - p' \cos\theta = p \cos\phi$$

$$p' \sin\theta = p \sin\phi$$

Squaring both the sides and adding,

$$p_o^2 - 2p_o p' \cos\theta + p'^2 = p^2$$

Substituting this p^2 in the equation :

$(E_o - E')^2 + 2(E_o - E')\, mc^2 = p^2c^2$, one gets

$$(E_o - E')^2 + 2(E_o - E')\, mc^2 = (p_o^2 - 2p_o p' \cos\theta + p'^2)c^2$$

Substituting photon energies and photon momenta one gets

$$\left(\frac{hc}{\lambda_o} - \frac{hc}{\lambda'}\right)^2 + 2\left(\frac{hc}{\lambda_o} - \frac{hc}{\lambda'}\right) mc^2 = \left(\frac{hc}{\lambda_o}\right)^2 - 2\left(\frac{hc}{\lambda_o}\right)\left(\frac{hc}{\lambda'}\right)\cos\theta + \left(\frac{hc}{\lambda'}\right)^2$$

Simplifying one gets

$$\left(\frac{hc}{\lambda_o}\right)^2 - 2\left(\frac{hc}{\lambda_o}\right)\left(\frac{hc}{\lambda'}\right) + \left(\frac{hc}{\lambda'}\right)^2 + 2\,hc\left(\frac{1}{\lambda_o} - \frac{1}{\lambda'}\right)mc^2 = \left(\frac{hc}{\lambda_o}\right)^2 - 2\left(\frac{hc}{\lambda_o}\right)\left(\frac{hc}{\lambda'}\right)cos\,\theta + \left(\frac{hc}{\lambda'}\right)^2$$

i.e., $$-\frac{hc}{\lambda_o\lambda'} + \left(\frac{1}{\lambda_o} - \frac{1}{\lambda'}\right)mc^2 = -\frac{hc}{\lambda_o\lambda'}cos\,\theta$$

OR, $$\left(\frac{\lambda'-\lambda_o}{\lambda_o\lambda'}\right)mc^2 = \frac{hc}{\lambda_o\lambda'}(1 - cos\,\theta)$$

Compton shift:

$$\boldsymbol{\lambda' - \lambda_o = \frac{h}{mc}(1 - cos\,\theta)}$$

2.4 DUAL NATURE OF LIGHT

- Light exhibits diffraction and interference phenomena that are only explicable in terms of wave properties.
- Photoelectric effect and Compton Effect can only be explained taking light as photons / particle.
- This means true nature of light is not describable in terms of any single picture, instead both wave and particle nature have to be considered. In short, the particle model and the wave model of light complement each other.

2.5 DE BROGLIE HYPOTHESIS - WAVE PROPERTIES OF PARTICLES

We have seen that light comes in discrete units (photons) with particle properties (energy E and momentum p) that are related to the wave-like properties of frequency and wavelength. Louis de Broglie postulated that because photons have both wave and particle characteristics, perhaps all forms of matter have wave-like properties, with the wavelength λ related to momentum p in the same way as for light.

de Broglie wavelength: $\boldsymbol{\lambda = \frac{h}{p} = \frac{h}{mv}}$ **2-9**

where h is Planck's constant and p is momentum of the quantum particle, m is mass of the particle, and v is speed of the particle. The electron accelerated through a potential difference of ΔV, has a non-relativistic kinetic energy $\frac{1}{2}m\,v^2 = e\,\Delta V$ where e is electron charge.

Hence, the momentum (p) of an electron accelerated through a potential difference of ΔV is

$$\boldsymbol{p = m\,v = \sqrt{2\,m\,e\,\Delta V}} \qquad \textbf{2-10}$$

Frequency of the matter wave associated with the particle is $\frac{E}{h}$, where E is total relativistic energy of the particle

2.5.1 Davisson and Germer experiment

This is an experiment on diffraction of accelerated electrons by crystals to establish the wave nature of electrons. The regular spacing of the atoms in a crystal act as a grating for an electron beam producing a diffraction pattern by electron scattering. The result of this experiment resembles the result of the x ray diffraction giving experimental evidence for de Broglie hypothesis.

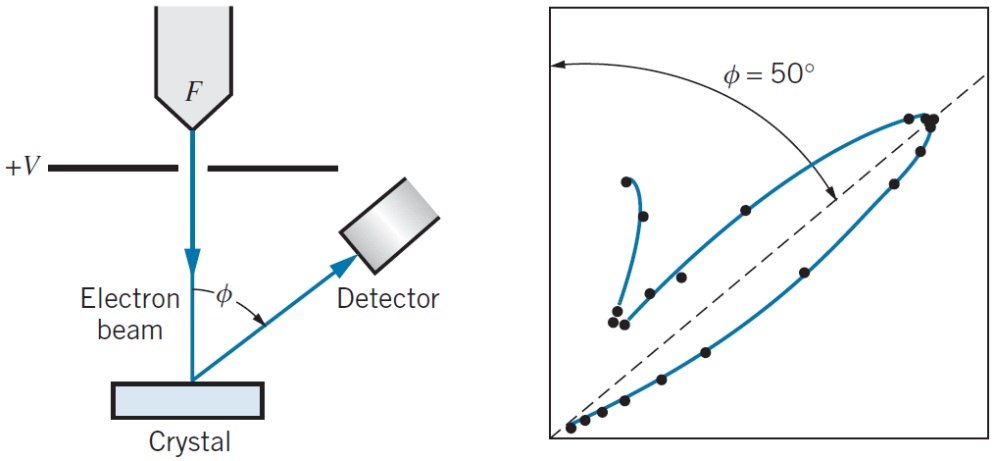


Figure 2.11: Schematic of Davisson-Germer experiment (left). Polar plot of intensities after scattering for incident electron energy of 54 eV. (right).

A beam of electrons from a heated filament accelerated to a potential V is collimated and allowed to strike a single crystal of nickel. Electrons are scattered in all directions by the atoms in the crystal. The intensity of the scattered electron beam is measured by a detector which can be moved to any angle ϕ relative to the incident beam and is shown in the polar plot of ϕ verses the intensity ($\propto$ r, the radius vector). The most intense reflection of electron beam at an angle $\phi = 50^{\circ}$, for an accelerating potential 54 V is observed.

An electron of mass m accelerated to a potential V has

kinetic energy, $\frac{1}{2}mv^2 = e\,V$ where , $v =$ velocity.

And the momentum, $p = m\,v = \sqrt{2\,m\,e\,V}$

$\therefore$ de Broglie wavelength, $\lambda = \frac{h}{p} = \frac{h}{m\,v} = \frac{h}{\sqrt{2\,m\,e\,V}}$

Substituting V = 54 V, we get the value for the de Broglie wavelength as **λ = 0.167nm.**

Wavelength can also be obtained, in this experiment, using grating equation. According to which the wavelength is given by the following relationship:

$d\ sin\phi = n\lambda$

Substituting n = 1 (for the first order diffraction maximum), $\phi = 50^{\circ}$ (scattering angle corresponding to the first order diffraction maximum), $d = 0.215\ nm$ (inter-atomic spacing in nickel), **we get λ = 0.165 nm**

The values of the wavelength from both the approaches agree very well. This verifies de Broglie's hypothesis and establishes the wave nature of electrons.

Davisson-Germer experiment and G P Thomson's electron diffraction experiment confirmed de Broglie relationship $p = h/\lambda$. Subsequently it was found that atomic beams, and beams of neutrons, also exhibit diffraction when reflected from regular crystals. Thus de Broglie's formula seems to apply to any kind of matter. Now the dual nature of matter and radiation is an accepted fact and it is stated in the principle of complementarity. This states that wave and particle models of either matter or radiation complement each other.

2.6 THE QUANTUM PARTICLE

Quantum particle is a model by which particles having dual nature are represented. We must choose one appropriate behavior for the quantum particle (particle or wave) in order to understand a particular behavior.

To represent a quantum wave, we have to combine the essential features of both an ideal particle and an ideal wave. An essential feature of a particle is that it is localized in space. But an ideal wave is infinitely long (non-localized) as shown in Figure 2.12.

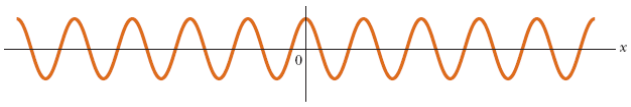


Figure 2.12: Section of an ideal wave of single frequency

Now to build a localized entity from an infinitely long wave, waves of same amplitude, but slightly different frequencies are superposed (Figure 2.13).

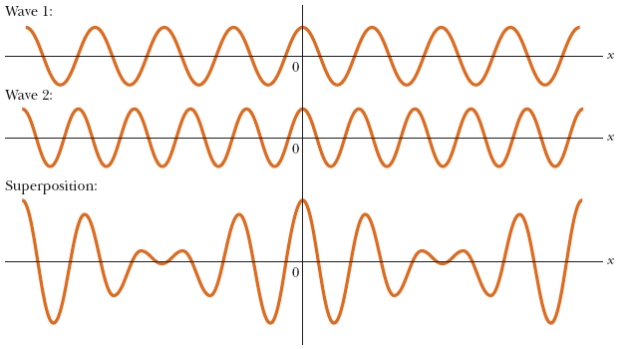


Figure 2.13: Superposition of two waves Wave 1 and Wave 2

If we add up large number of waves in a similar way, the small localized region of space where constructive interference takes place is called a **wave packet**, which represents a quantum particle (Figure 2.14).

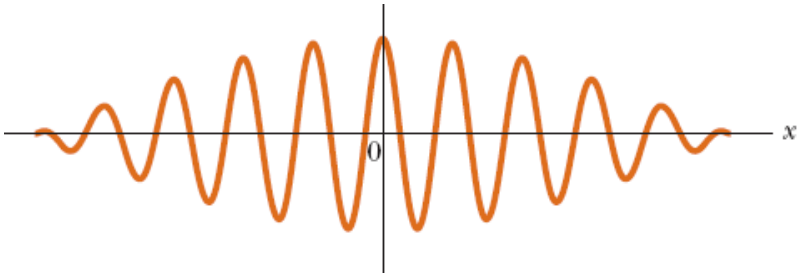


Figure 2.14: Wave packet

2.6.1 Mathematical representation of a wave packet

Superposition of two waves of equal amplitude, but with slightly different frequencies, f_1 and f_2, traveling in the same direction are considered. The waves are written as

$$y_1 = A\cos(k_1x - \omega_1 t) \quad \text{and} \quad y_2 = A\cos(k_2x - \omega_2 t)$$

$$\text{where } k = 2\pi/\lambda, \qquad \omega = 2\pi f$$

The resultant wave $y = y_1 + y_2$

$$y = 2A\left[\cos\left(\frac{\Delta k}{2}x - \frac{\Delta\omega}{2}t\right)\cos\left(\frac{k_1+k_2}{2}x - \frac{\omega_1+\omega_2}{2}t\right)\right]$$

where $\Delta k = k_1 - k_2$ and $\Delta\omega = \omega_1 - \omega_2$.

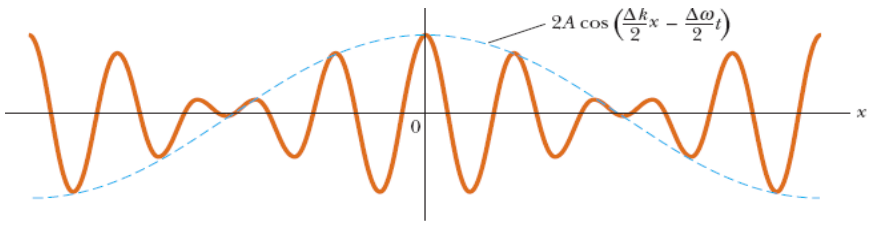


Figure 2.15: Beat pattern due to superposition of wave trains y1 and y2

The resulting wave oscillates with the average frequency, and its amplitude envelope (in square brackets, shown by the blue dotted curve in Figure 2.14) varies according to the difference

frequency. A realistic wave (one of finite extent in space) is characterized by two different speeds. The **phase speed**, the speed with which wave crest of individual wave moves, is given by

$$v_p = f\lambda \quad \text{or} \quad v_p = \frac{\omega}{k} \tag{2.11}$$

The envelope of group of waves can travel through space with a different speed than the individual waves. This speed is called the **group speed** or the speed of the wave packet which is given by

$$v_g = \frac{\left(\frac{\Delta\omega}{2}\right)}{\left(\frac{\Delta k}{2}\right)} = \frac{\Delta\omega}{\Delta k} \tag{2.12}$$

For a superposition of large number of waves to form a wave packet, this ratio is $v_g = \frac{d\omega}{dk}$

In general these two speeds are not the same.

2.6.2 Relation between group speed (v_g) and phase speed (v_p)

$$v_P = \frac{\omega}{k} = f\lambda \qquad \therefore \qquad \omega = k\,v_P$$

But $$v_g = \frac{d\omega}{dk} = \frac{d(kv_P)}{dk} = k\frac{dv_P}{dk} + v_P$$

Substituting for k in terms of λ, we get

$$v_g = v_P - \lambda\left(\frac{dv_P}{d\lambda}\right) \tag{2.13}$$

2.6.3 Relation between group speed (v_g) and particle speed (u)

$$\omega = 2\pi f = 2\pi\frac{E}{h} \quad \text{and} \quad k = \frac{2\pi}{\lambda} = \frac{2\pi}{h/p} = \frac{2\pi p}{h}$$

$$v_g = \frac{d\omega}{dk} = \frac{\frac{2\pi}{h}dE}{\frac{2\pi}{h}dp} = \frac{dE}{dp}$$

For a classical particle moving with speed u, the kinetic energy E is given by

$$E = \tfrac{1}{2}mu^2 = \frac{p^2}{2m} \quad \text{and} \quad dE = \frac{2p\,dp}{2m} \quad \text{or} \quad \frac{dE}{dp} = \frac{p}{m} = u$$

$$v_g = \frac{d\omega}{dk} = \frac{dE}{dp} = u \tag{2.14}$$

i.e., we should identify the group speed with the particle speed, speed with which the energy moves. To represent a realistic wave packet, confined to a finite region in space, we need the superposition of large number of harmonic waves with a range of k-values.

2.7 UNCERTAINTY PRINCIPLE

It is fundamentally impossible to make simultaneous measurements of a particle's position and momentum with infinite accuracy. This is known as **Heisenberg uncertainty principle**. The uncertainties arise from the quantum structure of matter.

For a particle represented by a single wavelength wave existing throughout space, λ is precisely known, and according to de Broglie hypothesis, its p is also known accurately. But the position of the particle in this case becomes completely uncertain.

This means $\Delta\lambda = 0$, $\Delta p = 0$; but $\Delta x = \infty$

In contrast, if a particle whose momentum is uncertain (combination of waves / a range of wavelengths are taken to form a wave packet), so that Δx is small, but $\Delta\lambda$ is large. If Δx is made zero, $\Delta\lambda$ and thereby Δp will become ∞.

In short $$(\Delta x)(\Delta p_x) \geq h/4\pi \qquad (2.15)$$

where Δx is uncertainty in the measurement of position x of the particle and Δp_x is uncertainty in the measurement of momentum p_x of the particle.

One more relation expressing uncertainty principle is related to energy and time which is given by

$$(\Delta E)(\Delta t) \geq h/4\pi \qquad (2.16)$$

where ΔE is uncertainty in the measurement of energy E of the system when the measurement is done over the time interval Δt.

2.8 QUESTIONS

1. Explain (a) Stefan's law (b) Wien's displacement law (c) Rayleigh-Jeans law.
2. Sketch schematically the graph of wavelength vs intensity of radiation from a blackbody.
3. Explain Planck's radiation law.
4. Write the assumptions made in Planck's hypothesis of blackbody radiation.
5. Explain photoelectric effect.
6. What are the observations in the experiment on photoelectric effect?
7. What are the classical predictions about the photoelectric effect?
8. Explain Einstein's photoelectric equation.
9. Which are the features of photoelectric effect-experiment explained by Einstein's photoelectric equation?

10. Sketch schematically the following graphs with reference to the photoelectric effect: (a) photoelectric current vs applied voltage (b) kinetic energy of most-energetic electron vs frequency of incident light.

11. Explain Compton effect.

12. Explain the experiment on Compton effect.

13. Derive the Compton shift equation.

14. Explain the wave properties of the particles.

15. How Davison and Germer experiment validated the de Broglie hypothesis? Explain.

16. Explain a wave packet and represent it schematically.

17. Explain (a) group speed (b) phase speed, of a wave packet.

18. Show that the group speed of a wave packet is equal to the particle speed.

19. (a) Name any two phenomena which confirm the particle nature of light.

(b) Name any two phenomena which confirm the wave nature of light.

20. Explain Heisenberg uncertainty principle.

21. Write the equations for uncertainty in (a) position and momentum (b) energy and time.

22. Mention two situations which can be well explained by the uncertainty relation.

2.9 PROBLEMS

1 Find the peak wavelength of the blackbody radiation emitted by each of the following.
   A. The human body when the skin temperature is 35°C
   B. The tungsten filament of a light bulb, which operates at 2000 K
   C. The Sun, which has a surface temperature of about 5800 K.
   Ans: 9.4 μm, 1.4 μm, 0.50 μm

2 A 2.0- kg block is attached to a spring that has a force constant of k = 25 N/m. The spring is stretched 0.40 m from its equilibrium position and released.
   A. Find the total energy of the system and the frequency of oscillation according to classical calculations.
   B. Assuming that the energy is quantized, find the quantum number n for the system oscillating with this amplitude.
   C. Suppose the oscillator makes a transition from the $n = 5.4 \times 10^{33}$ state to the state corresponding to $n = 5.4 \times 10^{33} - 1$. By how much does the energy of the oscillator change in this one-quantum change.
   Ans: 2.0 J, 0.56 Hz, 5.4×10^{33}, 3.7×10^{-34} J

3 The human eye is most sensitive to 560 nm light. What is the temperature of a black body that would radiate most intensely at this wavelength?

Ans: 5180 K

4 **A blackbody at 7500 K consists of an opening of diameter 0.050 mm, looking into an oven. Find the number of photons per second escaping the hole and having wavelengths between 500 nm and 501 nm.**

Ans: 1.30×10^{15}/s

5 **The radius of our Sun is 6.96×10^{8} m, and its total power output is 3.77×10^{26} W. (a) Assuming that the Sun's surface emits as a black body, calculate its surface temperature. (b) Using the result, find λ_{max} for the Sun.**

Ans: 5750 K, 504 nm

6 Calculate the energy in electron volts, of a photon whose frequency is (a) 620 THz, (b) 3.10 GHz, (c) 46.0 MHz. (d) Determine the corresponding wavelengths for these photons and state the classification of each on the electromagnetic spectrum.

Ans: 2.57 eV, 1.28×10^{-5} eV, 1.91×10^{-7} eV, 484 nm, 9.68 cm, 6.52 m

7 An FM radio transmitter has a power output of 150 kW and operates at a frequency of 99.7 MHz. How many photons per second does the transmitter emit?

Ans: 2.27×10^{30} photons/s

8 A sodium surface is illuminated with light having a wavelength of 300 nm. The work function for sodium metal is 2.46 eV. Find

A. The maximum kinetic energy of the ejected photoelectrons and
B. The cutoff wavelength for sodium.

Ans: 1.67 eV, 504 nm

9 Molybdenum has a work function of 4.2eV. (a) Find the cut off wavelength and cut off frequency for the photoelectric effect. (b) What is the stopping potential if the incident light has wavelength of 180 nm?

Ans: 296 nm, 1.01×10^{15} Hz, 2.71 V

10 Electrons are ejected from a metallic surface with speeds up to 4.60×10^{5} m/s when light with a wavelength of 625 nm is used. (a) What is the work function of the surface? (b) What is the cut-off frequency for this surface?

Ans: 1.38 eV, 3.34×10^{14} Hz

11 **The stopping potential for photoelectrons released from a metal is 1.48 V larger compared to that in another metal. If the threshold frequency for the first metal is 40.0 % smaller than for the second metal, determine the work function for each metal.**

Ans: 3.70 eV, 2.22 eV

12 Two light sources are used in a photoelectric experiment to determine the work function for a metal surface. When green light from a mercury lamp (λ = 546.1 nm) is used, a stopping potential of 0.376 V reduces the photocurrent to zero. (a) Based on this what is the work function of this metal? (b) What stopping potential would be observed when using the yellow light from a helium discharge tube (λ = 587.5 nm)?

Ans: 1.90 eV, 0.215 V

13 X-rays of wavelength λ_o = 0.20 nm are scattered from a block of material. The scattered X-rays are observed at an angle of 45° to the incident beam. Calculate their wavelength. What if we move the detector so that scattered X-rays are detected at an angle larger than 45°? Does the wavelength of the scattered X-rays increase or decrease as the angle θ increase?

Ans: 0.200710 nm, INCREASES

14 Calculate the energy and momentum of a photon of wavelength 700 nm.

Ans: 1.78 eV, 9.47×10^{-28} kg.m/s

15 **A 0. 00160 nm photon scatters from a free electron. For what photon scattering angle does the recoiling electron have kinetic energy equal to the energy of the scattered photon?**

Ans: 70°

16 **A 0.880 MeV photon is scattered by a free electron initially at rest such that the scattering angle of the scattered electron is equal to that of the scattered photon ($\theta = \phi$). (a) Determine the angles θ & ϕ. (b) Determine the energy and momentum of the scattered electron and photon.**

Ans: 43°, 43°, 0.602 MeV, 3.21×10^{-22} kg.m/s, 0.278 MeV, 3.21×10^{-22} kg.m/s

17 Calculate the de- Broglie wavelength for an electron moving at 1.0×10^7 m/s.

Ans: 7.28×10^{-11} m

18 A rock of mass 50 g is thrown with a speed of 40 m/s. What is its de Broglie wavelength?

Ans: 3.3×10^{-34} m

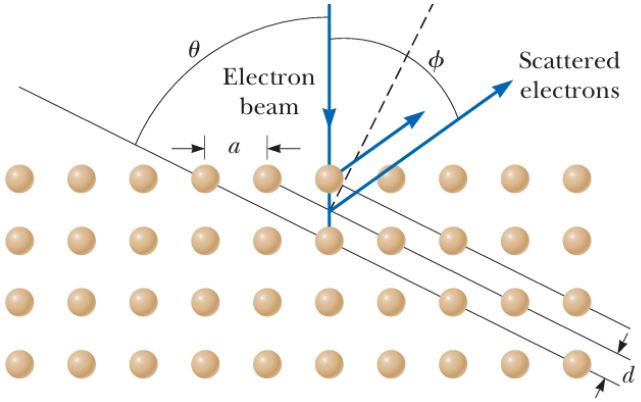
19 A particle of charge q and mass m has been accelerated from rest to a nonrelativistic speed through a potential difference of ΔV. Find an expression for its de Broglie wavelength.

Ans: $\lambda = \frac{h}{\sqrt{2\,m\,q\,\Delta v}}$

20 **(a) An electron has a kinetic energy of 3.0 eV. Find its wavelength. (b) Also find the wavelength of a photon having the same energy.**

Ans: 7.09×10^{-10} m, 4.14×10^{-7} m

21 In the Davisson-Germer experiment, 54.0 eV electrons were diffracted from a nickel lattice. If the first maximum in the diffraction pattern was observed at $\phi = 50.0°$, what was the lattice spacing a between the vertical rows of atoms in the figure?



Ans: 2.18×10^{-10} m

22 Consider a freely moving quantum particle with mass m and speed u. Its energy is E= K= $mu^2/2$. Determine the phase speed of the quantum wave representing the particle and show that it is different from the speed at which the particle transports mass and energy.

Ans: $v_{GROUP} = u \neq v_{PHASE}$

23 The speed of an electron is measured to be 5.00×10^3 m/s to an accuracy of 0.0030%. Find the minimum uncertainty in determining the position of this electron.

Ans: 0.383 mm

24 **The lifetime of an excited atom is given as 1.0×10^{-8} s. Using the uncertainty principle, compute the line width Δf produced by this finite lifetime?**

Ans: 8.0×10^6 Hz

25 **Use the uncertainty principle to show that if an electron were confined inside an atomic nucleus of diameter 2×10^{-15} m, it would have to be moving relativistically, while a proton confined to the same nucleus can be moving nonrelativistically.**

Ans: $v_{ELECTRON} \cong 0.99996\ c$, $v_{PROTON} \cong 1.8 \times 10^7$ m/s

26 Find the minimum kinetic energy of a proton confined within a nucleus having a diameter of 1.0×10^{-15} m.

Ans: 5.2 MeV

3 QUANTUM MECHANICS

OBJECTIVES:

- To learn the application of Schrödinger equation to a bound particle and to learn the quantized nature of the bound particle, its expectation values and physical significance.
- To understand the tunnelling behaviour of a particle incident on a potential barrier.
- To understand the behaviour of quantum oscillator.

3.1 AN INTERPRETATION OF QUANTUM MECHANICS

The amplitude of the de Broglie wave associated with a quantum particle/system is called the probability amplitude, or the wave function, and is denoted by ψ. The wave function ψ contains within it all the information that can be known about a quantum particle/system. In quantum mechanics, the state of the system is completely specified by its wave function ψ. In general, the complete wave function Ψ for a system depends on the positions of all the particles in the system and on time. This can be written as

$$\Psi(r_1, r_2, \dots, r_j, t) = \psi(r_1, r_2, \dots, r_j)e^{i\omega t}$$

where r_j is the position vector of the jth particle in the system. For any system in which the potential energy is time-independent and depends only on the position of particles within the system, the important information about the system is contained within the space part of the wave function.

The wave function for a quantum system will in general be a complex function. Hence it cannot represent a physical observable or quantity. If ψ represents the wave function (probability amplitude) of a single particle, then $|\psi|^2$ is always real and positive. $|\psi|^2$ is called **probability density** – it is the relative probability per unit volume that the particle will be found at any given point in the volume.

3.2 ONE-DIMENSIONAL WAVE FUNCTIONS AND EXPECTATION VALUES

Let ψ be the wave function for a particle moving along the x axis. Then $\boldsymbol{P(x)\,dx = |\psi|^2 dx}$ is the probability to find the particle in the infinitesimal interval dx around the point x. The probability of finding the particle in the arbitrary interval $a \le x \le b$ is

$$P_{ab} = \int_a^b |\psi|^2 \, dx \tag{3.1}$$

The probability of a particle being in the interval $a \leq x \leq b$ is the area under the probability density curve from a to b. The total probability of finding the particle is one. Forcing this condition on the wave function is called **normalization**.

$$\int_{-\infty}^{+\infty} |\psi|^2 \, dx = 1 \tag{3.2}$$

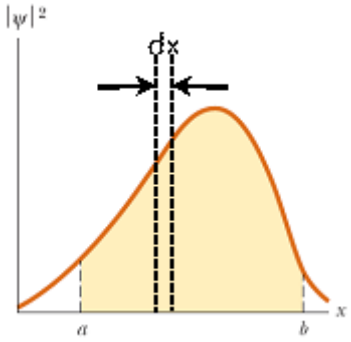


Figure 3.1: An arbitrary probability density curve for a particle

All the measurable quantities of a particle, such as its position, momentum and energy can be derived from the knowledge of ψ. e.g., the average position at which one expects to find the particle after many measurements is called the **expectation value of x** and is defined by the equation

$$\langle x \rangle \equiv \int_{-\infty}^{+\infty} \psi^* \, x \, \psi \, dx \tag{3.3}$$

The important mathematical features of a physically reasonable wave function $\psi(x)$ for a system are

- $\psi(x)$ may be a complex function or a real function, depending on the system.
- $\psi(x)$ must be finite, continuous and single valued everywhere.
- The space derivatives of ψ, must be finite, continuous and single valued everywhere.
- ψ must be normalizable.

3.3 TIME-DEPENDENT SCHRODINGER EQUATION

The Schrödinger equation is a differential equation that governs the wave function of a non-relativistic quantum-mechanical system. It is named after Erwin Schrödinger, an Austrian physicist, who postulated the equation in 1925.

We have studied that matter exhibits wave-like behaviour under suitable conditions. When the momentum of a particle is well defined, the wave can be of infinite extent. Therefore, a free particle moving along x-axis with a well-defined momentum is described by an infinite plane wave $\psi(x,t)$ given by

$$\psi(x,t) = A\exp[i(kx-\omega t)] \tag{3.4}$$

where $k = 2\pi/\lambda$ is wave vector and ω is the angular frequency. In the case of electromagnetic waves, the electromagnetic field varies in space and time. Similarly, sound waves can be described by the pressure variation in space and time. In other words, to describe wave motion, one requires a quantity which varies in space and time. In analogy with these, to describe matter waves associated with particles in motion, one requires a quantity which varies in space and time. This variable quantity, called the wave function $\psi(x,t)$, must be large in regions where the particle is likely to be found and small in the region where it is less likely to be found. That is, the wave function of a particle in conformity with the uncertainty principle must be localized in a small region around it. The wave function of the matter wave which is confined to a small region of space is termed as a wave packet or wave group (Fig. 3.2).

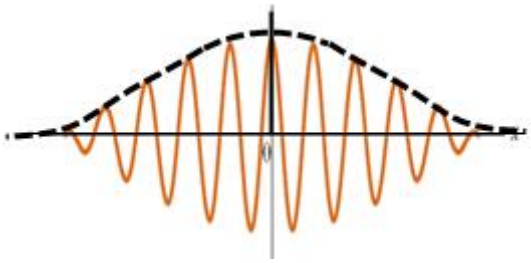


Figure 3.2: Wave packet

Mathematically, a wave packet can be constructed by the superposition of an infinite number of plane waves (as in eq. (3.4)) with slightly differing k values:

$$\Psi(x,t) = \int_{-\infty}^{+\infty} A(k)\exp[i(kx-\omega t)]\ dk \tag{3.5}$$

For a free particle, the classical expression for energy is

$$E = \frac{p^2}{2m} \tag{3.6}$$

Now from the de Broglie relation $\lambda = h/p$, and using $k = 2\pi/\lambda$, we can write

$$p = \hbar k \tag{3.7}$$

Further, the energy E of a quantum particle is related to the frequency ω through the relation

$$E = \hbar\omega \tag{3.8}$$

Using eqns. (3.7 and 3.8) we can rewrite eq. (3.5) as

$$\Psi(x,t) = \int_{-\infty}^{+\infty} A(p)\exp[\frac{i}{\hbar}(px - Et)]\ dp \tag{3.9}$$

Differentiating equation 3.9 with respect to t, we get

$$\frac{d\Psi}{dt} = \int_{-\infty}^{+\infty} A(p)\exp\left[\frac{i}{\hbar}(px - Et)\right]\left(-\frac{i}{\hbar}E\right)dp$$

Or

$$i\hbar\frac{d\Psi}{dt} = \int_{-\infty}^{+\infty} E\ A(p)\exp\left[\frac{i}{\hbar}(px - Et)\right]\ dp \tag{3.10}$$

Differentiating with respect to x, we get

$$\frac{d\Psi}{dx} = \int_{-\infty}^{+\infty} A(p)\exp\left[\frac{i}{\hbar}(px - Et)\right]\left(\frac{i}{\hbar}p\right)dp$$

Differentiating again,

$$\frac{d^2\Psi}{dx^2} = \int_{-\infty}^{+\infty} A(p)\exp\left[\frac{i}{\hbar}(px - Et)\right]\left(\frac{i}{\hbar}p\right)^2 dp$$

Or

$$\frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2} = \int_{-\infty}^{+\infty} -\frac{p^2}{2m}A(p)\exp\left[\frac{i}{\hbar}(px - Et)\right]dp \tag{3.11}$$

Adding (3.10) and (3.11)

$$i\hbar\frac{d\Psi}{dt} + \frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2} = \int_{-\infty}^{+\infty}\left(E - \frac{p^2}{2m}\right)A(p)\exp\left[\frac{i}{\hbar}(px - Et)\right]dp \tag{3.12}$$

Using eq. (3.6) in the above equation, we get

$$i\hbar\frac{d\Psi}{dt} + \frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2} = 0$$

Or

$$i\hbar\frac{d\Psi}{dt} = -\frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2} \tag{3.13}$$

This equation is called the **free-particle Schrodinger equation**.

Now for a particle moving in a region where the potential energy is $U(x,t)$, the classical expression for energy is

$$E = \frac{p^2}{2m} + U(x,t) \tag{3.14}$$

Substituting eq. (3.14) in eq. (3.12),

$$i\hbar\frac{d\Psi}{dt} + \frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2} = \int_{-\infty}^{+\infty}\left(\frac{p^2}{2m} + U(x,t) - \frac{p^2}{2m}\right)A(p)\exp\left[\frac{i}{\hbar}(px - Et)\right]dp$$

Or

$$i\hbar\frac{d\Psi}{dt} + \frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2} = \int_{-\infty}^{+\infty} U(x,t)\,A(p)\exp\left[\frac{i}{\hbar}(px - Et)\right]dp$$

$$= U(x,t)\int_{-\infty}^{+\infty} A(p)\exp\left[\frac{i}{\hbar}(px - Et)\right]dp = U(x,t)\,\Psi(\mathrm{x},\mathrm{t})$$

Thus, for a particle moving in a region where the potential energy is $U(x,t)$, the wavefunction $\Psi(\mathrm{x},\mathrm{t})$ satisfies the differential equation

$$\boldsymbol{i\hbar\frac{d\Psi}{dt} = -\frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2} + U(x,t)\,\Psi(\mathbf{x},\mathbf{t})} \tag{3.15a}$$

Equation $(3.15a)$ can also be written in the form

$$i\hbar\frac{d\Psi}{dt} = \left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x,t)\right)\Psi(\mathrm{x},\mathrm{t})$$

$$\boldsymbol{i\hbar\frac{d\Psi}{dt} = H\,\Psi(\mathbf{x},\mathbf{t})} \tag{3.15b}$$

where

$$H = \left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x,t)\right) \tag{3.16}$$

is an (differential) operator called the Hamiltonian (also called the energy operator). Equation $3.15a$ or $3.15b$ is called the **time-dependent Schrodinger equation**.

3.3.1 Operator for momentum

Consider a free unlocalized quantum particle. According to the uncertainty principle, such a particle will have a precise (well-defined) value for momentum (or wavelength) but not a well-defined value for position. Hence the wave function of this particle can be described by a plane wave of the form given in eq. (3.4)

$$\psi(x,t) = A\exp[i/\hbar(px - Et)] \tag{3.17}$$

where we have used equations (3.7) and (3.8). It can be shown that (3.17) solves the Schrodinger equation (3.15) and hence is valid wave function. Differentiating (3.17) with respect to x we get

$$\frac{d\psi}{dx} = A\exp[i/\hbar(px - Et)]\left(\frac{ip}{\hbar}\right)$$

$$-i\hbar\frac{d\psi}{dx} = p\,\psi(x,t) \tag{3.18}$$

Equation (3.18) suggests that momentum p can be represented by the differential operator

$$p \to -i\hbar\frac{d}{dx} \tag{3.19}$$

3.3.2 Hamiltonian (H) - the energy operator:

For a particle, total energy

$$E = KE + PE = \frac{p^2}{2m} + U(x,t) = \frac{1}{2m}p.p + U(x,t)$$

Using eq.(3.19) in RHS of the above equation

$$E = \frac{p^2}{2m} + U(x,t) = \frac{1}{2m}\left(-i\hbar\frac{d}{dx}\right)\left(-i\hbar\frac{d}{dx}\right) + U(x,t) = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x,t) = H$$

Also, the operator $-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$ is called the *kinetic energy operator*.

3.4 TIME-INDEPENDENT SCHRODINGER EQUATION

The potential energy $U(x,t)$ appearing in the Schrodinger equation (3.15a) is in general a function of both space x and time t. For some systems in quantum mechanics, potential energy is a function of space coordinate x only. For such cases, we can reduce the time-dependent

Schrodinger equation into time-independent form as follows. We start with the Schrodinger equation:

$$i\hbar\frac{d\Psi}{dt} = -\frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2} + U(x,t)\,\Psi(\mathrm{x},\mathrm{t}) \tag{3.20}$$

But now the potential energy $U(x,t) \equiv U(x)$, that is U is independent of time t. We try to solve the above equation by the method of separation of variables. We look for solutions of the form

$$\Psi(x,\mathrm{t}) = \psi(x)\phi(t) \tag{3.21}$$

Here $\psi(x)$ depends only on x and $\phi(t)$ depends only on t. If solutions of the form (3.21) exist, then it should solve the Schrodinger equation (eq. (3.20)). Let us calculate the following:

$$i\hbar\frac{d\Psi}{dt} = i\hbar\frac{d\psi(x)\phi(t)}{dt} = i\hbar\,\psi(x)\frac{d\phi(t)}{dt} \tag{3.22a}$$

$$-\frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2} = -\frac{\hbar^2}{2m}\frac{d^2\psi(x)\phi(t)}{dx^2} = -\frac{\hbar^2}{2m}\phi(t)\frac{d^2\psi(x)}{dx^2} \tag{3.22b}$$

Substituting (3.21), (3.22a) and (3.22b) in eq. (3.20), we get

$$i\hbar\,\psi(x)\frac{d\phi(t)}{dt} = -\frac{\hbar^2}{2m}\phi(t)\frac{d^2\psi(x)}{dx^2} + U(x)\psi(x)\phi(t)$$

Dividing throughout by $\psi(x)\phi(t)$ we get

$$i\hbar\frac{1}{\phi(t)}\frac{d\phi(t)}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{d^2\psi(x)}{dx^2} + U(x) \tag{3.23}$$

In equation (3.23), the LHS is a function of t alone, while the RHS is a function of x alone. The only way this can possibly be true is if both sides are in fact constant – otherwise, by varying t, one could change the left side without touching the right side, and the two would no longer be equal. For reasons that will appear in a moment, we shall call the separation constant E (It can be proved that the constant E represents the total energy of the particle). Then

$$i\hbar\frac{1}{\phi(t)}\frac{d\phi(t)}{dt} = E \tag{3.24a}$$

$$-\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{d^2\psi(x)}{dx^2} + U(x) = E \tag{3.24b}$$

Equation (3.24a) can be written as

$$i\hbar\frac{d\phi(t)}{dt} = E\phi(t)$$

$$\frac{d\phi(t)}{dt} = -\frac{iE}{\hbar}\,\phi(t)$$

The above differential equation can be easily solved to obtain:

$$\phi(t) = \exp(-iEt/\hbar) \tag{3.25}$$

Equation (3.24b) can be rewritten as

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + U(x)\psi(x) = E\psi(x) \tag{3.26}$$

This equation (Equation 3.26) is known as the **time-independent Schrödinger equation** for a particle of mass m moving in a time-independent potential $U(x)$. We can go no further with it until the potential $U(x)$ is specified.

Thus, the complete solution (wave function) of the time-dependent Schrodinger equation for potentials $U(x)$ which are time-independent will be of the form (see eqns. 3.21 & 3.25)

$$\Psi(\mathrm{x},\mathrm{t}) = \psi(x)\exp(-iEt/\hbar) \tag{3.27}$$

where $\psi(x)$ is obtained by solving the time-independent Schrödinger equation (eq. (3.26)). The solution $\psi(x)$ can be obtained once the potential $U(x)$ is specified.

3.5 APPLICATION OF TIME-INDEPENDENT SCHRÖDINGER EQUATION

1. Particle in an infinite potential well (particle in a box)
2. Particle in a finite potential well
3. Tunnelling
4. Quantum oscillator

3.5.1 Particle in an Infinite Potential Well (Particle In A "Box")

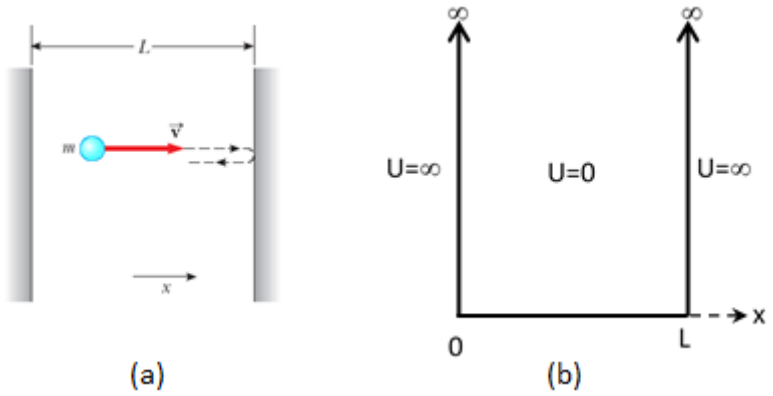


Figure 3.3: (a) Particle in a potential well of infinite height, (b) Sketch of potential well

Consider a particle of mass m and velocity v, confined to bounce between two impenetrable walls separated by a distance L, as shown in Figure 3.3(a). Figure 3.3(b) shows the potential energy function for the system.

$U(x) = 0$ for $0 < x < L$,

$U(x) = \infty$ for $x \leq 0$ and $x \geq L$

Since $U(x) = \infty$, for $x < 0$ and $x > L$, $\psi(x) = 0$ in these regions. Also, $\psi(0) = 0$ and $\psi(L) = 0$. Only those wave functions that satisfy these boundary conditions are allowed.

In the region $0 < x < L$, where $U = 0$, the Schrödinger equation takes the form,

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} E \psi = 0$$

Or $\frac{d^2\psi}{dx^2} = -k^2\psi$, where $k^2 = \frac{2mE}{\hbar^2}$ or $k = \frac{\sqrt{2mE}}{\hbar}$

The most general form of the solution to the above equation is

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

where A and B are constants determined by the boundary and normalization conditions.

Applying the first boundary condition,

i.e., at $x = 0$, $\psi = 0$ leads to

$0 = A \sin 0 + B \cos 0$ or $B = 0$,

And at $x = L$, $\psi = 0$,

$0 = A\sin(kL) + B\cos(kL) = A\sin(kL) + 0$,

Since $A \neq 0$, $\sin(kL) = 0$. $\therefore kL = n\pi$; $(n = 1, 2, 3, \cdots\cdots)$

Now the wave function reduces to $\psi_n(x) = A \sin\left(\frac{n\pi x}{L}\right)$

To find the constant A, apply normalization condition

$$\int_{-\infty}^{+\infty}|\psi|^2\,dx = 1 \qquad \text{or} \qquad \int_0^L A^2\left[sin\left(\frac{n\,\pi\,x}{L}\right)\right]^2 dx = 1\,.$$

$$A^2\int_0^L \tfrac{1}{2}\left[1-cos\left(\frac{2\,n\,\pi\,x}{L}\right)\right]dx = 1$$

Solving we get $A = \sqrt{\frac{2}{L}}$

Thus $\psi_n(x) = \sqrt{\frac{2}{L}}sin\left(\frac{n\,\pi\,x}{L}\right)$ is the wave function for particle in a box.

Since $k = \frac{\sqrt{2\,m\,E}}{\hbar}$ and $kL = n\,\pi$

We get, $\frac{\sqrt{2\,m\,E}}{\hbar}\,L = n\,\pi\,.$

$$\therefore \quad E_n = \left(\frac{h^2}{8\,m\,L^2}\right)n^2\,, \qquad n = 1,\ 2,\ 3,\ \cdots\cdots \tag{3.28}$$

Each value of the integer n corresponds to a quantized energy value, E_n.

The lowest allowed energy ($n = 1$), $E_1 = \frac{h^2}{8\,m\,L^2}$.

This is the ground state energy for the particle in a box. Excited states correspond to n = 2, 3, 4,…which have energies given by $4E_1$, $9E_1$, $16E_1$…. respectively. Energy level diagram, wave function and probability density sketches are shown in Figure 3.4 and 3.5 respectively. Since ground state energy $E_1 \neq 0$, the particle can never be at rest.

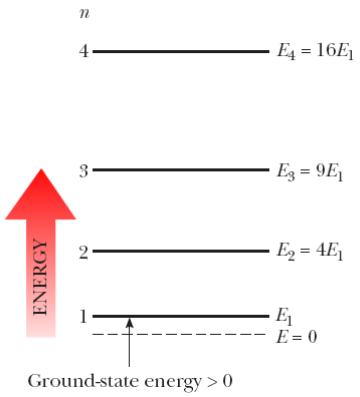


Figure 3.4 Energy level diagram for a particle in potential well of infinite height

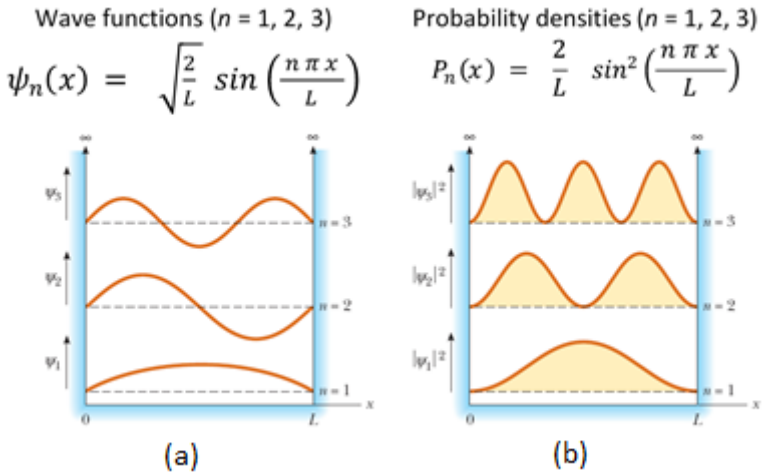


Figure 3.5: Sketch of (a) wave function, (b) Probability density for a particle in potential well of infinite height

3.5.2 A Particle in a Potential Well of Finite Height

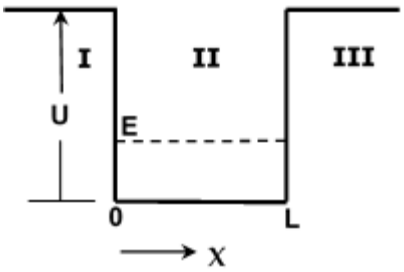


Figure 3.6: Potential well of finite height U and length L

Consider a particle with the total energy E, trapped in a finite potential well of height U such that

$U(x) = 0$ for $0 < x < L$

$U(x) = U$ for $x \leq 0, \quad x \geq L$

Classically, for energy $E<U$, the particle is permanently bound in the potential well. However, according to quantum mechanics, a finite probability exists that the particle can be found outside the well even if $E<U$. That is, the wave function is generally nonzero in the regions I and III. In region II, where $U = 0$, the allowed wave functions are again sinusoidal. But the boundary conditions no longer require that the wave function must be zero at the ends of the well.

Schrödinger equation outside the finite well in regions I & III

$$\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}(U-E)\psi \quad \text{or} \quad \frac{d^2\psi}{dx^2} = C^2\psi$$

where $C^2 = \frac{2m}{\hbar^2}(U-E)$

General solution of the above equation is,

$$\psi(x) = Ae^{Cx} + Be^{-Cx}$$

where A and B are constants.

A must be zero in Region III and B must be zero in Region I, otherwise, the probabilities would be infinite in those regions. For solution to be finite,

$$\psi_I = Ae^{Cx} \qquad \text{for } x \le 0$$

$$\psi_{III} = Be^{-Cx} \qquad \text{for } x \ge L$$

This shows that the wave function outside the potential well decay exponentially with distance.

Schrodinger equation inside the square well potential in region II, where $U = 0$

$$\frac{d^2\psi_{II}}{dx^2} + \left(\frac{2m}{\hbar^2}E\right)\psi_{II} = 0, \qquad \frac{2mE}{\hbar^2} = k^2$$

General solution of the above equation

$$\psi_{II} = F\sin(kx) + G\cos(kx)$$

To determine the constants A, B, F, G and the allowed values of energy E, apply the four boundary conditions and the normalization condition:

At $x = 0$, $\psi_I(0) = \psi_{II}(0)$ and $\left[\frac{d\psi_I}{dx}\right]_{x=0} = \left[\frac{d\psi_{II}}{dx}\right]_{x=0}$

At $x = L$, $\psi_{II}(L) = \psi_{III}(L)$ and $\left[\frac{d\psi_{II}}{dx}\right]_{x=L} = \left[\frac{d\psi_{III}}{dx}\right]_{x=L}$

$$\int_{-\infty}^{+\infty} |\psi|^2\, dx = 1$$

Figure 3.7 shows the plots of wave functions and their respective probability densities.

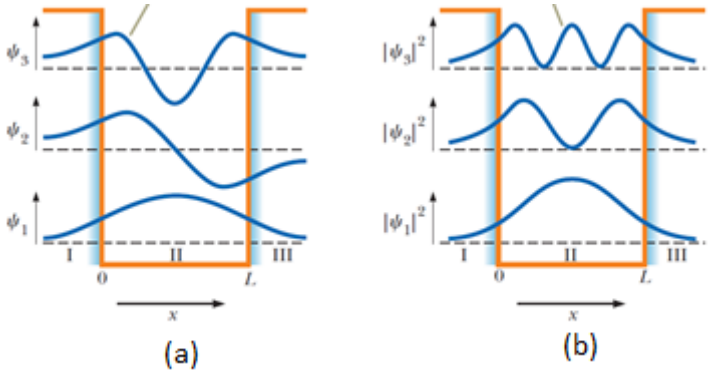


Figure 3.7: Sketch of (a) wave function, (b) Probability density for a particle in potential well of finite height

It is seen that wavelengths of the wave functions are longer than those of wave functions of infinite potential well of same length and hence the quantized energies of the particle in a finite well are lower than those for a particle in an infinite well.

3.5.3 Tunnelling Through a Potential Energy Barrier

Consider a particle of energy E approaching a potential barrier of height U, ($E < U$). Potential energy has a constant value of U in the region of width L and is zero in all other regions. This is called a square barrier and U is called the barrier height. Since $E < U$, classically the regions II and III shown in the figure are forbidden to the particle incident from left. But according to quantum mechanics, all regions are accessible to the particle, regardless of its energy.

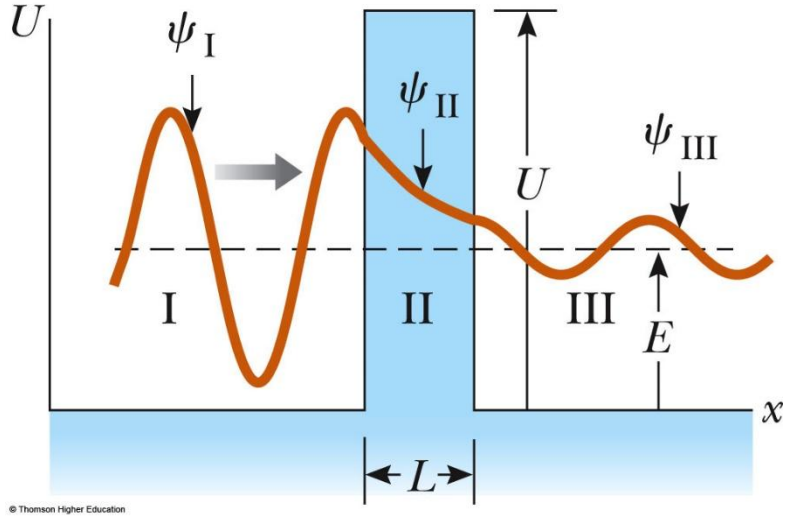


Figure 3.8: Tunnelling through a potential barrier of finite height

By applying the boundary conditions, i.e. ψ and its first derivative must be continuous at boundaries (at $x = 0$ and $x = L$), full solution to the Schrödinger equation can be found which is shown in figure. The wave function is sinusoidal in regions I and III but exponentially decaying in region II. The probability of locating the particle beyond the barrier in region III is nonzero. The movement of the particle to the far side of the barrier is called tunnelling or barrier penetration. The probability of tunnelling can be described with a transmission coefficient T and a reflection coefficient R.

The transmission coefficient represents the probability that the particle penetrates to the other side of the barrier, and reflection coefficient is the probability that the particle is reflected by the barrier. Because the particles must be either reflected or transmitted,

we have, $$R + T = 1$$

An approximate expression for the transmission coefficient, when $T << 1$ is

$$T \approx e^{-2CL} \qquad \text{where } C = \frac{\sqrt{2\,m\,(U-E)}}{\hbar} \tag{3.29}$$

3.5.4 Quantum Harmonic Oscillator

We now turn our attention to arguably the most important system in all of quantum mechanics, the quantum harmonic oscillator (QHO). Here we will study the quantum mechanics of a particle whose force is of the form $F(x) = -kx$. That is, a restoring force proportional to the displacement from the origin. Quantum harmonic oscillators are among the most important systems in physics; it provides an approximation to the behaviour of almost all potential wells, the quantum oscillator pops up in a huge range of situations, from describing light in quantum field theory, to describing phonons in condensed matter physics and far beyond.

We know that in classical mechanics, systems with such restoring forces lead to simple harmonic motion. Our goal now is to understand the quantum mechanics of such systems.

For a harmonic oscillator, the potential energy is

$$U(x) = \frac{1}{2}m\omega^2x^2 \tag{3.30}$$

Classically, the total energy of the harmonic oscillator is given by

$$E = KE + PE = \frac{1}{2}kA^2 \tag{3.31}$$

where A is the amplitude of oscillation. In the classical model, any continuous value of E is allowed, including $E = 0$, which is the total energy when the particle is at rest at $x = 0$ and $p = 0$.

The dynamics of a quantum harmonic oscillator can be investigated by solving the Schrodinger equation corresponding to it. The time-independent Schrödinger equation for a quantum harmonic oscillator is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2x^2\,\psi(x) = E\psi(x) \tag{3.32}$$

The wave function and energy of a quantum harmonic oscillator can be obtained by solving the above differential equation by applying appropriate boundary conditions. After some calculations, we can show that the wave functions that are physically acceptable and that satisfy eq. (3.32) are given by

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(x') e^{-x'^2/2} \quad ; \quad n = 0, 1, 2 \ldots \tag{3.33}$$

where $x' = \sqrt{m\omega/\hbar}\, x$ and $H_n(x')$ are called the *Hermite's polynomials*.

The energies corresponding to these wave functions are:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega \quad ; \quad n = 0, 1, 2 \ldots \tag{3.34}$$

The first few Hermite's polynomials are given by:

$$H_0(x') = 1 \qquad H_1(x') = 2x' \qquad H_2(x') = 4x'^2 - 2$$

$$H_3(x') = 8x'^3 - 12x' \qquad H_4(x') = 16x'^4 - 48x'^2 + 12$$

In equations (3.33 & 3.34), n is called the quantum number which characterizes the energy state of a quantum oscillator. The state $n = 0$ corresponds to the ground state, $n = 1$ corresponds to the first excited state and so on.

For example, the ground state wavefunction for a QHO is obtained by substituting $n = 0$ in eq. (3.33):

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \frac{1}{\sqrt{2^0 0!}} H_0(x') \exp\left(-\frac{x'^2}{2}\right)$$

Substituting $H_0(x') = 1$ and $x' = \sqrt{m\omega/\hbar}\, x$, we get the normalized ground state wave function as

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \exp\left(-\frac{m\omega}{2\hbar} x^2\right) \tag{3.35}$$

The ground state energy is obtained by substituting $n = 0$ in eq. (3.34), which gives

$$E_0 = \frac{1}{2}\hbar\omega \tag{3.36}$$

This energy is called the ground state energy or the zero point energy of a quantum harmonic oscillator. The $n = 1$ state corresponds to the first excited state, whose energy is $E_1 = \frac{3}{2}\,\hbar\omega$, and so on. The energy-level diagram and wave functions for a QHO are shown in Figure 3.9. The energy spectrum of the harmonic oscillator consists of energy levels that are equally spaced: $\Delta E = E_{n+1} - E_n = \hbar\omega$.

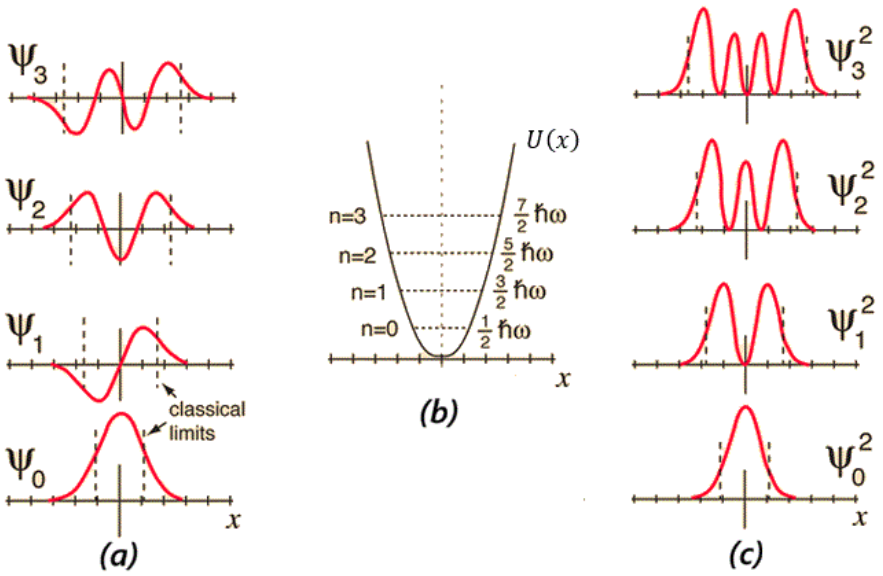


Fig 3.9: (a) Wave functions for n = 0, 1, 2, 3 states (b) energy-level diagram for a QHO, superimposed on the potential energy function (c) probability density ($|\psi|^2$) for n = 0, 1, 2, 3 states.

Once again, as in the case of the infinite square well potential, we encounter the zero-point energy phenomenon: the lowest energy eigenvalue of the oscillator is not zero but is instead equal to $E_0 = (1/2)\hbar\omega$. The zero-point energy of bound state systems cannot be zero, otherwise it would violate the uncertainty principle. For the harmonic oscillator, for instance, the classical minimum energy corresponds to $x = 0$ and $p = 0$; there would be no oscillations in this case. This would imply that we know simultaneously and with absolute precision both the position and the momentum of the system. This would contradict the uncertainty principle.

3.6 QUESTIONS

1 Using the following expression for a wave packet corresponding to a free particle, arrive at the free particle Schrodinger equation.

$$\Psi(x,t) = \int_{-\infty}^{+\infty} A(k)\exp[i(kx-\omega t)]\ dk$$

2 Using the following expression for a wave packet, arrive at the time-dependent Schrodinger equation.

$$\Psi(x,t) = \int_{-\infty}^{+\infty} A(p)\exp[\frac{i}{\hbar}(px-Et)]\ dp$$

3 Reduce the time-dependent Schrodinger equation to the time-independent form.

4 Consider the following wave function for a free particle.

$$\psi(x,t) = A\exp[i/\hbar(px-Et)]$$

Using the above wave function, derive the operator corresponding to momentum.

5 Using the expression for momentum operator, derive the kinetic energy operator and Hamiltonian operator (energy operator).

6 What is a wave function? What is its physical interpretation?

7 What are the mathematical features of a wave function?

8 By solving the Schrödinger equation, obtain the wave-functions for a particle of mass m in a one-dimensional "box" of length L.

9 Apply the Schrödinger equation to a particle in a one-dimensional "box" of length L and obtain the energy values of the particle.

10 Sketch the lowest three energy states, wave-functions, probability densities for the particle in a one-dimensional "box".

11 The wave-function for a particle confined to moving in a one-dimensional box is $\psi(x) = A\sin\left(\frac{n\pi x}{L}\right)$. Use the normalization condition on ψ to show that $A = \sqrt{\frac{2}{L}}$.

12 The wave-function of an electron is $\psi(x) = A\sin\left(\frac{n\pi x}{L}\right)$. Obtain an expression for the probability of finding the electron between $x = a$ and $x = b$.

13 Sketch the potential-well diagram of finite height U and length L, obtain the general solution of the Schrödinger equation for a particle of mass m in it.

14 Sketch the wave-functions and the probability densities for the lowest three energy states of a particle in a potential well of finite height.

15 Give a brief account of tunnelling of a particle through a potential energy barrier.

16 Give a brief account of the quantum mechanical treatment of a simple harmonic oscillator.

17 How does the energies of a quantum harmonic oscillator compare with its classical counterpart?

18 Explain the concept of zero-point energy of a quantum harmonic oscillator.

3.7 PROBLEMS

1 **A particle wave function is given by the equation $y(x) = A e^{-ax^2}$**

(A) What is the value of A if this wave function is normalized?

(B) What is the expectation value of x for this particle?

Ans: $A = (2a/\pi)^{1/4}$, $\langle x \rangle = 0$

2 **A free electron has a wave function $\psi(x) = A \exp[i(5.0 \times 10^{10})x]$**

where x is in meters. Find (a) its de Broglie wavelength, (b) its momentum, and (c) its kinetic energy in electron volts.

Ans: 1.26×10^{-10}m, 5.27×10^{-24}kg.m/s, 95.5 eV

3 An electron is confined between two impenetrable walls 0.20 nm apart. Determine the energy levels for the states n =1 ,2 , and 3.

Ans: 9.2 eV, 37.7 eV, 84.8 eV

4 A 0.50 kg baseball is confined between two rigid walls of a stadium that can be modeled as a "box" of length 100 m. Calculate the minimum speed of the baseball. If the baseball is moving with a speed of 150 m/s, what is the quantum number of the state in which the baseball will be?

Ans: 6.63×10^{-36} m/s, 2.26×10^{37}

5 **A proton is confined to move in a one-dimensional "box" of length 0.20 nm. (a) Find the lowest possible energy of the proton. (b) What is the lowest possible energy for an electron confined to the same box? (c) Account for the great difference in results for (a) and (b).**

Ans: 5.13×10^{-3} eV, 9.41 eV

6 **(A) Using the simple model of a particle in a box to represent an atom, estimate the energy (in eV) required to raise an atom from the state n =1 to the state n =2. Assume the atom has a radius of 0.10 nm and that the moving electron carries the energy that has been added to the atom.**

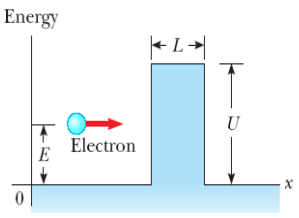
(B) Atoms may be excited to higher energy states by absorbing photon energy. Calculate the wavelength of the photon that would cause the transition from the state n =1 to the state n =2.

Ans: 28.3 eV, 43.8 nm

7 A 30-eV electron is incident on a square barrier of height 40 eV. What is the probability that the electron will tunnel through the barrier if its width is (A) 1.0 nm? (B) 0.10 nm?

Ans: 8.5×10^{-15}, 0.039

8 **An electron with kinetic energy E = 5.0 eV is incident on a barrier with thickness L = 0.20 nm and height U = 10.0 eV as shown in the figure. What is the probability that the electron (a) will tunnel through the barrier? (b) will be reflected?**



Ans: 0.0103, 0.990

9 **A quantum simple harmonic oscillator consists of an electron bound by a restoring force proportional to its position relative to a certain equilibrium point. The proportionality constant is 8.99 N/m. What is the longest wavelength of light that can excite the oscillator?**

Ans: 600 nm

10 A quantum simple harmonic oscillator consists of a particle of mass m bound by a restoring force proportional to its position relative to a certain equilibrium point. The proportionality constant is k. What is the longest wavelength of light that can excite the oscillator? Ans: $2\pi c\sqrt{\frac{m}{k}}$

11 The ground state wavefunction of a harmonic oscillator is given by

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp(-\frac{m\omega}{2\hbar}x^2)$$

Prove that the above wavefunction solves the Schrodinger equation for the harmonic oscillator.

Hint: Substitute the given wavefunction in the Schrodinger equation for the harmonic oscillator and verify RHS = LHS.

12 Construct the wavefunction corresponding to the first excited state of a quantum harmonic oscillator.

Ans:

$$\psi_1(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \sqrt{\frac{2m\omega}{\hbar}}\, x \exp\left(-\frac{m\omega}{2\hbar}x^2\right)$$

4 SOLID STATE PHYSICS

OBJECTIVES:

- To understand the cohesion of solid metals using bonding in solids
- To comprehend the electrical properties of metals, semiconductors and insulators
- To understand the effect of doping on electrical properties of semiconductors
- To understand the properties of dielectrics, superconductors and their engineering applications.

4.1 BONDING IN SOLIDS

A crystalline solid consists of a large number of atoms arranged in a regular array, forming a periodic structure.

4.1.1 Ionic Solids

Many crystals are formed by ionic bonding, in which the dominant interaction between ions is the Coulomb force. Consider a portion of the NaCl crystal shown in Figure 4.1a. The red spheres are sodium ions, and the blue spheres are chlorine ions. As shown in Figure 4.1b, each Na^+ ion has six nearest-neighbor Cl^- ions. Similarly, in Figure 4.1c, each Cl^- ion has six nearest-neighbor Na^+ ions. Each Na^+ ion is attracted to its six Cl^- neighbors. The corresponding potential energy is $-6k_e e^2/r$, where k_e is the Coulomb constant and r is the separation distance between each Na^+ and Cl^-. In addition, there are 12 next-nearest-neighbor Na^+ ions at a distance of $\sqrt{2}r$ from the Na^+ ion, and these 12 positive ions exert weaker repulsive forces on the central Na^+. Furthermore, beyond these 12 Na^+ ions are more Cl_2 ions that exert an attractive force, and so on. The net effect of all these interactions is a resultant negative electric potential energy

$$U_{\text{attractive}} = -\alpha k_e \frac{e^2}{r} \tag{4.1}$$

where α is a dimensionless number known as the **Madelung constant.** The value of α depends only on the particular crystalline structure of the solid ($\alpha = 1.747$ for the NaCl structure). When the constituent ions of a crystal are brought close together, a repulsive force exists because of electrostatic forces and the exclusion principle. The potential energy term B/r_m in Equation 4.2 accounts for this repulsive force. The repulsive forces occur only for ions that are very close together.

Therefore, we can express the total potential energy of the crystal as

$$U_{\text{total}} = -\alpha k_e \frac{e^2}{r} + \frac{B}{r^m} \tag{4.2}$$

where *m* in this expression is some small integer.

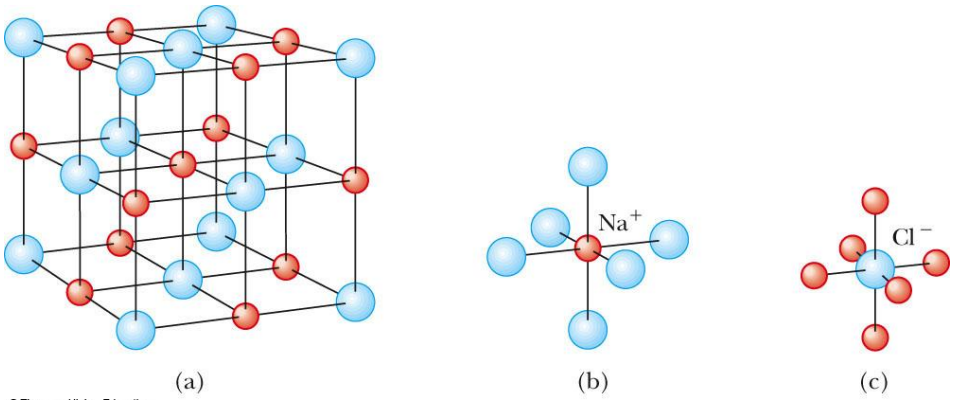


Figure 4.1: (a) Crystalline structure of NaCl. (b) Each positive sodium ion is surrounded by six negative chlorine ions. (c) Each chlorine ion is surrounded by six sodium ions.

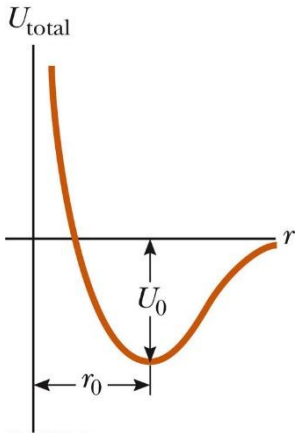


Figure 4.2: Total potential energy versus ion separation distance for an ionic solid, where U_0 is the ionic cohesive energy and r_0 is the equilibrium separation distance between ions.

A plot of total potential energy versus ion separation distance is shown in Figure 4.2. The potential energy has its minimum value U_0 at the equilibrium separation, when $r = r_0$.

$$U_0 = -\alpha k_e \frac{e^2}{r_0}\left(1 - \frac{1}{m}\right) \tag{4.3}$$

This minimum energy U_0 is called the **ionic cohesive energy** of the solid, and its absolute value represents the energy required to separate the solid into a collection of isolated positive and negative ions.

4.1.2 Covalent Solids

Solid carbon, in the form of diamond, is a crystal whose atoms are covalently bonded. In the diamond structure, each carbon atom is covalently bonded to four other carbon atoms located at four corners of a cube as shown in Figure 4.3(a). The crystalline structure of the diamond is shown in Figure 4.3(b). The basic structure of diamond is called tetrahedral (each carbon atom

is at the center of a regular tetrahedron), and the angle between the bonds is 109.5°. Other crystals, such as silicon and germanium, have the same structure. Covalently bonded solids are usually very hard, have high bond energies and high melting points, and are good electrical insulators.

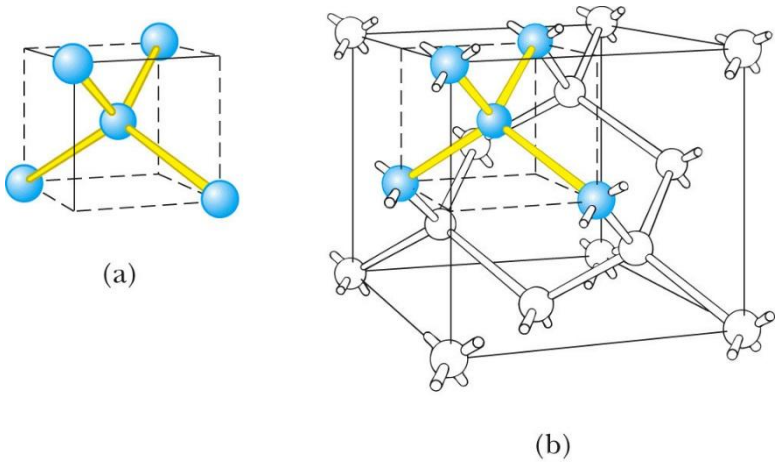


Figure 4.3: (a) Each carbon atom in a diamond crystal is covalently bonded to four other carbon atoms so that a tetrahedral structure is formed. (b) The crystal structure of diamond, showing the tetrahedral bond arrangement.

4.1.3 Metallic Solids

Metallic bonds are generally weaker than ionic or covalent bonds. The outer electrons in the atoms of a metal are relatively free to move throughout the material, and the number of such mobile electrons in a metal is large. The metallic structure can be viewed as a "sea" or a "gas" of nearly free electrons surrounding a lattice of positive ions (Figure 4.4). The bonding mechanism in a metal is the attractive force between the entire collection of positive ions and the electron gas.

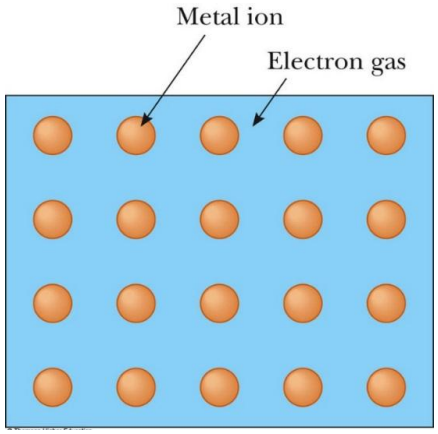


Figure 4.4: Schematic diagram of a metal showing an electron gas and metal ions.

Light interacts strongly with the free electrons in metals. Hence, visible light is absorbed and re-emitted quite close to the surface of a metal, which accounts for the shiny nature of metal surfaces. Because the bonding in metals is between all the electrons and all the positive ions, metals tend to bend when stressed.

4.2 FREE-ELECTRON THEORY OF METALS

Quantum based free electron theory of metals is centered on wave nature of electrons. In this model, one imagines that the outer-shell electrons are free to move through the metal but are trapped within a three-dimensional box formed by the metal surfaces. Therefore, each electron is represented as a particle in a box and is restricted to quantized energy levels. Each energy state can be occupied by only two electrons (one with spin up & the other with spin down) as a consequence of exclusion principle. In quantum statistics, it is shown that the probability of a particular energy state E being occupied by an electrons is given by

$$f(E) = \frac{1}{exp\left(\frac{E - E_F}{kT}\right) + 1} \tag{4.4}$$

where $f(E)$ is called the **Fermi-Dirac distribution function** and E_F is called the **Fermi energy**. Plot of $f(E)$ versus E is shown in Figure 4.5.

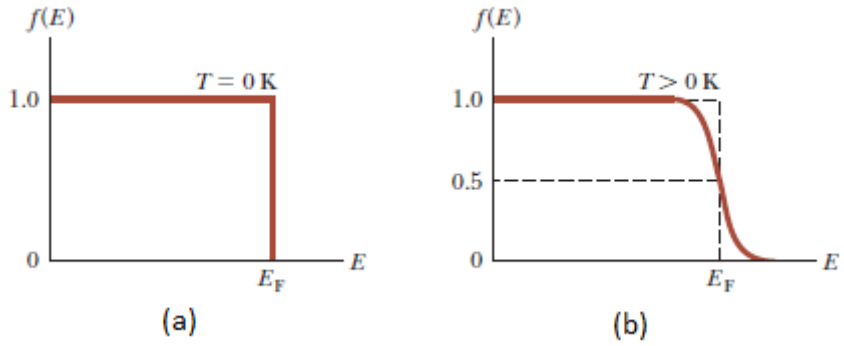


Figure. 4.5: Plot of Fermi-Dirac distribution function f(E) Vs energy E at (a) T = 0K and (b) T > 0K.

At zero kelvin (0 K), all states having energies less than the Fermi energy are occupied, and all states having energies greater than the Fermi energy are vacant. i.e. Fermi energy is the highest energy possessed by an electron at 0 K (Figure 4.5a). As temperature increases (T > 0K), the distribution rounds off slightly due to thermal excitation and probability of Fermi level being occupied by an electron becomes half (Figure 4.5b). In other words, Fermi energy is that energy

state at which probability of electron occupation is half. The Fermi energy E_F also depends on temperature, but the dependence is weak in metals.

4.2.1 Density of states

From particle in a box problem, for a particle of mass m is confined to move in a one-dimensional box of length L, the allowed states have quantized energy levels given by,

$$E_n = \frac{h^2}{8\,m\,L^2}\,n^2 = \frac{\hbar^2\pi^2}{2\,m\,L^2}\,n^2 \qquad n = 1, 2, 3 \ldots \tag{4.5}$$

An electron moving freely in a metal cube of side L, can be modeled as particle in a three-dimensional box. It can be shown that the energy for such an electron is

$$E = \frac{\hbar^2\pi^2}{2\,m\,L^2}\left(n_x^2 + n_y^2 + n_z^2\right) \tag{4.6}$$

where m is mass of the electron and n_x, n_y, n_z are quantum numbers(positive integers). Each allowed energy value is characterized by a set of three quantum numbers (n_x, n_y, n_z - one for each degree of freedom). Imagine a three-dimensional **quantum number space** whose axes represent n_x, n_y, n_z. The allowed energy states in this space can be represented as dots located at positive integral values of the three quantum numbers as shown in the Figure 4.6.

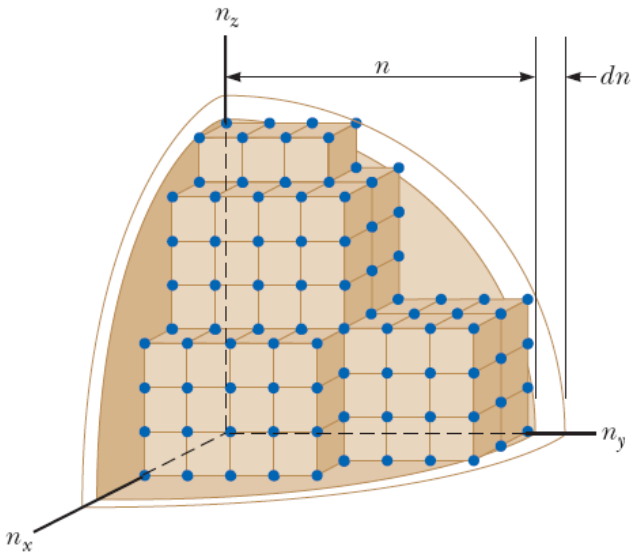


Figure 4.6: Representation of the allowed energy states in a quantum number space (dots represent the allowed states).

Eq. 4.6 can be written as,

$$n_x^2 + n_y^2 + n_z^2 = \frac{E}{E_o} = n^2 \tag{4.7}$$

where $E_o = \dfrac{\hbar^2\pi^2}{2\ m\ L^2}$ and $n = \sqrt{\dfrac{E}{E_o}}$

Eq. 4.7 represents a sphere of radius *n*. Thus, the number of allowed energy states having energies between *E* and *E+dE* is equal to the number of points in a spherical shell of radius *n* and thickness *dn*. In this quantum number space each point is at the corners of a **unit cube** and each corner point is shared by eight unit cubes and as such the contribution of each point to the cube is 1/8 th. Because a cube has eight corners, the effective point per unit cube and hence unit volume is one. In other words, **number of points is equal to the volume of the shell**. The "volume" of this shell, denoted by *G(E)dE*.

$$G(E)\ dE\ =\ \left(\frac{1}{8}\right)(4\pi\ n^2\mathrm{dn})\ \ =\ \ \left(\frac{1}{2}\right)\pi\ n^2 dn$$

$$G(E)\ dE\ \ =\ \ \tfrac{1}{2}\ \pi\left(\tfrac{E}{E_o}\right)\ d\left[\left(\tfrac{E}{E_o}\right)^{\frac{1}{2}}\right] \qquad \text{using the relation } n\ =\ \sqrt{\tfrac{E}{E_o}}$$

$$G(E)\ dE\ \ =\ \ \frac{1}{2}\ \pi\left(\frac{E}{E_o}\right)\ E_o^{-\frac{1}{2}}\ \frac{1}{2}\ E^{-\frac{1}{2}}\ dE\ \ =\ \ \frac{1}{4}\ \pi\ E_o^{-\frac{3}{2}}\ E^{\frac{1}{2}}\ dE$$

$$G(E)\ dE\ \ =\ \ \frac{1}{4}\ \pi\ \left(\frac{\hbar^2\pi^2}{2mL^2}\right)^{-\frac{3}{2}}\ E^{\frac{1}{2}}\ dE$$

$$G(E)\ dE\ \ =\ \ \frac{\sqrt{2}}{2}\ \frac{m^{\frac{3}{2}}\ L^3}{\pi^2\ \hbar^3}\ E^{\frac{1}{2}}\ dE, \qquad L^3\ =\ V$$

Number of states per unit volume per unit energy range, called **density of states**, *g(E)* is given by,

$$g(E) = G(E)/V$$

$$g(E)\ dE\ \ =\ \ \frac{G(E)}{V}\ dE\ \ =\ \ \frac{\sqrt{2}}{2}\ \frac{m^{\frac{3}{2}}}{\pi^2\ \hbar^3}\ E^{\frac{1}{2}}\ dE$$

$$g(E)\ dE\ \ =\ \ \frac{4\sqrt{2}\ \pi\ m^{\frac{3}{2}}}{h^3}\ E^{\frac{1}{2}}\ dE \qquad \because\quad \hbar = \frac{h}{2\pi}$$

Finally, we multiply by 2 for the two possible spin states of each particle.

$$g(E)\,dE \;=\; \frac{8\sqrt{2}\,\pi\,m^{\frac{3}{2}}}{h^3}\,E^{\frac{1}{2}}\,dE \tag{4.8}$$

g(E) is called the **density-of-states function**.

4.2.2 Electron density

For a metal in thermal equilibrium, the number of electrons *N(E) dE*, per unit volume, that have energy between *E* and *E+dE* is equal to the product of the density of states and the probability that a state is occupied. that is,

$$N(E)dE = [\,g(E)dE\,]\;f(E)$$

$$N(E)\,dE \;=\; \frac{8\sqrt{2}\,\pi\,m^{\frac{3}{2}}}{h^3}\;\frac{E^{\frac{1}{2}}\,dE}{exp\left(\frac{E-E_F}{k\,T}\right)\;+\;1} \tag{4.9}$$

Plots of *N(E)* versus *E* for two temperatures are given in Figure 4.7.

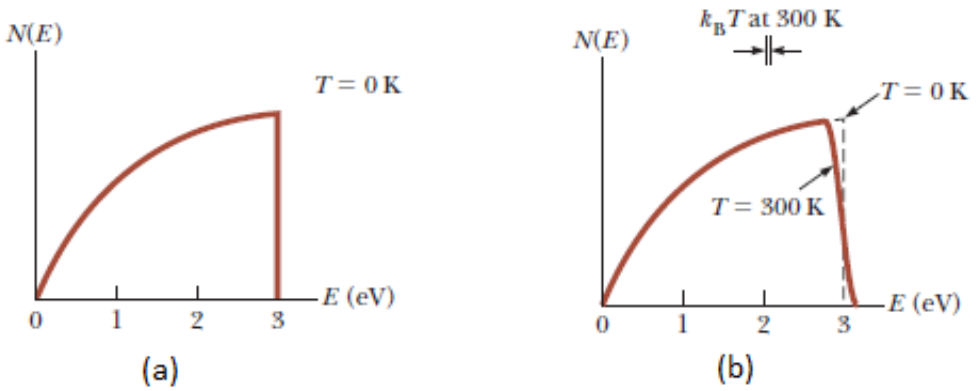


Figure 4.7: Plots of N(E) versus E for (a) T = 0K (b) T = 300K.

If n_e is the total number of electrons per unit volume, we require that

$$n_e \;=\; \int_0^{\infty} N(E)\,dE \;=\; \frac{8\sqrt{2}\,\pi\,m^{\frac{3}{2}}}{h^3}\;\int_0^{\infty}\frac{E^{\frac{1}{2}}\,dE}{exp\left(\frac{E-E_F}{k\,T}\right)\;+\;1} \tag{4.10}$$

At T = 0K, the Fermi-Dirac distribution function *f(E)* = 1 for $E < E_F$ and *f(E)* = 0 for $E > E_F$. Therefore, at T = 0K,

$$n_e \;=\; \frac{8\sqrt{2}\,\pi\,m^{\frac{3}{2}}}{h^3}\;\int_0^{E_F} E^{\frac{1}{2}}\,dE \;=\; \frac{8\sqrt{2}\,\pi\,m^{\frac{3}{2}}}{h^3}\left(\frac{2}{3}\right)E_F^{\frac{3}{2}} \;=\; \frac{16\sqrt{2}\,\pi\,m^{\frac{3}{2}}}{3\,h^3}\,E_F^{\frac{3}{2}} \tag{4.11}$$

Solving for Fermi energy at 0K, we obtain

$$E_F(0) = \frac{h^2}{2\,m}\left(\frac{3\,n_e}{8\,\pi}\right)^{\frac{2}{3}} \tag{4.12}$$

The average energy of a free electron in a metal at 0 K is $E_{av} = \frac{3}{5}E_F$.

4.3 BAND THEORY OF SOLIDS

When a quantum system is represented by wave function, probability density $|\Psi|^2$ for that system is physically significant while the probability amplitude Ψ not. Consider an atom such as sodium that has a single *s* electron outside of a closed shell. Both the wave functions $\psi_S^+(r)$ and $\psi_S^-(r)$ are valid for such an atom [$\psi_S^+(r)$and $\psi_S^-(r)$ are symmetric and anti symmetric wave functions]. As the two sodium atoms are brought closer together, their wave functions begin to overlap. Figure 4.8 represents two possible combinations : i) symmetric - symmetric and ii) symmetric – antisymmetric. These two possible combinations of wave functions represent two possible states of the two-atom system. Thus, the states are split into two energy levels. The energy difference between these states is relatively small, so the two states are close together on an energy scale.

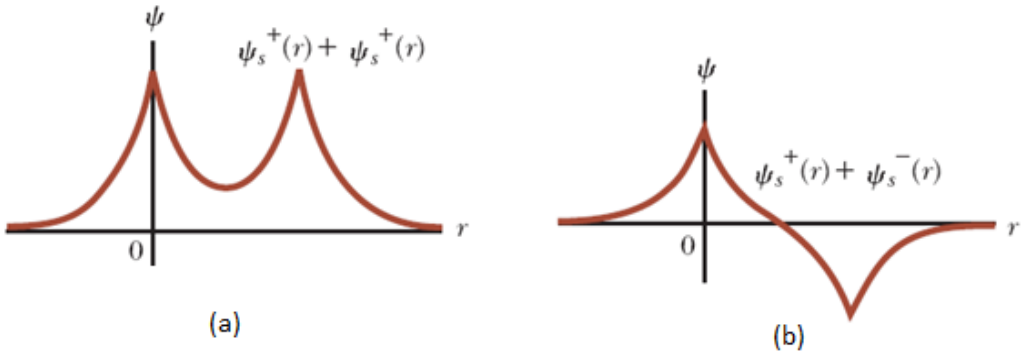


Figure. 4.8: The wave functions of two atoms combine to form a composite wave function : a) symmetric-symmetric b) symmetric-antisymmetric.

When two atoms are brought together, each energy level will split into 2 energy levels. (In general, when N atoms are brought together N split levels will occur which can hold 2N electrons). The split levels are so close that they may be regarded as a continuous band of energy levels. Figure 4.9 shows the splitting of *1s* and *2s* levels of sodium atom when : (a) two sodium atoms are brought together (b)five sodium atoms are brought together (c) a large number of sodium atoms are assembled to form a solid. The close energy levels forming a band are seen clearly in (c).

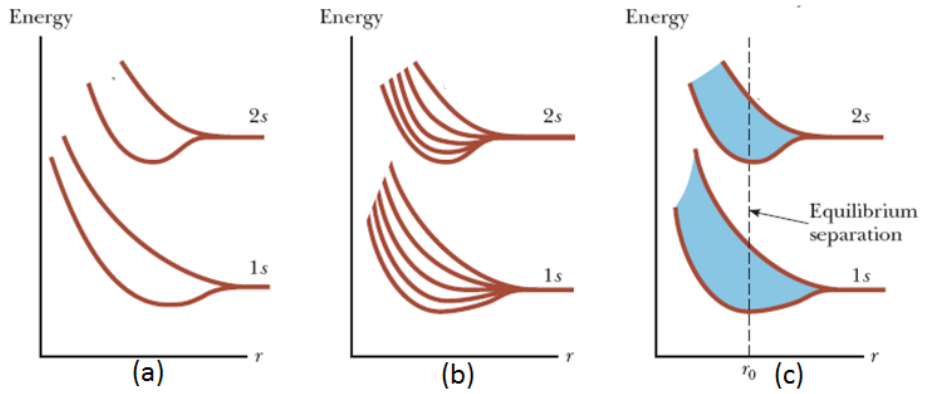


Figure 4.9: Splitting of 1s and 2s levels of sodium atoms due to interaction between them.

Some bands may be wide enough in energy so that there is an overlap between the adjacent bands. Some other bands are narrow so that a gap may occur between the allowed bands, and is known as **forbidden energy gap**. The 1s, 2s, and 2p bands of solid sodium are filled completely with electrons. The 3s band (2N states) of solid sodium has only N electrons and is partially full; The 3p band, which is the higher region of the overlapping bands, is completely empty as shown in Figure 4.10.

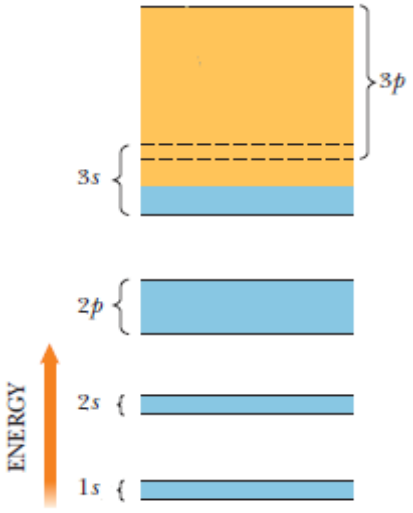


Figure 4.10: Energy bands of a sodium crystal.

4.4 ELECTRICAL CONDUCTION IN METALS, INSULATORS AND SEMICONDUCTORS

Good electrical conductors contain a high density of free charge carriers, and the density of free charge carriers in insulators is nearly zero. In semiconductors, free-charge-carrier densities are intermediate between those of insulators and those of conductors.

Metals: Metals have a partially filled energy band (Figure 4.11 (a)). At 0 K, Fermi level is the highest electron-occupied energy level. If a potential difference is applied to the metal,

electrons having energies near the Fermi energy require only a small amount of additional energy to reach nearby empty energy states above the Fermi-level. Therefore, electrons in a metal experiencing a small force (from a weak applied electric field) are free to move because many empty levels are available close to the occupied energy levels. The model of metals based on band theory demonstrates that metals are excellent electrical conductors.

Insulators: Consider the two outermost energy bands of a material in which the lower band is filled with electrons and the higher band is empty at 0 K (Figure 4.11 (b)). The lower, filled band is called the valence band, and the upper, empty band is the conduction band. The energy separation between the valence and conduction band, called energy gap E_g, is large for insulating materials. The Fermi level lies somewhere in the energy gap. Due to a larger energy gap compared to thermal energy kT (26 meV) at room temperature, excitation of electrons from the valence band to the conduction band is hardly possible. Since the free-electron density is nearly zero, these materials are bad conductors of electricity.

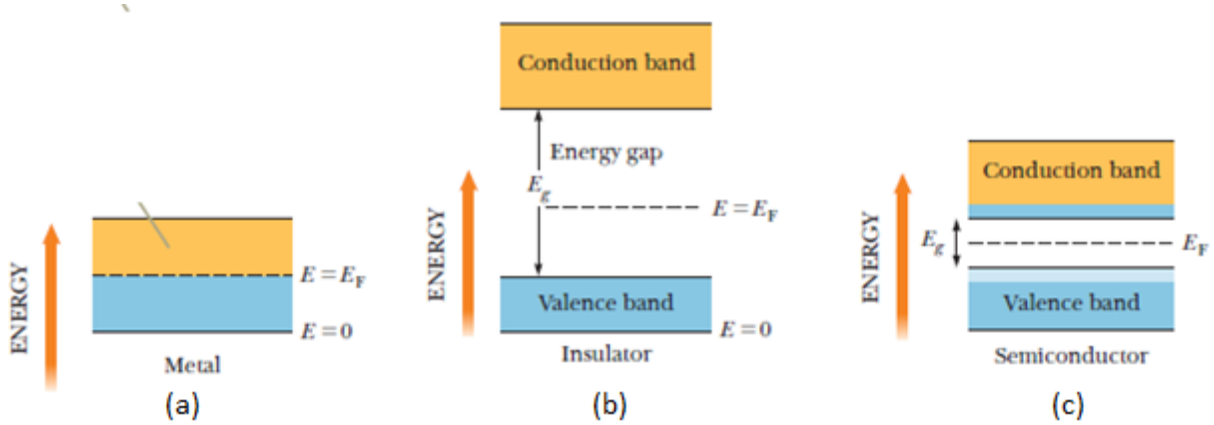


Figure 4.11: Band structure of (a) Metals (b) Insulators (c) Semiconductors.

Semiconductors: Semiconductors have the same type of band structure as an insulator, but the energy gap is much smaller, of the order of 1 eV. The band structure of a semiconductor is shown in Figure 4.11c. Because the Fermi level is located near the middle of the gap for a semiconductor and E_g is small, appreciable numbers of electrons are thermally excited from the valence band to the conduction band. Because of the many empty levels above the thermally filled levels in the conduction band, a small applied potential difference can easily raise the energy of the electrons in the conduction band, resulting in moderate conduction. At T = 0 K, all electrons in these materials are in the valence band, and no energy is available to excite them across the energy gap. Therefore, semiconductors are poor conductors at very low temperatures. Because the thermal excitation of electrons across the narrow gap is more probable at higher temperatures, the conductivity of semiconductors increases rapidly with temperature. This is in sharp contrast to the conductivity of metals, which decreases with increasing temperature. Charge carriers in a semiconductor can be negative, positive, or both. When an electron moves from the valence band into the conduction band, it leaves behind a vacant site, called a hole, in the otherwise filled valence band.

In an intrinsic semiconductor (pure semiconductor) there are equal number of conduction electrons and holes. In the presence of an external electric field, the holes move in the direction

of the field, and the conduction electrons move opposite to the direction of the field. Both these motions correspond to the current in the same direction (Figure 4.12).

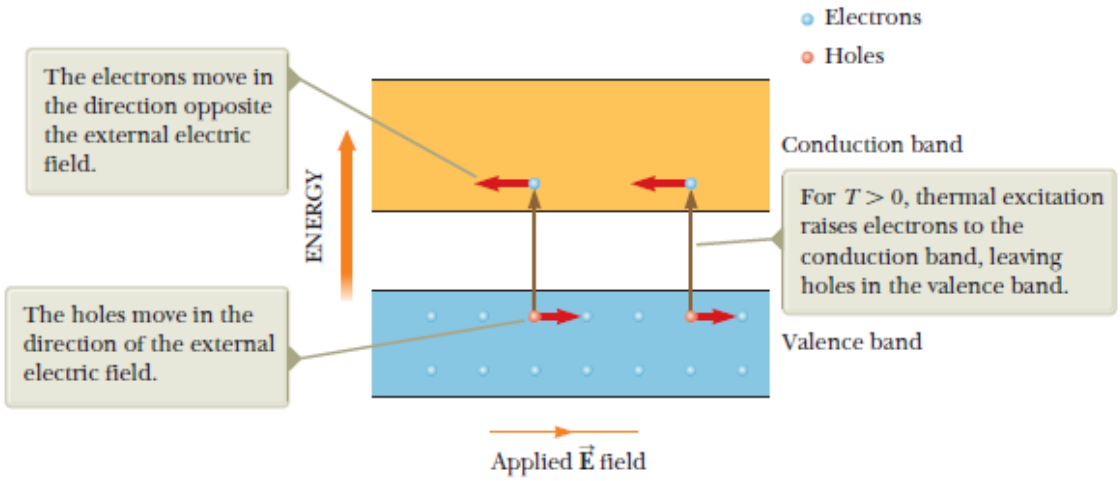


Figure 4.12: Movement of electrons and holes in an intrinsic semiconductor.

Doped Semiconductors (Extrinsic semiconductors): Semiconductors in their pure form are called intrinsic semiconductors, while doped semiconductors are called extrinsic semiconductors. Doping is the process of adding impurities to a semiconductor. By doping, both the band structure of the semiconductor and its resistivity are modified. If a tetravalent semiconductor (Si or Ge) is doped with a pentavalent impurity atom (donor atom), four of the electrons form covalent bonds with atoms of the semiconductor, and one is left over (Figure 4.13). At zero K, this extra electron resides in the donor levels, which lie in the energy gap, just below the conduction band. Since the energy E_d between the donor levels and the bottom of the conduction band is small, at room temperature, the extra electron is thermally excited to the conduction band. This type of semiconductor is called an n-type semiconductor because the majority of charge carriers are electrons (negatively charged).

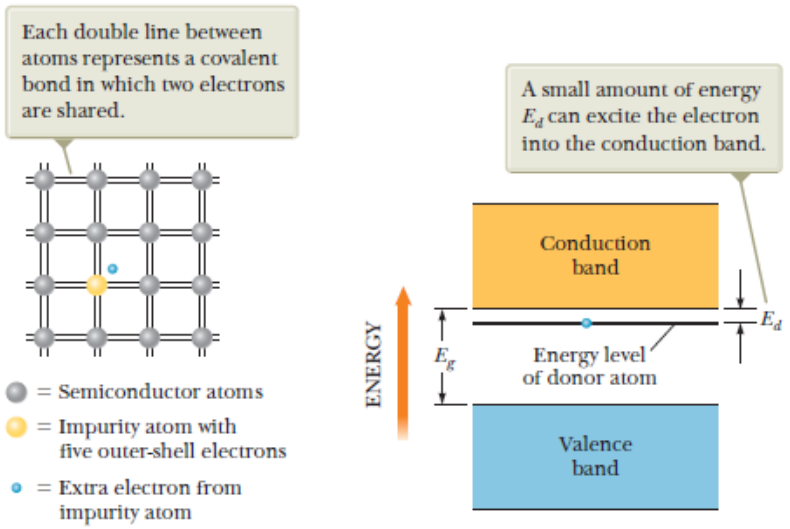


Figure 4.13: n-type semiconductor – two-dimensional representation and band structure.

If a tetravalent semiconductor is doped with a trivalent impurity atom (acceptor atom), the three electrons form covalent bonds with neighbouring semiconductor atoms, leaving an electron deficiency (a hole) at the site of fourth bond (Figure 4.14). At zero K, this hole resides in the acceptor levels that lie in the energy gap just above the valence band. Since the energy E_a between the acceptor levels and the top of the valence band is small, at room temperature, an electron from the valence band is thermally excited to the acceptor levels leaving behind a hole in the valence band. This type of semiconductor is called p-type semiconductor because the majority of charge carriers are holes (positively charged).

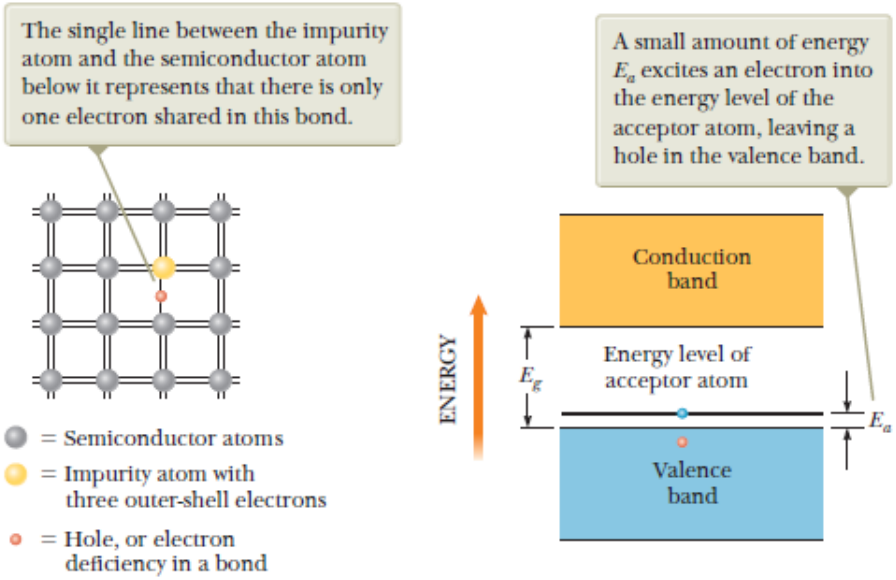


Figure 4.14: p-type semiconductor: two-dimensional representation and band structure.

When conduction in a semiconductor is the result of acceptor or donor impurities, the material is called an extrinsic semiconductor. The typical range of doping densities for extrinsic semiconductors is 10^{13} to 10^{19} cm^{-3}, whereas the electron density in a typical semiconductor is about 10^{21} cm^{-3}.

4.5 ELECTRICAL CONDUCTIVITY OF A SEMICONDUCTOR

Under the influence of an electric field, the charge carrier in a semiconductor drifts with a speed v, which is proportional to the applied field E,

$$v \propto E, \quad v = \mu E \qquad \text{i.e.,} \qquad \mu = \frac{v}{E}$$

where, μ = mobility of the charge carrier (drift speed per unit applied electric field)

The drift speed and hence the mobility are not the same for conduction electrons and positive holes. This is because the effective masses of electrons and holes differ.

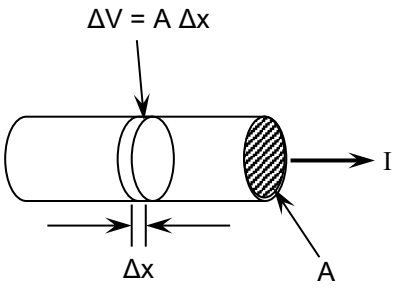


Figure 4.15: A current carrying cylindrical semiconductor.

If $\boldsymbol{n}$ is the conduction electron density (number of conduction electrons per unit volume of the semiconductor), the charge available for conduction of electric current per unit volume of the semiconductor is $\boldsymbol{ne}$, where $\boldsymbol{e}$ = electronic charge.

The charge in the volume, $\Delta V = A\,\Delta x$, shown in Figure 4.15 of a cylindrical semiconductor is,

$$\Delta q = nA\,\Delta x\,e$$

which drifts a distance Δx in a time Δt in the electric field $\boldsymbol{E}$ set up in the semiconductor.

Then the current I is given by,

$$I = \frac{\Delta q}{\Delta t} = e\,n\,A\left(\frac{\Delta q}{\Delta t}\right) = e\,n\,A\,v$$

where the drift velocity, $v = \frac{\Delta x}{\Delta t}$.

The current density (J) is the current per unit cross-sectional area,

$$J = \frac{I}{A} = n\,e\,v$$

This current density due to the conduction electrons is written as,

$$J_n = n\,e\,v \tag{4.13}$$

The conductivity of the semiconductor due to the conduction electrons is

$$\sigma_n = \frac{J_n}{E} = \frac{n\,\mathrm{e}\,\mathrm{v}}{E} = n\,\mathrm{e}\,\mu_n \tag{4.14}$$

where m_n is the mobility of the conduction electrons.

[$J = \sigma E$ is the microscopic form of the Ohm's law.]

Similarly the conductivity of the semiconductor due to the positive holes is

$$\sigma_p = p\,\mathrm{e}\,\mu_p \tag{4.15}$$

where p = positive hole density, μ_p= mobility of the holes.

The total **electrical conductivity of** the **semiconductor** is,

$$\sigma = \sigma_p + \sigma_n$$

$$\sigma = e\left(n\mu_n + p\mu_p\right)$$

\ The conductivity of a semiconductor depends on the charge carrier density and the mobility of the charge carriers.

The conductivity of an intrinsic semiconductor is,

$$\sigma_i = e\left(n_i\mu_n + n_i\mu_p\right)$$

since $n = p = n_i$ (n_i = intrinsic electron density)

$$\sigma_i = n_i e\left(\mu_n + \mu_p\right) \tag{4.16}$$

In the n-type semiconductor the conduction electron density n_n is much more than the positive hole density p_n.

i. e., $$n_n >> p_n$$

\ The conductivity of a n-type semiconductor is,

$$\sigma_n = n_n e\mu_n \tag{4.17}$$

In the p-type semiconductor the positive hole density p_p is much more than the conduction electron density n_p.

i. e., $$p_p >> n_p$$

\ The conductivity of a p-type semiconductor is,

$$\sigma_p = p_p e\mu_p \tag{4.18}$$

Variation of conductivity of an intrinsic semiconductor with temperature is given by,

$$\sigma_i = (constant)\, T^{\frac{3}{2}}\ exp\ \left(\frac{-E_g}{2kT}\right) \tag{4.19}$$

For a short range of temperature the variation of σ_i with $T^{\frac{3}{2}}$ is much less compared to the variation with $\exp\left(\frac{1}{T}\right)$. Thus, one can write

$$\sigma_i = \sigma_0 \exp\left(\frac{-E_g}{2kT}\right) \tag{4.20}$$

where σ_0 is a constant. This expression (Eqn. 4.20) can be used to determine the value of E_g.

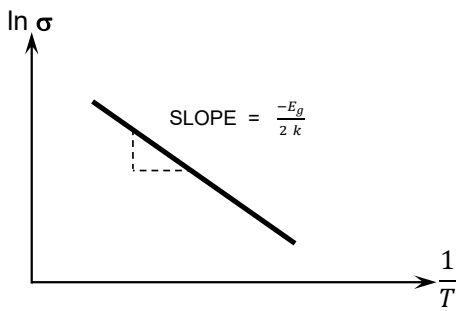


Figure 4.16: Variation of conductivity in an intrinsic semiconductor with the inverse of temperature.

4.6 HALL EFFECT

Hall effect is a process in which a transverse electric field is developed in a solid material when the material carrying an electric current is placed in a magnetic field that is perpendicular to the current. The Hall effect was discovered by Edwin Herbert Hall in 1879. When a current-carrying conductor or a semiconductor is introduced to a perpendicular magnetic field, a voltage can be measured at the right angle to the current path. This effect of obtaining a measurable voltage is known as the Hall effect.

Consider a semiconductor (assumed to be n-type) in the form of a rectangular strip of width ***w***, thickness t and electron density n. Let a current ***I*** flow along its length in X direction and a transverse magnetic field ***B*** be applied across its thickness **t** along the Y direction (Figure 4.17).

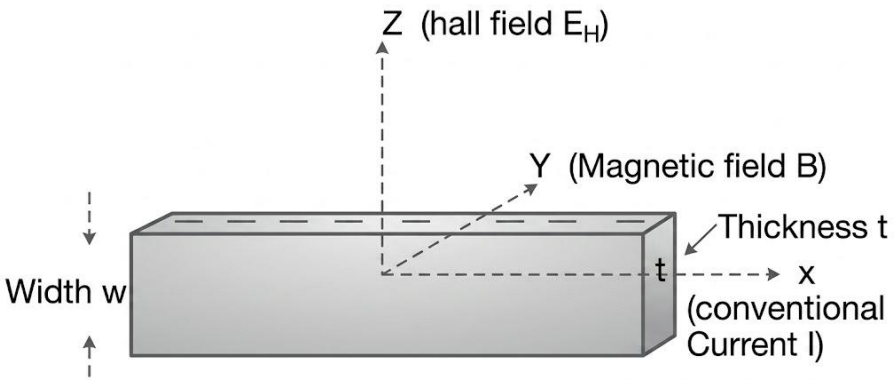


Figure 4.17: Hall Effect in *n*-type semiconductor.

The moving electrons experience a force F_M due to the magnetic field. Due to F_M , the electrons tend to move in the Z direction leaving behind the positive charges.

$$F_M = e\, v_D B \tag{4.21}$$

where, v_D is the drift speed of the electrons and e is the charge on the electron.

This separation of charges results in an electric field E_H across the width of the specimen (in Z direction). E_H exerts a force on the electrons given by

$$F_E = -e\, E_H \tag{4.22}$$

Under equilibrium conditions $e\, E_H = e\, v_D B$ \ $E_H = v_D B$

We have $E_H = \frac{V_H}{w}$ and $v_D = \frac{I}{n\ e\ w\ t}$

Substituting these values, we get

$$V_H = \frac{I\ B}{n\ e\ t} \quad \text{or} \quad V_H = \frac{R_H I\ B}{t}$$

Where, the quantity $\frac{1}{n\ e} = R_H$ is called the Hall coefficient of the specimen.

$$\therefore R_H = \left(\frac{V_H}{B}\right)\left(\frac{t}{I}\right) \tag{4.23}$$

The hall coefficient is positive if the number of positive charges is more than the negative charges. Similarly, it is negative when electrons are more than holes.

Applications of Hall Effect:

1. Magnetic field sensing equipment (Gauss Meter)
2. For the measurement of direct current, Hall effect Tong Tester is used.
3. It is used in phase angle measurement.
4. Proximity detectors.
5. Hall effect Sensors and Probes.
6. For detecting wheel speed and accordingly assist the anti-lock braking system.

4.7 RESISTIVITY MEASUREMENT BY FOUR PROBE METHOD

A material's electrical resistivity is a measure of how strongly it opposes the flow of current. The Ω-m is the SI unit of electrical resistivity. The electric resistance (R) of a material is given by $R = V/I$ where V is electrical potential difference across the material and I is the current through the material. The resistance depends on the dimension of the material. For a wire $R = \rho L/A$, where L is the length of the area, A is the cross sectional area of the wire and ρ is the electrical resistivity of the material of the wire. Thus, for a wire, the resistivity,

$$\rho = \frac{R \times A}{L} \tag{4.24}$$

When two probes are used for measurement of voltage and current, and hence for the calculation of resistance in the case of a semiconductor, the probe resistance, the contact resistance, the spreading resistance, and the sheet resistance (resistance of a square piece of a thin material with contacts made to two opposite sides of the square) due to the semiconductor will come into effect. Using four probes one can eliminate the probe resistance, contact resistance (the electrical resistance that arises at the contact point when components are

connected) and the spread resistance (refers to the phenomenon in which current spreads through a material or device, resulting in a decrease in resistance. in the measurement).

The current is passed through the outer probes and the voltage is measured across the inner probes. The resistivity measured using the four probes may vary with the geometry (thickness and shape) of the semiconductor sample and placement of the probes on the surface of the sample. It may also vary with the probe gap, nature of the surface of the sample, the size of the sample.

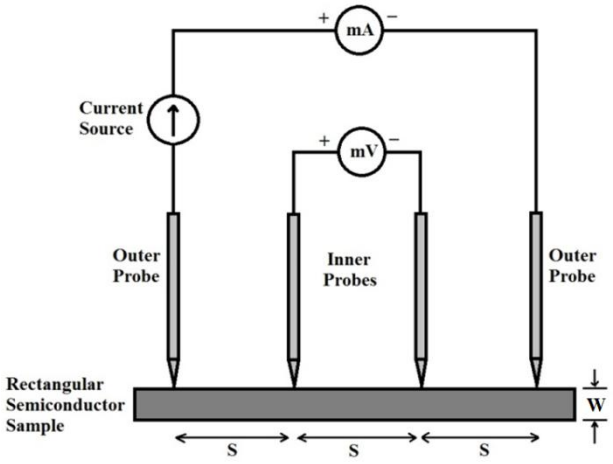


Figure 4.18: Experimental setup for resistivity measurement.

Mathematical calculations show that the resistivity of a semi-infinite sample is given by

$$\rho_0 = \frac{V}{I} \times 2\pi S \tag{4.25}$$

where I is the current through the outer probes, V is the voltage across the inner probes, S is the distance between the probes (which is taken to be the same here) and $2\pi S$ is the geometric factor of the semi-infinite sample.

For a finite rectangular sample of thickness W, a dimension correction function $f\,(W/S)$ is introduced in the resistivity expression as,

$$\rho = \frac{\rho_0}{f(W/S)}$$

i.e.,

$$\rho = \frac{2\pi S}{f(W/S)} \times \frac{V}{I} \tag{4.26}$$

The resistivity is estimated form the above expression.

4.8 JUNCTION DIODE

A p-n junction is formed when a p-type semiconductor is joined to an n-type semiconductor. A p-n junction has three distinct regions: a p-region, an n-region, and a depletion region at the junction (Figure 4.19a). The depletion region has no movable charges (conduction electrons and holes) because the conduction electrons on the n-side have crossed the junction due to diffusion and neutralized the holes on the p-side.

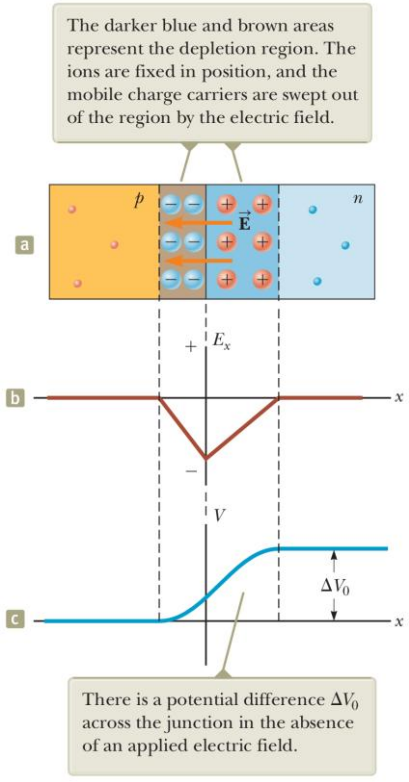


Figure 4.19: a) Schematic diagram of a typical p-n junction. b) Variation of the electric field across the depletion region. c) Variation of the potential difference across the depletion region.

Because of the fixed ion cores in the depletion region, an electric field exists (Figure 4.19b), due to which there is a potential difference ΔV_0 across the junction (Figure 4.19c). In the forward bias, the p-side of the junction is made positive with respect to the n-side, by application of an external voltage ΔV. Then the internal potential difference ΔV_0 across the junction decreases. This gives rise to a current I, which increases exponentially with the increase in forward bias ΔV. In the reverse bias, the p-side of the junction is made negative with respect to the n-side, by application of a negative voltage ΔV. Then the internal potential difference ΔV_0 across the junction increases. This gives rise to a reverse current that quickly reaches a saturation value I_0 (reverse saturation current). The current in an ideal diode under the biasing voltage ΔV at temperature T is,

$$I = I_o \left[exp\left(\frac{e\ \Delta V}{k\ T}\right) - 1 \right] \tag{4.27}$$

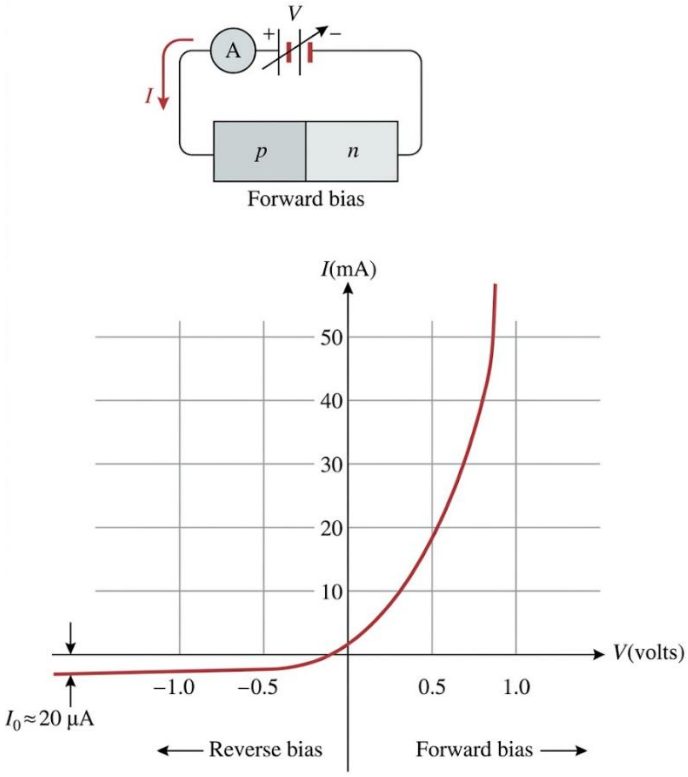


Figure 4.20: Characteristic I-V curve of a p-n junction diode.

4.9 PHOTOVOLTAIC EFFECT AND SOLAR CELLS

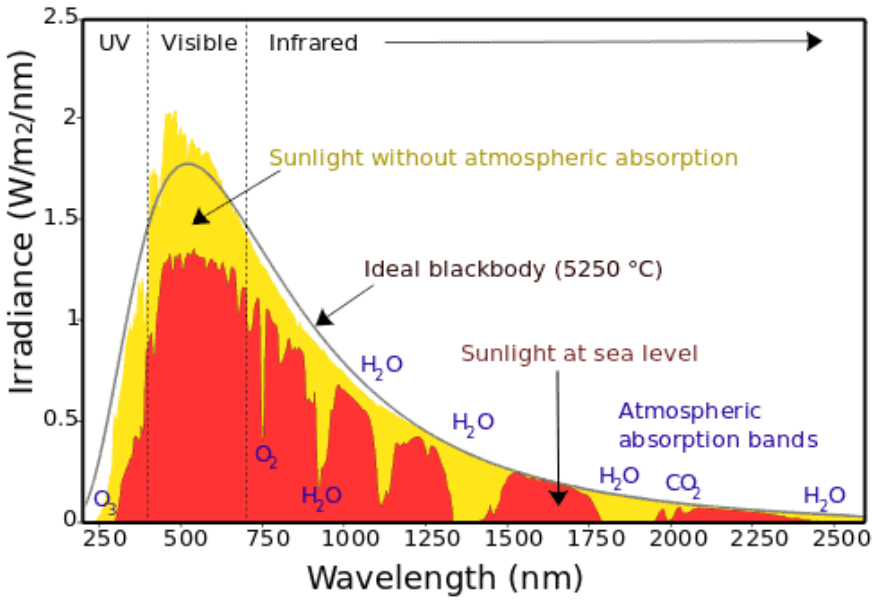
Solar cells, also referred to as photovoltaic cells, are pivotal in the realm of renewable energy. Their functionality hinges on the photovoltaic effect, where semiconductor materials within the cell absorb photons from sunlight. These absorbed photons energize electrons, enabling them to flow as an electric current. This process occurs without moving parts, noise, or greenhouse gas emissions, rendering solar cells an environmentally friendly and sustainable energy source. The significance of solar cells lies in their contribution to mitigating climate change and reducing dependence on finite fossil fuels. By harnessing the virtually infinite energy of the sun, solar cells offer a clean alternative to traditional energy sources. They aid in shrinking carbon footprints, thereby combating global warming and environmental degradation. Moreover, solar cells have versatile applications, ranging from powering residential homes and commercial buildings to providing electricity in remote locations

without access to the grid. They also find use in spacecraft, satellites, and portable electronic devices, showcasing their adaptability and reliability in various settings.

4.9.1 Solar spectrum and Air Mass (AM)

The solar spectrum and air mass are crucial concepts in understanding how sunlight interacts with the Earth's atmosphere and influences the performance of solar energy systems. The solar spectrum refers to the distribution of electromagnetic radiation emitted by the sun across different wavelengths. Sunlight contains a broad range of wavelengths, from ultraviolet (UV) to infrared (IR), with visible light occupying the middle portion of the spectrum. Each wavelength carries a specific amount of energy, and the distribution of this energy determines the spectral composition of sunlight. However, as sunlight travels through the Earth's atmosphere, it encounters various molecules and particles that can scatter and absorb certain wavelengths. This interaction alters the solar spectrum reaching the Earth's surface. For instance, shorter wavelengths like UV are more prone to scattering by molecules such as oxygen and nitrogen, while longer wavelengths like IR can be absorbed by greenhouse gases like water vapor and carbon dioxide.

Air mass (AM) quantifies the path length of sunlight through the Earth's atmosphere before reaching the surface. It is expressed numerically as the ratio of the actual path length to the shortest possible path length through the atmosphere (when the sun is directly overhead). Air mass values vary depending on the angle of the sun relative to the observer's location. For example, AM1 corresponds to solar radiation perpendicular to the Earth's surface at sea level, while higher AM values represent sunlight traveling through a thicker atmospheric layer at greater angles. Understanding the solar spectrum and air mass is crucial for designing and optimizing solar energy systems. The standard spectrum for space applications is referred to as AM0. It has an integrated power of 1366.1 W/m^2. The AM1.5 Global spectrum is designed for flat plate modules and has an integrated power of 1000 W/m^2.



(Source: https://g2voptics.com/)

Figure 4.21. Solar spectrum (Red: AM1.5)

4.9.2 Solar Cell: p-n junction under illumination

At the heart of a solar cell is a semiconductor material, often silicon, with additional elements added to create either a positive (p-type) or negative (n-type) charge. When sunlight strikes the solar cell, it excites electrons in the semiconductor material, causing them to break free from their atoms. This process generates electron-hole pairs. However, for these electron-hole pairs to produce an electric current, they need to be separated and directed towards an external circuit. This separation occurs at the junction between the p-type and n-type regions of the solar cell, known as the p-n junction. At this junction, the free electrons from the n-type region are attracted to the positive charge in the p-type region, while the holes from the p-type region are attracted to the negative charge in the n-type region. This movement of charge creates an electric field across the junction.

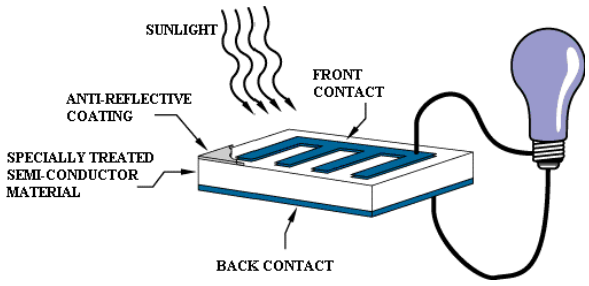


Figure 4.22: Schematic of a solar cell.

When an external circuit is connected to the solar cell, the separated electrons and holes can flow in opposite directions through the circuit, generating an electric current. This flow of

electrons constitutes electricity, which can be harnessed to power electrical devices or stored in batteries for later use.

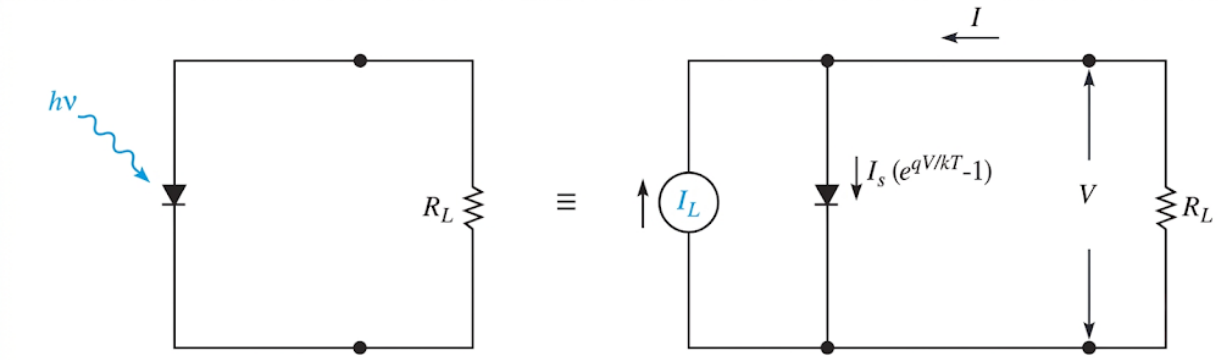


Figure 4.23: Solar cell under illumination.

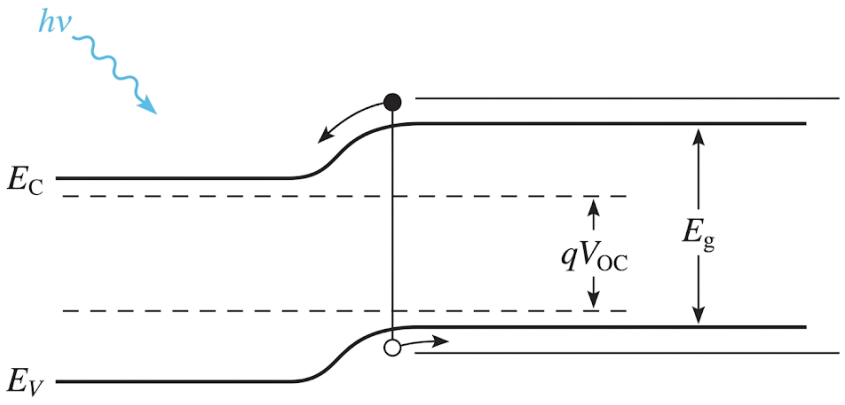


Figure 4.24: Energy level diagram showing carrier transport under illumination of a p-n junction.

4.9.3 Solar Cell Parameters

(i) *Maximum Power Point Power* (P_{MPP}): The semiconductor material of the solar cell has its internal resistance. This resistance varies with the light intensity and attains a minimum value at maximum intensity. At this saturation value the power output is maximum. This maximum power output is termed as maximum power point power and denoted as P_{MPP}. At this point maximum power transfer takes from the solar cell to the load, and it is given by,

$$P_{MPP} = I \times V \tag{4.28}$$

P_{MPP} is measured by finding the product of voltage and current at the maximum power point.

(ii) *Short Circuit Current* (I_{SC}) : When the solar cell terminals are shorted ($R_L = 0$) together maximum current flows through the terminals. This maximum current is called short-circuited current and denoted as I_{SC}.

(iii) *Open Circuit Voltage* (V_{OC}) : The voltage measured at the terminals of the solar cell without load ($R_L = \infty$) is called open circuit voltage and denoted as V_{OC}. At both the cases of open circuit or short circuit the power dissipated by the solar cell is zero.

(iv) *Conversion Efficiency* (η) : In a solar cell light energy is converted in to electrical energy. In this process of conversion not all input solar radiation is converted in to electric current. The conversion efficiency denoted in percentage and it is given by,

$$\eta = \frac{Output\ Power}{Input\ power}$$

$$\eta = \frac{P_{MPP}}{A_C E} \tag{4.29}$$

where A_C is the surface area of the solar cell, E is the light intensity in watts/cm^2.

(v) *Fill Factor*: The fill factor (FF) of a solar cell is a parameter used to quantify its efficiency in converting sunlight into electrical energy. It represents the ratio of the maximum power output of the solar cell to the product of its open-circuit voltage (Voc) and short-circuit current (Isc). Mathematically, the fill factor (FF) is defined as:

$$FF = \frac{Maximum\ power\ output}{V_{OC} I_{SC}} \tag{4.30}$$

The fill factor essentially measures how effectively a solar cell can utilize the available sunlight to produce electricity. It takes into account losses within the solar cell, including those due to recombination of charge carriers, series resistance, and shunt resistance.

Ideally, a solar cell would have a fill factor of 1, indicating perfect efficiency. However, in practice, fill factors for commercial solar cells typically range from 0.5 to 0.8, depending on various factors such as the materials used, manufacturing processes, and operating conditions.

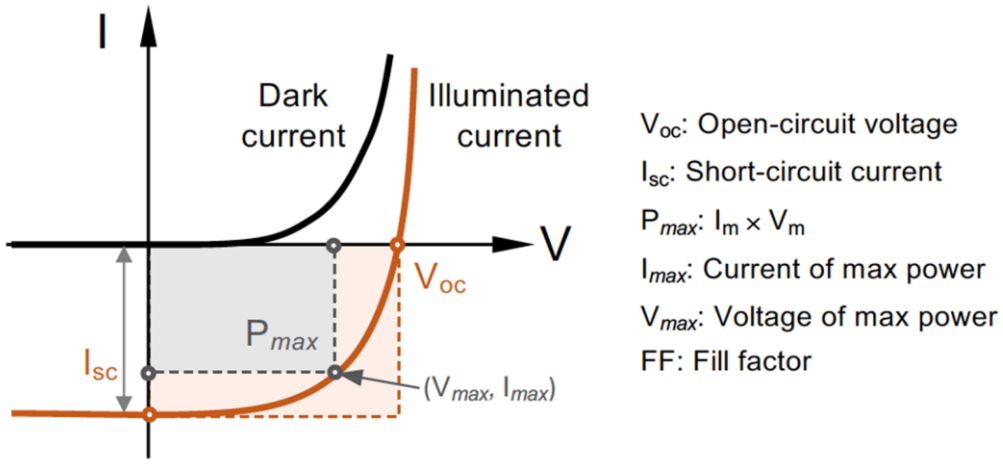
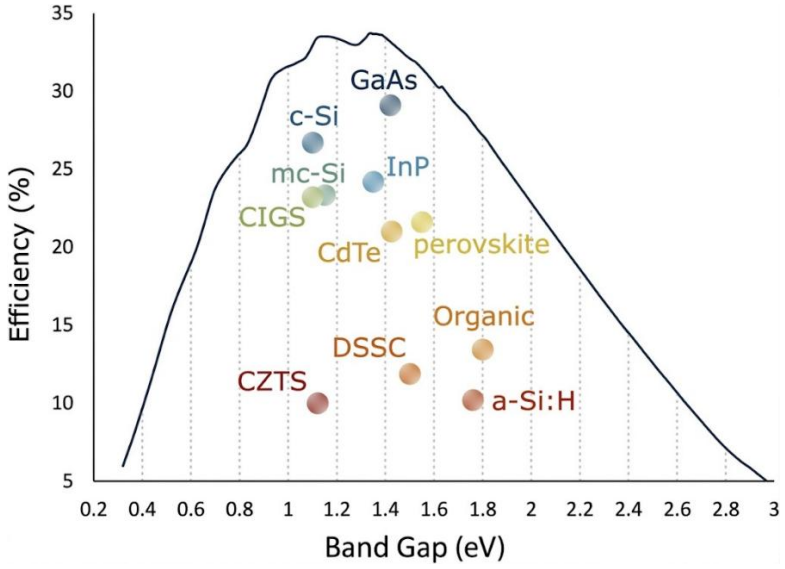


Figure 4.25: I-V curve for a typical solar cell. P_{max} is the maximum achievable power from the cell.

4.9.4 Solar cell materials

The Shockley-Queisser limit, named after scientists William Shockley and Hans-Joachim Queisser, is a theoretical maximum efficiency limit for a single-junction solar cell. It establishes the maximum efficiency a solar cell can achieve based on fundamental principles of thermodynamics and semiconductor physics. It takes into account factors such as the bandgap energy of the semiconductor material used in the solar cell, the energy of photons in sunlight, and the loss mechanisms within the cell. According to the Shockley-Queisser limit, the maximum efficiency of a single-junction solar cell operating under one sun illumination (AM1.5G spectrum) at room temperature is approximately 33.7%. For single-junction solar cells, the bandgap energy should be such that it matches the solar spectrum to achieve maximum absorption of photons. Typically, bandgap energies in the range of 1.1 to 1.7 electron volts (eV) are considered suitable for solar cells.



Source: https://doi.org/10.1016/j.joule.2020.05.001

Figure 4.26: Shockley-Queisser limit.

Choosing the right material for solar cells involves a strategic approach that considers several factors to optimize performance, cost-effectiveness, and sustainability.

Efficiency: Prioritize materials with high conversion efficiency to maximize electricity generation per unit area. Silicon-based solar cells offer proven high efficiency, but emerging materials like perovskites show potential for even greater efficiency gains.

Cost-effectiveness: Balance efficiency with cost to ensure affordability and scalability. Silicon remains a popular choice due to its abundance, but thin-film technologies like cadmium telluride (CdTe) and copper indium gallium selenide (CIGS) offer lower manufacturing costs per watt.

Availability: Consider the availability and accessibility of raw materials to ensure long-term sustainability and reduce supply chain risks. Choose materials with ample global reserves and minimal geopolitical constraints.

Stability and Reliability: Select materials with proven long-term stability and reliability under various environmental conditions to ensure consistent performance over the solar cell's lifetime.

Toxicity and Environmental Impact: Assess the environmental impact and toxicity of materials throughout their lifecycle, from extraction and manufacturing to disposal or

recycling. Choose materials with minimal environmental footprint and recyclability to promote sustainability.

Technological Feasibility: Evaluate the technological readiness and scalability of materials for large-scale manufacturing. Consider factors such as ease of fabrication, compatibility with existing infrastructure, and potential for integration into commercial production processes.

4.10 LIGHT-EMITTING DIODE (LED)

An electron excited electrically into the conduction band (injected into a p-region of an LED) can easily recombine with a hole (Figure 4.27). Then, a photon of energy E_g is emitted. This is the principle of light-emission from an LED or a semiconductor laser.

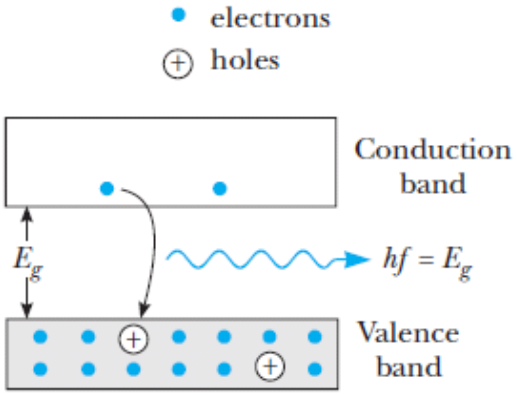


Figure 4.27: Light emission in a typical semiconductor

4.11 MOSFET

A MOSFET is a metal-oxide-semiconductor field-effect transistor (Figure 4.28). In this, there are three metal connections to the transistor: source, drain, and gate. The source and drain are connected to n-type semiconductor regions. These regions are connected by a narrow channel of n-type material (n-channel). The source and drain regions, as well as the n-channel, are embedded in a p-type substrate.

This forms a depletion region along the bottom of the n-channel. The gate is separated from the n-channel by a layer of insulating silicon dioxide. Thus, it does not make electrical contact with the rest of the semiconducting material.

If a voltage ΔV_{SD} is applied across the source and drain (Figure 4.28b), the electrons flow through the upper region of the n-channel, because the lower part of the n-channel is depleted of charge carriers. Now, if a second voltage ΔV_{SG} is applied across the source and gate with

the positive potential on the gate (Figure 4.28), the depletion region in the n-channel becomes narrower and hence the source-drain current increases.

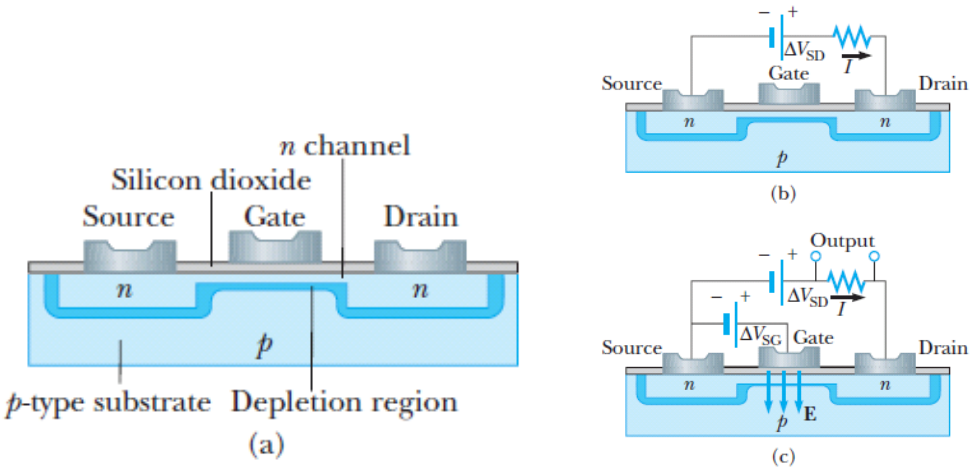


Figure 4.28: Schematic diagram of MOSFET.

If a varying voltage is applied to the gate, the source-drain current also varies. A small variation in gate voltage ΔV_{SG} results in a large variation in source-drain current, and a correspondingly large output-voltage across the resistor in Figure 4.28c. Therefore, the MOSFET acts as a voltage amplifier. If a negative potential is applied to the gate, the n-channel decreases in size when ΔV_{SG} increases. This reduces the source-drain current and stops the current when ΔV_{SG} is large. Thus, MOSFET can be used as an on-off switch by changing the polarity of ΔV_{SG}.

Advantages of MOSFET:

- High Input Impedance: The insulated gate ensures near-zero gate current, preventing circuit loading.
- Low Power Loss: Extremely low power consumption in CMOS logic, ideal for modern portable electronics.
- Fast Switching: Capable of operating at very high frequencies for efficient power and signal control.
- High Scalability: Compact design allows billions of units to fit on a single chip for microprocessors.

4.12 SUPERCONDUCTIVITY

Superconductors are a class of metals and compounds whose electrical resistance decreases to virtually zero below a certain temperature T_c called the critical temperature. The critical temperature is different for different superconductors. Mercury loses its resistance completely

and turns into a superconductor at 4.2 K. Critical temperatures for some important elements/compounds are listed below.

Element/Compound	*Tc* (*K*)
La	6.0
NbNi	10.0
Nb_3Ga	23.8

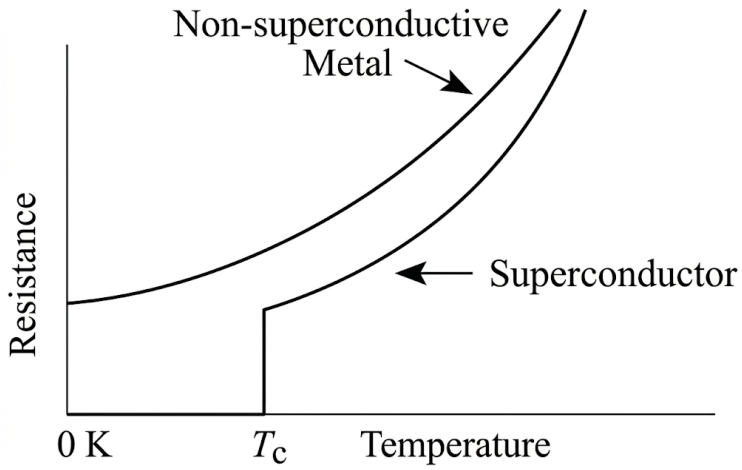


Figure 4.29: Plot of Resistance Vs Temperature for a normal metal and a superconductor.

4.12.1 Meissner Effect

In the presence of magnetic field, as the temperature of superconducting material is lowered below T_c, the field lines are spontaneously expelled from the interior of the superconductor (B = 0, Figure 4.30). Therefore, a superconductor is more than a perfect conductor; it is also a perfect diamagnet. The property that $B = 0$ in the interior of a superconductor is as fundamental as the property of zero resistance. If the magnitude of the applied magnetic field exceeds a critical value B_c, defined as the value of B that destroys a material's superconducting properties, the field again penetrates the sample. The Meissner effect can be explained as follows.

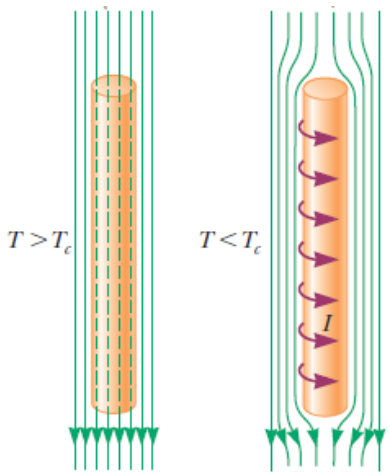


Figure 4.30: A superconductor in the form of a long cylinder in the presence of an external magnetic field.

A good conductor expels static electric fields by moving charges to its surface. In effect, the surface charges produce an electric field that exactly cancels the externally applied field inside the conductor. In a similar manner, a superconductor expels magnetic fields by forming surface currents. Consider the superconductor shown in Figure 4.30. Let's assume the sample is initially at a temperature $T > T_c$ so that the magnetic field penetrates the cylinder. As the cylinder is cooled to a temperature $T < T_c$, the field is expelled. Surface currents induced on the superconductor's surface produce a magnetic field that exactly cancels the externally applied field inside the superconductor. As expected, the surface currents disappear when the external magnetic field is removed.

4.12.2 BCS Theory

In 1957, Bardeen, Cooper and Schrieffer gave a successful theory to explain the phenomenon of superconductivity, which is known as BCS theory. According to this theory, two electrons can interact via distortions in the array of lattice ions so that there is a net attractive force between the electrons. As a result, the two electrons are bound into an entity called a **Cooper pair**, which behaves like a single particle with integral spin. Particles with integral spin are called bosons. An important feature of bosons is that they do not obey the Pauli exclusion principle. Consequently, at very low temperatures, it is possible for all bosons in a collection of such particles to be in the lowest quantum state and as such the entire collection of Cooper pairs in the metal is described by a single wave function. There is an energy gap equal to the binding energy of a Cooper pair between this lowest state and the next higher state. Under the action of an applied electric field, the Cooper pairs experience an electric force and move

through the metal. A random scattering event of a Cooper pair from a lattice ion would represent resistance to the electric current. Such a collision would change the energy of the Cooper pair because some energy would be transferred to the lattice ion. There are no available energy levels below that of the Cooper pair (it is already in the lowest state), however, and none available above because of the energy gap. As a result, collisions do not occur, and there is no resistance to the movement of Cooper pairs.

4.12.3 Types of Superconductors

Type - I Superconductors: Type I superconductors are those superconductors that lose their superconductivity very easily or abruptly when placed in the external magnetic field. As you can see from the graph of the intensity of magnetization (M) versus applied magnetic field (H), when the Type I superconductor is placed in the magnetic field, it suddenly or easily loses its superconductivity at the critical magnetic field (H_c) (point A).

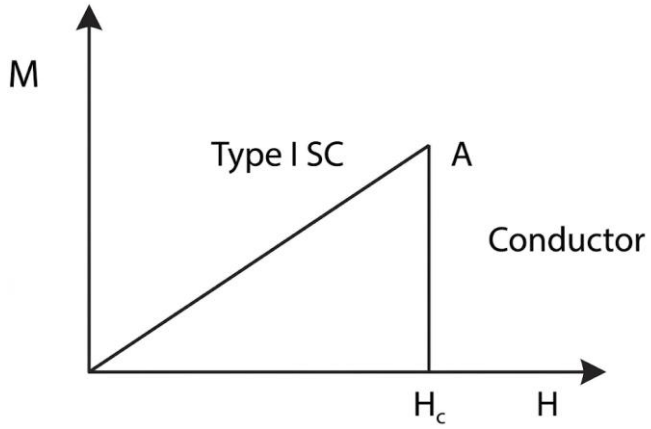


Figure 4.31: Graph of the intensity of magnetization (M) versus applied magnetic field (H).

After H_c, the Type I superconductor will become a conductor. Type I superconductors are also known as soft superconductors because they lose their superconductivity easily. Type I superconductors are perfectly diamagnetic in nature and obey the Meissner effect. Lead (Pb) at 7.196 K, Lanthanum (La) at 4.88 K and Tantalum (Ta) at 4.47 K are Type-I superconductors.

Type-II Superconductors: Type II superconductors are those superconductors that lose their superconductivity gradually but not easily or abruptly when placed in the external magnetic field. As seen from the graph of the intensity of magnetization (M) versus applied magnetic field (H), when the Type II superconductor is placed in the magnetic field, it gradually loses its superconductivity. Type II superconductors start to lose their superconductivity at the lower

critical magnetic field (H_{c1}) and completely lose their superconductivity at the upper critical magnetic field (H_{c2}).

The state between the lower critical magnetic field (H_{c1}) and upper critical magnetic field (H_{c2}) is known as **vortex state or intermediate state**. After H_{c2}, the Type II superconductor will become a conductor.

Type II superconductors are also known as hard superconductors because they lose their superconductivity gradually but not easily. Type II superconductors obey the Meissner effect but not completely. Example of Type II superconductors: YBCO (Yttrium barium copper oxide), niobium nitride (NbN), Barium bismuth ($BaBi_3$). Type II superconductors are used for strong field superconducting magnets.

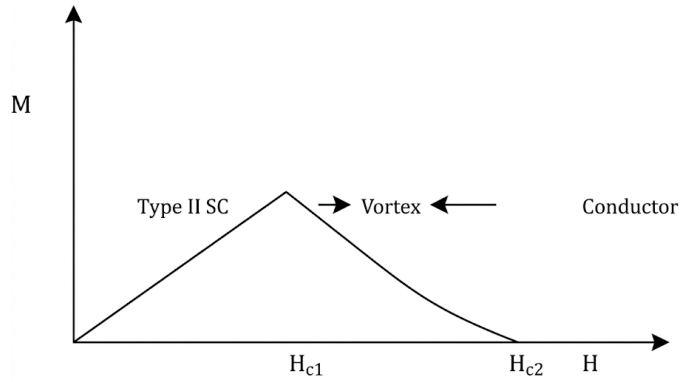


Figure 4.32: Graph of the intensity of magnetization (M) versus applied magnetic field (H).

4.12.4 Applications

Most important and basic application of superconductors is in high field solenoids which can be used to produce intense magnetic field. Superconducting magnets are used in magnetic resonance imaging (MRI) technique. Magnetic levitation, based on Meissner effect, is another important application of superconductors. This principle is used in maglev vehicles. Detection of a weak magnetic field and lossless power transmission are some other important applications of superconductors.

4.13 DIELECTRICS

In the realm of electrical engineering, dielectrics stand as a crucial class of materials, distinct in their properties and applications. These materials do not conduct electricity like metals do. Instead, when placed them in an electric field of sufficient magnitude, something interesting happens. The positive and negative charges inside them move a little bit. That means the centres

of the positive charges are displaced slightly along the direction of the applied field and that of the negative charges displaced in the opposite direction. This produces local *electric dipole* throughout the material. An *electric dipole* is composed of two opposite charges of equal magnitudes, $+q$ and $-q$, separated by a certain distance $\vec{r}$ apart as shown in Figure 4.33. The *electric dipole moment* ($\vec{p}$) is defined as the total charge (q) of one kind (positive or negative, as both are equal) multiplied by the distance ($\vec{r}$) between the two kinds of charges i.e., $\vec{p} = q\vec{r}$.

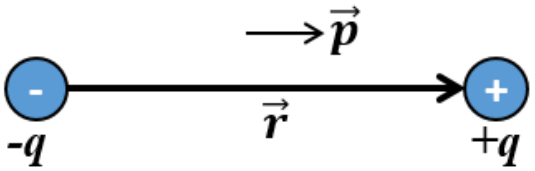


Figure 4.33**:** An electric dipole.

Dielectric materials are characterized by their non-conductivity, lacking free charge carriers within them. When subjected to an electric field, dielectrics alter the field and themselves undergo significant changes, enabling them to store electrical charges. For a material to be a good dielectric, it must be an *insulator*. The function of an insulating material is to obstruct the flow of electric current while that of a dielectric material is to store electrical energy. Thus, insulating materials and dielectric materials differ in their functions. The dielectric material may be polar or nonpolar.

Polar dielectrics: In this case of dielectrics the center of the gravity of the positive charges in the molecule *does not* coincident with the center of gravity of the negative charges, and thus, they possess a permanent electric dipole moment. When an electric field is applied all the molecules in the polar dielectric almost align itself in the direction of the applied electric field. Example of a polar dielectric is water.

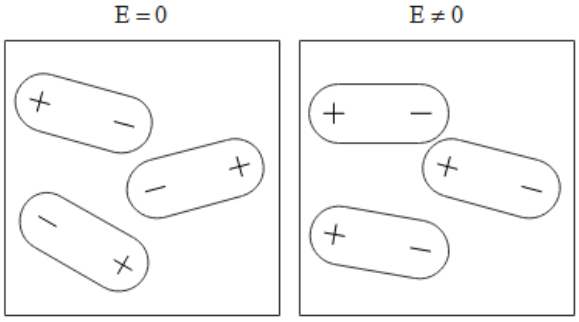


Figure 4.34**:** Polar dielectric

Non-polar dielectrics: In this case of dielectrics the center of gravity of positive charges coincide with the center of gravity of negative charges, hence their dipole moment is zero. When a non-zero electric field is applied all the molecules in the non-polar dielectric align themselves in the direction of the applied electric field and produce net dipole moment. Example of a non-polar dielectric is methane.

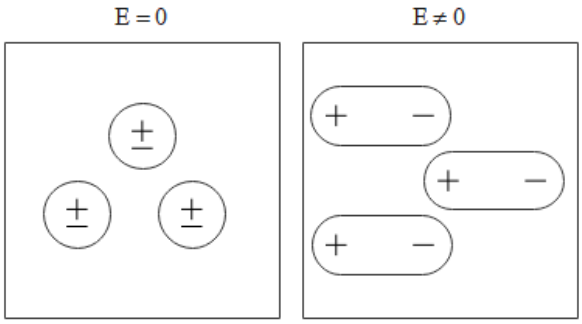


Figure 4.35**:** Non-polar dielectric

4.14 POLARIZATION IN A DIELECTRIC MATERIAL

Polarization in a dielectric material refers to the process by which the material develops an internal electric field when placed in an external electric field. This phenomenon arises due to the alignment or reorientation of electric dipoles within the dielectric material. When an external electric field is applied to a dielectric, it exerts forces on the electric charges within the material, causing them to shift slightly. Positive charges tend to move in the direction of the field, while negative charges move in the opposite direction. This displacement of charges creates an induced electric dipole moment in each atom or molecule of the dielectric. As a result, the dielectric becomes polarized, meaning that its overall electric field no longer aligns perfectly with the external field. Instead, the induced dipoles within the material oppose the external field to some extent. This polarization effect reduces the net electric field experienced by the dielectric material.

Upon the application of external electric field, polarization can occur through various mechanisms, including *electronic (or atomic) polarization* (due to the displacement of electron clouds in the atom), *ionic (or displacement) polarization* (due to the displacement of ions within the material), and *orientational (or dipolar) polarization* (due to the rotation of permanent dipoles). The combined effect of these mechanisms determines the overall polarization response of the dielectric material. Polarization is a crucial concept in dielectric materials because it influences their electrical properties, such as capacitance, permittivity, and

energy storage capabilities. Understanding polarization is essential for designing and optimizing the performance of devices and systems that utilize dielectric materials, such as capacitors, insulators, and electrical transmission lines.

The important parameters which characterize the dielectric materials are described below:

4.14.1 Dielectric Constant (ϵ_r)

A dielectric is primarily characterized by its dielectric constant. *Dielectric constant* of a dielectric is defined and measured as the ratio of capacitance of a capacitor containing the dielectric medium to the capacitance of the same capacitor with air as the medium.

$$\epsilon_r = \frac{C}{C_0} \tag{4.31}$$

where C_0 is the capacitance with air as the medium between the plates and C is the capacitance with dielectric as medium. ϵ_r is called *dielectric constant* or *relative permittivity*. It is a dimensionless quantity, which is always greater than unity in case of dielectrics, and it is independent of the size or shape of the dielectric. In fact, ϵ_r gives the measure of the *ability of a substance to store electrical energy* when placed in an electric field. At times another quantity known as the *permittivity of the medium* (ϵ) is used. It is given by,

$$\epsilon = \epsilon_0 \, \epsilon_r \tag{4.32}$$

where $\epsilon_0 = 8.854 \times 10^{-12}$ F/m represents the *permittivity of free space*. Permittivity is a material property that affects the Coulomb force between two point charges in the material. Permittivity of the medium is a measure of the degree to which a medium can resist the flow of charge.

4.14.2 Dielectric Polarization ($\vec{P}$)

The *dielectric polarization (also called polarization or polarization density)* $\vec{P}$ at any point of the dielectric is defined as the *electric dipole moment per unit volume* at that point. Polarization is a macroscopic property of the material or in other words, polarization takes place in all the molecules of a dielectric, that is, it is a bulk property. This causes the formation of an electric dipole moment in the entire volume of the dielectric and in each separate polarizing particle such as atom, ion, molecule, and so on and $\vec{P}$ is given by,

$$\vec{P} = \sum_i n_i \vec{p_i} = N\vec{p} \tag{4.33}$$

where n_i represents number of individual particles (atom, ion, molecules etc), p_i is the dipole moment of these particles, N is the number of particles per unit volume of the dielectric and p is the net dipole moment of the dielectric.

4.14.3 Electric susceptibility (χ)

How easily a dielectric material can be polarized when subjected to an electric field is measured by the parameter called *electric susceptibility*. Thus, the electric susceptibility indicates the degree of polarization of a dielectric material in response to an applied electric field. In a homogeneous isotropic dielectric material the polarization $\vec{P}$ is proportional to the applied weak electric field $\vec{E}$. Hence, the polarization is written as,

$$\vec{P} = \epsilon_0 \chi \vec{E} \tag{4.34}$$

where ϵ_0 is the *permittivity of free space* and χ is the *electric susceptibility* of the material. Thus, greater the electric susceptibility, the greater the ability of a material to polarize in response to the applied electric field, and thereby reduce the total electric field inside the material (and hence store energy).

The susceptibility χ is related to its relative permittivity ϵ_r (dielectric constant) as

$$\epsilon_r = 1 + \chi \tag{4.35}$$

4.14.4 Electric displacement field or Displacement vector ($\vec{D}$)

The *electric displacement field* is a vector field that describes the displacement effects of an electric field on the charges within a dielectric material, such as polarisation charges or bound charges. It is defined by the equation,

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \tag{4.36}$$

It has a unit of coulombs per square meter (C/m²). The electric displacement field has no real physical meaning but is used for easier calculations involving dielectric materials.

Since $\vec{P} = \epsilon_0 \chi \vec{E}$ and $\epsilon_r = 1 + \chi$ we have,

$$\vec{D} = \epsilon_0 \vec{E} + \epsilon_0 \chi \vec{E} = (1 + \chi)\epsilon_0 \vec{E} = \epsilon_r \epsilon_0 \vec{E} = \epsilon \vec{E} \tag{4.37}$$

4.14.5 Polarizability (α)

There is another important property connected with dielectrics, that is, *polarizability*. Polarizability usually refers to the tendency of matter, when subjected to an electric field, to acquire an electric dipole moment in proportion to that applied field. When the dielectric is subjected to external electric field, an electric dipole moment is induced on every particle (atom, ion, dipole), which is given by

$$\vec{p} = \alpha \vec{E} \tag{4.38}$$

where $\vec{p}$ is the induced dipole moment, $\vec{E}$ is the electric field, and α is the constant of proportionality, called the *polarizability* of that particle. Polarizability reflects the properties of an individual particle of matter and not of a certain very large volume of it. It is a microscopic electrical parameter of a dielectric. In *cgs* system, the polarizability α is expressed as cm^3 and its SI unit is Fm^2. For anisotropic molecules, polarizability is different in different directions.

4.15 COMPONENTS OF POLARIZABILITY

Polarization of a dielectric takes place due to the polarizability of the dielectric medium on the application of the electric field. There are *three* main components of polarizability:

4.15.1 Electronic (or atomic) polarizability (α_{el})

In all circumstances, the application of an electric field causes some distortion of the electronic cloud of the atom relative to the nucleus. Hence *electronic polarization* results from the displacement of the electron clouds of atoms with respect to heavy 'fixed' nuclei to a distance that is less than the dimensions of the atoms. It occurs in all dielectrics for any state of aggregation. The phenomenon is illustrated in Figure.

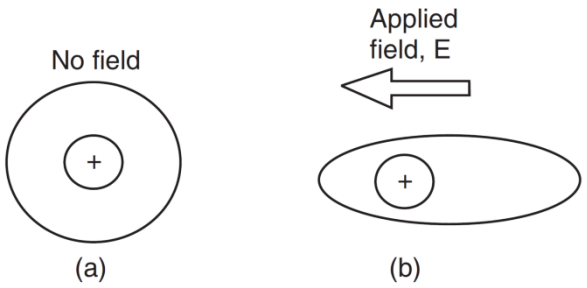


Figure 4.36: Electronic polarization a) Atoms are not polarized in the absence of the electric field, b) Electronic polarization results from the distortion of electron cloud by an applied electric field.

When an external *static electric field* $\vec{E}$ is applied across the dielectric, the atoms having electron cloud which is assumed to be a spherical region of uniform charge of radius R, will get polarised with electronic (or atomic) polarizability,

$$\alpha_{el} = 4\pi\epsilon_0 R^3 \tag{4.39}$$

On the other hand, when an *alternating electric field* of angular frequency ω is applied across the dielectric then electronic (or atomic) polarizability is given by,

$$\alpha_{el}(\omega) = \frac{Ze^2/m}{(\omega_0^2 - \omega^2)} \tag{4.40}$$

where ω_0 is the natural frequency of oscillation of electrons, m is the electron mass, e is the electronic charge and Z is the atomic number.

The frequency ω_0 lies in the UV region. As $\omega \to \omega_0$, atoms of the dielectric starts absorbing energy from the source and at $\omega = \omega_0$, the maximum absorption of energy takes place at natural frequency. This phenomenon is commonly called *resonance absorption*.

4.15.2 Ionic (or displacement) polarizability (α_{ion})

Ionic polarizability α_{ion} is due to the displacement of adjacent ions of opposite sign. The *cations* (positive ions) are displaced parallel to the local field, and the *anions* (negative ions) are displaced in the opposite direction. These effects are most important in alkali halides and other ionic solids.

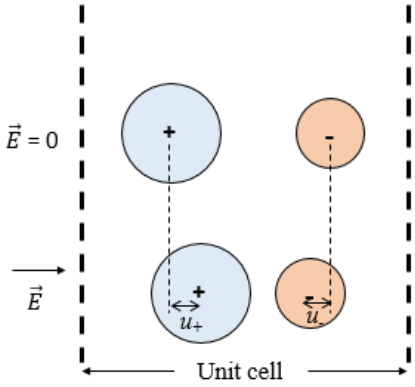


Figure 4.37: Illustrating polarization of ions in the electric field $\vec{E}$.

The torque produced by an electric filed lines up the dipole parallel to the field at absolute zero temperature. The field produces forces on the two charges of opposite sign so that the distance between them is changed by some amount. The balance between the electrostatic force and the interatomic force due to stretching or compression gives the value of the change in the distance between two ions of opposite charge. From this change in the distance, the ionic dipole moment and hence ionic polarizability α_{ion} can be calculated.

Let us treat the ions as rigid ions and by the application of electric field, the center of mass of each unit cell is not altered. Let $\overrightarrow{u_+}$ and $\overrightarrow{u_-}$ be the displacement of the positive ions of mass M_+ and negative ions of mass M_- respectively with respect to their equilibrium positions.

Let, $\frac{1}{M_+} + \frac{1}{M_-} = \frac{1}{M}$ where M is reduced mass of ion pair.

If an alternating electric field $E = E_0 e^{-i\omega t}$ of angular frequency ω is applied along x-direction. Then ionic polarizability,

$$\alpha_{ion} = \frac{e^2/M}{(\bar{\omega}^2 - \omega^2)} \tag{4.41}$$

where $\bar{\omega}$ is the natural frequency of the lattice vibration.

$$\text{If} \quad \omega << \bar{\omega} \quad \text{then} \quad \alpha_{ion} \cong \frac{e^2}{M\bar{\omega}^2} \tag{4.42}$$

$\bar{\omega}_{max} = \omega_D$ is called *Debye frequency*, which is in the IR region. Most of the ionic solids exhibit strong interaction in the IR region.

4.15.3 Dipolar (or orientational) polarizability (α_{dip})

The *orientation polarization* is characteristic of polar dielectrics, which consist of molecules having permanent dipole moment. Dipolar (or orientational) polarizability (α_{dip}) occurs in substances which have asymmetric molecules whose permanent dipole moments can be aligned (or oriented in a specific direction) by the electric field. In the absence of external electric field, the orientation of dipoles is random resulting in a complete cancellation of each other's effect, as illustrated in Fig. 4.38 (a). When the electric field is applied the molecular dipoles rotate about their axis of symmetry to align with the applied field. In case of electronic and ionic polarizations, the force due to the external field is balanced by a restoring force due to coulomb attraction, but for orientation polarization, restoring forces do not exist. However, the dipole alignment is counteracted by thermal agitation. The higher the temperature, the greater is the thermal agitation. The dipoles can turn only through a small angle, as illustrated in Fig. 4.38 (b).

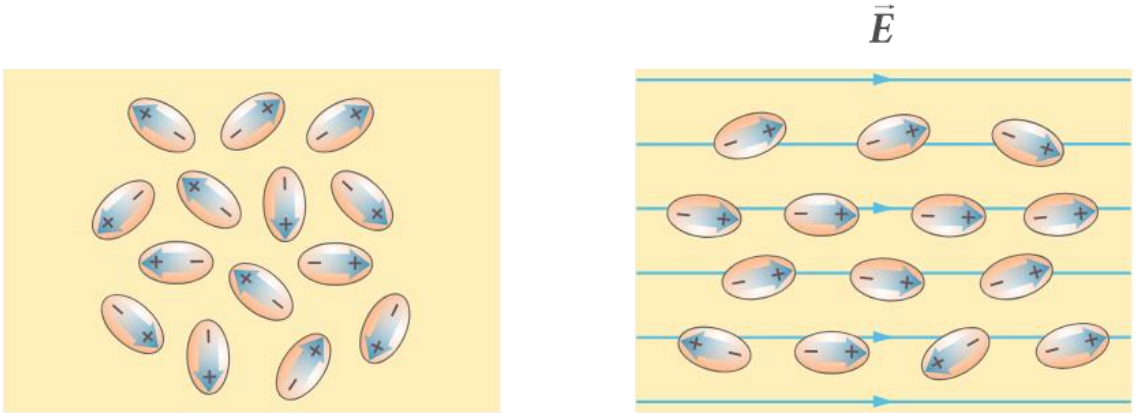


Figure 4.38: Orientation polarization a) molecular dipoles are randomly oriented when $E = 0$ and b) when the electric field E is applied the dipoles are partially aligned.

Even in case of liquids or gases, where molecules are free to rotate, a complete alignment cannot be achieved due to the randomizing effect of the temperature T. Thus, orientation polarization is strongly temperature dependent. This type of polarization occurs in gases, liquids and amorphous viscous substances. In case of solids, the molecules are fixed in their positions and their rotation is highly restricted by the lattice forces, leading to a great reduction in their contribution to orientation polarization. Because of this reason, while the dielectric constant of water is about 80, that of solid ice is about only 10. As the process of orientation polarization involves rotation of molecules, it takes relatively longer time than the electronic and ionic polarizations.

For a system of N permanent dipoles with dipole moment $\vec{p}$, subjected to an external electric field $\vec{E}$, the dipolar (or orientational) polarizability is given by,

$$\alpha_{dip} = \frac{p^2}{3k_B T} \tag{4.43}$$

where k_B is Boltzmann constant and T is the temperature.

4.15.4 Total Polarizability (α_{tot})

The total polarization of a polar dielectric is given by

$$P = N(\alpha_{el} + \alpha_{ion} + \alpha_{dip})E$$

Hence, the total polarizability in a dielectric material is given by,

$$\alpha_{tot} = \alpha_{el} + \alpha_{ion} + \alpha_{dip}$$

i.e., $$\alpha_{tot} = 4\pi\epsilon_0 R^3 + \frac{e^2/M}{(\bar{\omega}^2 - \omega^2)} + \frac{p^2}{3k_B T} \quad (4.44)$$

It is possible for one or more of the contributions to the polarization to be either absent or negligible in magnitude relative to the others. For instance, orientation polarization does not exist in non-polar dielectrics. Similarly, ionic polarization will not be found in covalently bonded materials. Electronic polarization will be negligible compared to orientation polarization in polar dielectrics.

Usually, $\alpha_{ion} << \alpha_{el}$, therefore $\alpha_{tot} = \alpha_{el} + \alpha_{dip}$

$$\alpha_{tot} = \alpha_{el} + \frac{p^2}{3k_B T} \quad (4.45)$$

This is known as *Langevin-Debye equation* and is of great importance to scientists in interpreting molecular structure.

4.16 FREQUENCY DEPENDENCE OF POLARIZATION

In many practical situations, dielectric materials are exposed to alternating electric field (ac field, which changes direction over time). As the field direction reverses, the electric dipoles within the dielectric must reorient themselves to align with the field. The ability of these dipoles to follow the changing field determines the total polarization *P* of the dielectric. With the increase in frequency, polarization of the dielectric material decreases not continuously but in big steps and these steps lie in the microwave, infrared, and ultraviolet regions. The dependence of Polarizability with frequency is shown in Fig. 4.39.

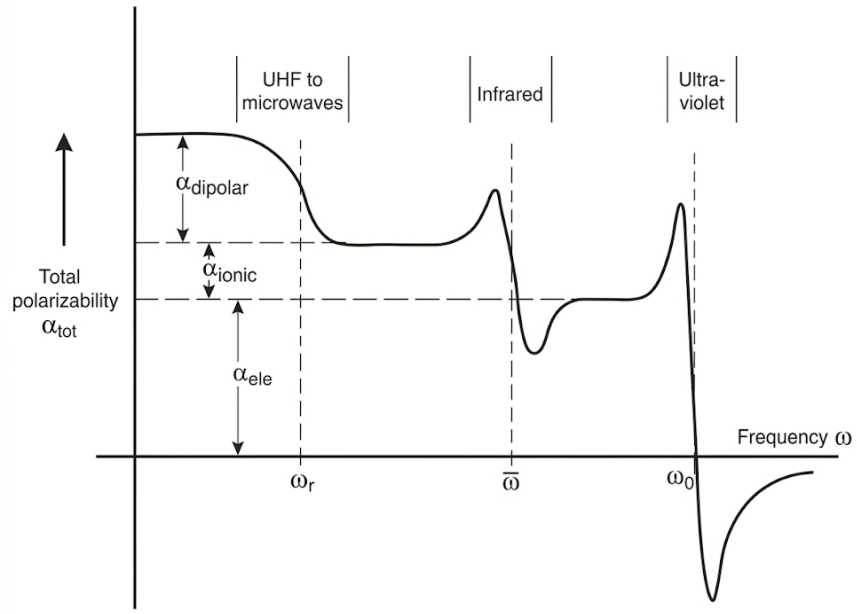


Figure 4.39: Frequency dependence of polarizability.

As shown in Figure 4.39, at low frequencies, all types of polarization are possible, allowing dipoles enough time to fully align with the field. The average time it takes for dipoles to adjust to the field direction is known as the *relaxation time* (τ) and its reciprocal is the *relaxation frequency* (ν). If the electric field frequency is higher than the relaxation frequency, dipoles can't keep up with the changes i.e., the dipoles cannot reverse fast enough. Thus, at very low frequencies of the applied electric field, all types of polarization are active. These include electronic polarization (distortion of electron clouds), ionic polarization (displacement of positive and negative ions), and dipolar polarization (rotation of permanent dipoles). Since the field changes slowly, all these mechanisms can follow it easily. Hence, the total dielectric polarizability is maximum and remains almost constant in this region.

As the frequency increases to the microwave region (around ω_r), dipolar polarization can no longer keep up with the rapidly alternating field because molecular dipoles are relatively slow. Therefore, dipolar polarization disappears, causing a sudden decrease in total polarizability. After this point, only ionic and electronic polarizations contribute to the dielectric response.

With further increase in frequency into the infrared region ($\bar{\omega}$), ionic polarization becomes dominant. Near the ionic resonance frequency, the ions vibrate strongly, producing a peak and dip in the curve. Beyond this region, ions also fail to follow the field. As a result, ionic polarization vanishes, leaving only electronic polarization.

At still higher frequencies in the ultraviolet region (ω_0), electronic polarization shows resonance behavior because electrons respond most rapidly. A sharp peak followed by a deep drop appears in the graph. Beyond this electronic resonance, even electrons cannot follow the oscillating field. In this region, the polarizability is characterized by sharp peaks corresponding to electronic transitions within the material. These peaks occur because the energy of the applied electric field matches the energy required to excite electrons from lower energy states to higher energy states, leading to significant changes in the electronic configuration and, consequently, the polarizability of the material. In this optical region, the material's relative permittivity is equal to the square of its refractive index i.e., $\varepsilon_r = n^2$.

Beyond the ultraviolet range, the electron cloud too fails to follow the field alternations and electronic contribution to the polarization ceases. Consequently, the total polarization becomes zero and the relative permittivity approaches unity i.e., $\varepsilon_r = 1$. Thus, polarizability and in turn dielectric constant of the material changes enormously with the frequency of the applied electric field. For example, water has a high static dielectric constant of about 80 at low frequencies, but it decreases to 1.9 in the optical frequency region. Thus, as frequency increases, polarization decreases step by step: first dipolar, then ionic, and finally electronic polarization.

4.17 TEMPERATURE DEPENDENCE OF POLARIZATION

The *electronic* and *ionic* polarization *do not* depend on temperature and remain constant at all temperatures. However, the *orientation* polarization is inversely proportional to the temperature and decreases as the temperature increases.

We have, $$P = N(\alpha_{el} + \alpha_{ion} + \alpha_{dip})E$$

i.e., $$P = N(\alpha_e + \alpha_i)E + \frac{Np^2E}{3k_BT} \tag{4.46}$$

For a particular electric field E, if polarization P is plotted as function of $1/T$, a straight line will be obtained, as shown in Fig. 4.40. The intercept of the line with y-axis at $1/T = 0$ gives the value of $N(\alpha_e+\alpha_i)$ from which $(\alpha_e+\alpha_i)$ can be evaluated. The dipole moment p can be computed from the slope of the straight line, knowing the value of N, the number of molecular dipoles per cubic meter.

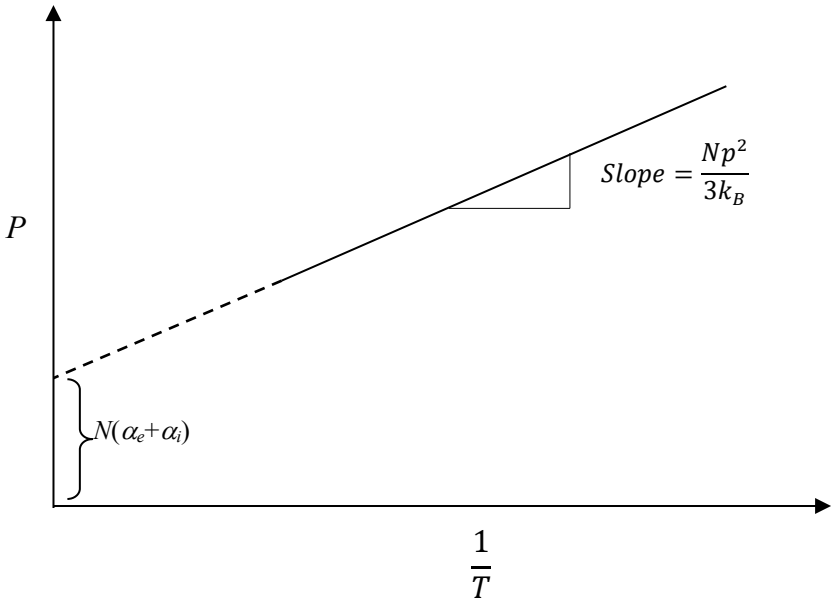


Figure 4.40: The variation of total polarization as a function of 1/T.

4.18 DIELECTRIC LOSS AND LOSS TANGENT

When an electric field, either AC or DC, is applied to a conductor, it loses some electrical energy as heat, which is called power loss or I^2R loss. In conductors, this loss is significant. In dielectrics (insulating materials), DC voltage causes very little power loss due to their high resistance. However, under AC voltage, the dipoles in the dielectrics tend to follow the field, flipping back and forth as the field reverses its direction during each cycle. Dipoles experiences some friction due to their collision with other molecules in the system. This means that some energy is absorbed from the field, and appears in the form of heat, which raises the temperature of the substance. Thus, when a dielectric material is subjected to an alternating voltage, the electrical energy is absorbed by the material and is dissipated in the form of heat. This dissipation of energy is called *dielectric loss*.

In the case of an ideal dielectric (perfect insulator), polarization is complete during each cycle and there is no consumption of energy and the charging current leads the applied voltage by 90º. But for commercial dielectric (lossy dielectric), the phase angle is less than 90º by an angle δ and is called *dielectric loss angle*. *Tan* is taken as measure of dielectric loss and is known as *loss tangent (tan δ)*. It represents the electrical power lost which is often in the form of heat. Hence *tan* δ is also called *dissipation factor*.

The lossy dielectric can be described with the aid of a *complex relative permittivity* ε_r^* given by,

$$\varepsilon_r^* = \varepsilon_r' - j\varepsilon_r^{''} \tag{4.47}$$

where the real part ε_r' is called *dielectric constant*, which represents the *stored energy* and the imaginary part $\varepsilon_r^{''}$ is called *dielectric loss*, which represents the *lost energy*.

The *dissipation factor* or *loss tangent (tan δ)* can also be written as,

$$\tan\delta = \frac{\varepsilon_r^{''}}{\varepsilon_r'} \tag{4.48}$$

The product $\varepsilon_r' \tan\delta$ is known as the *loss factor*.

4.19 DIELECTRIC BREAKDOWN AND DIELECTRIC STRENGTH

When an electric field is applied to a dielectric and the electric field strength is continuously increased then a stage will be reached where the field separates the electron cloud into single electrons. Hence free electrons are generated so that the dielectric become a conductor. That means when higher electric fields are applied, the insulator starts conducting and behaves as a conductor. In such conditions, dielectric materials lose their dielectric properties. This phenomenon is known as *Dielectric Breakdown*. It is an irreversible process, and it leads to the failure of dielectric materials.

The dielectric strength of a material is a measure of the ability of that material to withstand high electric fields. Maximum electric field that a dielectric can withstand before dielectric breakdown occurs is called *dielectric strength*. Hence, dielectric strength is the limiting electric field intensity, above which a breakdown occurs and the charge storage property of the dielectric disappears.

4.20 APPLICATIONS OF DIELECTRIC MATERIALS

Dielectric materials can be solids, liquids, gases, and vacuum. Solid dielectrics are highly used in electrical engineering. Some examples of solid dielectrics are porcelain, ceramics, glass, paper, etc. Dry air, nitrogen, sulfur hexafluoride and the oxides of various metals are examples of gaseous dielectrics. Distilled water, transformer oil are common examples of liquid dielectrics.

Dielectric materials are crucial for two main applications: as *insulators* and *in capacitors*. For insulation, dielectrics should have a low dielectric constant, minimal dielectric loss, high resistance, and strong dielectric strength. They also need to be chemically stable, resist moisture, and possess suitable mechanical properties.

Capacitors store energy using two metal plates separated by a dielectric. Based on the types of dielectric used, capacitors are classified as:

1. Paper Capacitors: Use thin paper soaked in oil or wax.
2. Plastic Capacitors: Utilize films of polyester, PTFE, etc.
3. Ceramic Capacitors: Employ ceramic materials like steatite or barium titanate.
4. Mica Capacitors: Use laminated mica sheets.
5. Glass Capacitors: Thin glass plates with aluminium foil.
6. Electrolytic Capacitors: Have a metal anode with an oxide film dielectric and an electrolyte as the cathode.

Liquid dielectrics (like mineral oil and synthetic oil) are used in transformers for insulation and cooling. These materials help to transfer power between circuits and keep transformers from overheating. They insulate high and low voltage windings and the transformer core, preventing electrical short circuits that could cause failures or fires. By maintaining temperature and providing high dielectric strength, these materials ensure reliable operation and prolong the transformer's life by preventing thermal and electrical damage.

4.21 QUESTIONS

1. Explain the expression for the total potential energy of a crystal. Sketch schematically the plot of the same.
2. Define (a) ionic cohesive energy (b) atomic cohesive energy, of a solid.
3. Write the expression for Fermi-Dirac distribution function. Sketch schematically the plots of this function for zero kelvin and for temperature above zero kelvin.
4. Derive an expression for density-of-states.
5. Assuming the Fermi-Dirac distribution function , obtain an expression for the density of free-electrons in a metal with Fermi energy E_F, at zero K and, hence obtain expression for Fermi energy E_F in a metal at zero K. [Given: density-of-states function $g(E)\,dE = \frac{8\sqrt{2}\,\pi\, m^{3/2}}{h^3}\, E^{1/2}\, dE$]
6. Explain the formation of energy bands in solids with necessary diagrams.
7. Distinguish between conductors, insulators and semiconductors on the basis of band theory.

8. Indicate the position of (a) donor levels (b) acceptor levels, in the energy band diagram of a semiconductor.
9. What is the difference between p-type and n-type semiconductors? Explain with band diagram.
10. With necessary diagrams, explain doping in semiconductors.
11. Why the electrical conductivity of an intrinsic semiconductor increases with increasing temperature?
12. Derive an expression for electrical conductivity of a semiconductor.
13. Explain Hall effect and derive an expression for Hall coefficient.
14. Explain measurement of resistivity of a semiconductor by four-probe method.
15. What are the advantages of four-probe method?
16. Define (i) Fill factor, (ii) Open circuit voltage, (iii) Short circuit current, (iv) Efficiency of a solar cell.
17. Give representative I-V plot of a solar cell under illumination and indicate V_{oc}, I_{sc}, maximum power point.
18. Write a note on junction diode.
19. Write a note on light emitting diodes (LEDs).
20. Write a note on MOSFET.
21. What are superconductors? Draw a representative graph of Resistance Vs Temperature for a superconductor.
22. Explain Meissner effect.
23. Explain briefly the BCS theory of superconductivity in metals.
24. Classify and explain types of superconductors.
25. What is meant by electric *dipole moment*?
26. What are *polar* and *non-polar* dielectrics?
27. Explain polarization in a dielectric material.
28. What do you understand by *dielectric constant*? Define *dielectric susceptibility*.
29. Define *dielectric polarization* and *polarizability* of a dielectric.
30. Explain *displacement vector* in dielectrics.
31. With relevant mathematical expressions explain *electronic*, *ionic* and *orientational polarizabilities*.
32. Discuss the frequency dependence of various polarization processes in dielectric materials.
33. Discuss the effect of temperature on electronic, ionic and orientational polarization.
34. Write a note on *dielectric loss*. Explain *loss tangent* and *loss factor*.

35. Explain *dielectric breakdown* and *dielectric strength*.

36. What is the role of dielectrics in *capacitors* and *transformers*?

4.22 PROBLEMS

1. Consider a one-dimensional chain of alternating positive and negative ions. Show that the potential energy associated with one of the ions and its interactions with the rest of this hypothetical crystal is

$$U(r) = -k_e \alpha \frac{e^2}{r}$$

where the Madelung constant is $\alpha = 2 \ln 2$ and r is the inter-ionic spacing.
Hint: Use the series expansion

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots$$

2. Each atom of gold (Au) contributes one free-electron to the metal. The concentration of free-electron in gold is $5.90 \times 10^{28}/m^3$. Compute the Fermi Energy of gold.
Ans: 5.53 eV

3. Sodium is a monovalent metal having a density of 971 kg/m^3 and a molar mass of 0.023 kg/mol. Use this information to calculate (a) the density of charge carriers and (b) the Fermi energy.
Ans: $2.54 \times 10^{28}/m^3$, 3.15 eV

4. Calculate the energy of a conduction electron in silver at 800 K, assuming the probability of finding an electron in that state is 0.950. The Fermi energy is 5.48 eV at this temperature.
Ans: 5.28 eV

5. Show that the average kinetic energy of a conduction electron in a metal at zero K is (3/5) E_F
Suggestion: In general, the average kinetic energy is

$$E_{AV} = \frac{1}{n_e} \int E\, N(E)\, dE$$

where the density of particles

$$n_e = \int_0^\infty N(E)\, dE$$

$$N(E)\, dE = \frac{8\sqrt{2}\, \pi\, m^{\frac{3}{2}}}{h^3} \frac{E^{\frac{1}{2}}\, dE}{exp\left(\frac{E - E_F}{k\, T}\right) + 1}$$

6. (a) Consider a system of electrons confined to a three-dimensional box. Calculate the ratio of the number of allowed energy levels at 8.50 eV to the number at 7.00 eV. (b) Copper has a Fermi energy of 7.0 eV at 300 K. Calculate the ratio of the number of occupied levels at an energy of 8.50 eV to the number at Fermi energy. Compare your answer with that obtained in part (a).
Ans: (a) 1.10 (b) 1.46×10^{-25}

7. Most solar radiation has a wavelength of 1 μm or less. What energy gap should the material in solar cell have in order to absorb this radiation ? Is silicon (E_g= 1.14 eV) appropriate?
   Ans: 1.24 eV or less; yes
8. The energy gap for silicon at 300 K is 1.14 eV. (a) Find the lowest-frequency photon that can promote an electron from the valence band to the conduction band. (b) What is the wavelength of this photon?
   Ans: 2.7×10^{14} Hz, 1090 nm
9. **A light-emitting diode (LED) made of the semiconductor GaAsP emits red light (λ= 650nm). Determine the energy-band gap in the semiconductor.**
   Ans: 1.91 eV

10. **Consider a semiconductor in the form of a rectangular strip of thickness 10 μm. Let a current of 1 mA flow along its length and a transverse magnetic field of strength 1T be applied across its thickness. If Hall voltage is 2mV, then calculate charge carrier density.**
    Ans: $3.1 \times 10^{23} m^{-3}$
11. **For what value of the bias voltage in the diode-current equation $I = I_o \left[exp\left(\frac{e\,\Delta V}{k\,T}\right) - 1\right]$ does (a) $I = 9.00\ I_o$? (b) $I = -0.900\ I_o$? Assume T= 300 K. Ans. (a) 59.5 mv (b) -59.5mv**
12. The current in a diode under forward bias of 100 mV is 200 mA at a temperature of 300 K. What is the current in the diode if it is under reverse bias of 100 mV ? Ans. -4.19 mA
13. The Hall coefficient of certain Silicon specimen was found to be $-7.35 \times 10^5\ m^3C^{-1}$. Calculate the charge carrier density. Mention type of the charge carrier.
    Ans: $8.455 \times 10^{22}\ m^{-3}$; electrons

14. In a drop of water of radius 10^{-3}m, the molecular dipoles are pointed in the same direction. If the dipole moment of the water molecule is 6×10^{-30}C.m, calculate the polarization. Molecular weight of water (H_2O) is 18 and density of water (H_2O) is 1000 kg/m^3. (Ans: $8.40 \times 10^{-10}\ C/m^2$)

15. **Two parallel plates have equal and opposite charges. They are separated by a dielectric material with dielectric constant 3. If the electric field intensity in the dielectric is 10^6 V/m, calculate the polarization in the dielectric. Permittivity of free space $\epsilon_0 = 8.854 \times 10^{-12}\ m^{-3}\ kg^{-1}\ s^4\ A^2$. (Ans: $17.7 \times 10^{-6}\ C/m^2$)**

16. When NaCl crystal is subjected to an electric field of 50 V/cm, the resulting polarization is 2.215×10^{-7} C/m^2. Calculate relative permittivity of NaCl. (Ans: 6.006)

17. Calculate the induced dipole moment per unit volume (i.e., polarization) of helium gas when it is placed in a field of 6×10^5 V/m. The polarizability of helium is 0.18×10^{-40} Fm^2 and the concentration of helium atom is 2.6×10^{25} /m^3. Also calculate the separation of positive and negative charges in each atom.
(Ans: 2.808×10^{-10} C/m^2, $3.375 \times 10^{-17}\ m$)

18. If all the Carbon atoms of diamond are polarized alike, and produce a polarization 1.32×10^{-5} C/m^2, calculate the shift of the center of negative cloud of 6 electrons on each atom from the nucleus. The density of Carbon is 3500 kg/m^3 and its atomic weight is 12.
(Ans: $7.82 \times 10^{-17}\ m$)

19. **The capacitance of a parallel plate capacitor increases from 0.085μF to 0.195μF when a sheet of polythene is placed between the plates. Determine the dielectric constant of the polythene. (Ans: 2.29)**

20. Assuming that there are 10^{27} molecules/m^3 in HCl vapour, calculate the orientational polarization at room temperature if the vapour is subjected to an electric field of 10^6 V/m. The dipole moment of HCl molecule is 3.46×10^{-30} C.m.
(Ans: $9.64 \times 10^{-7}\ C/m^2$)

21. **The polarizability of NH_3 molecule is found experimentally by the measurement of dielectric constant as 2.42×10^{-39} Fm^2 at 309K and 1.74×10^{-39} Fm^2 at 448K respectively. Calculate for each temperature the polarizability due to permanent dipole moment and due to deformation of molecules.**
(Ans: $\alpha_{el} = 2.28 \times 10^{-40}\ Fm^2$, $\alpha_{dip(309)} = 2.19 \times 10^{-39}\ Fm^2$, $\alpha_{dip(448)} = 1.51 \times 10^{-39}\ Fm^2$)

5 MATERIALS CHARACTERIZATION TECHNIQUES

OBJECTIVES:

- To understand fundamentals of X-ray diffraction and its applications.
- To understand different scanning probe microscopic techniques and applications.
- To learn about high-resolution electron microscopes.

5.1 X-RAY DIFFRACTOMETRY

5.1.1 Basics of X-Ray Diffraction (XRD)

X-rays were discovered in 1895 by the German physicist Rontgen and were so named because their nature was unknown at that time. Unlike ordinary light, these rays were invisible, but they travelled in straight lines and affected photographic film in the same way as light. On the other hand, they were much more penetrating than light and could easily pass through the human body, wood, metal and other "opaque" objects. X-rays are electromagnetic radiation with wavelengths of the order of 0.1 nm which is comparable with interatomic spacing in crystals. Bragg's law predicts the conditions under which diffraction of X-ray beams from a crystal are possible.

In order to carry out X-ray studies based on Bragg's law, a crystal is mounted such that the X-ray beam is incident on the crystal at an angle θ and a detector scans through various angles for the diffracted X-rays (Fig. 5.1). The detector shows peaks for those angles at which **constructive interference** occurs.

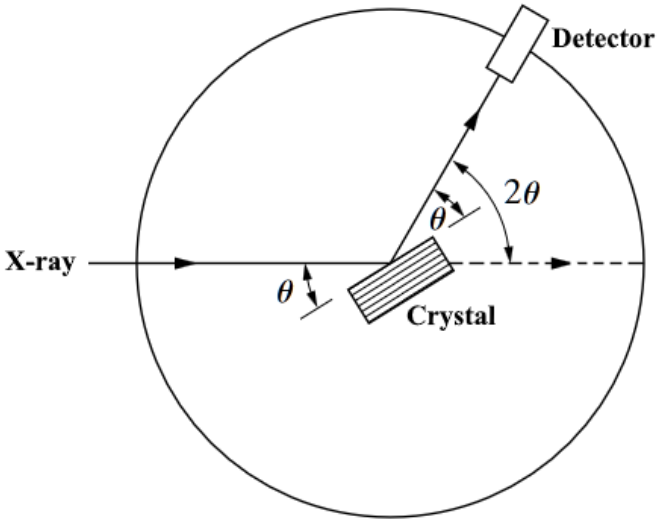


Fig. 5.1: Bragg's diffraction setup showing the incident X-ray, the crystal at angle θ and the detector at angle 2θ.

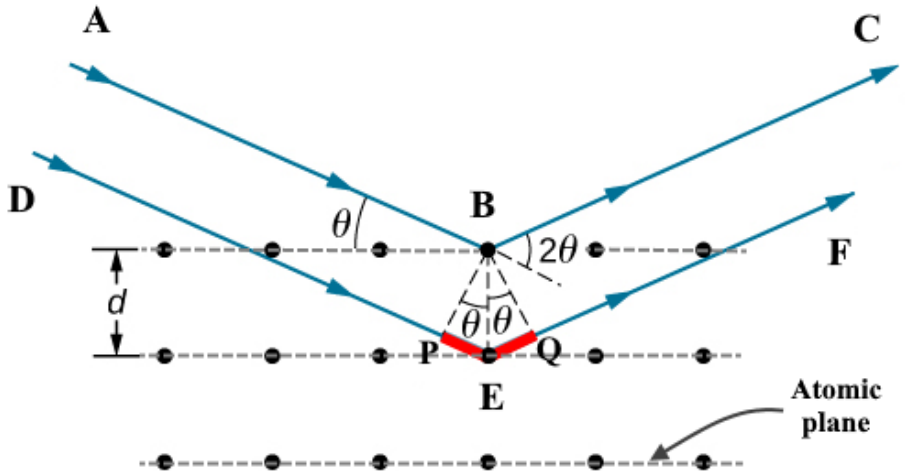


Fig. 5.2: X-ray scattering by a crystal.

Fig. 5.2 shows a crystal in which the dashed lines indicate atomic planes, or Bragg planes, and the dots indicate the position of the atoms. Let a monochromatic parallel beam of X-rays of wavelength λ be incident upon the crystal at an angle θ as shown in Fig. 2. The ray along **AB**, which is part of the incident beam, is scattered by the atom **B** in the first plane. Another ray along **DE** which is part of the same incident beam is scattered by the atom **E** in the next adjacent parallel plane. In the crystal, each of the atoms scatters the incident rays in all directions. Among the rays scattered from **B** and **E**, those which are parallel to each other, constructive interference can be observed only for a path difference of $n\lambda$ (where n is an integer).

From Fig. 5.2, the path difference, δ, between the two rays is,

$$\delta = PE + EQ$$
$$\delta = BE \sin\theta + BE \sin\theta \; = \; 2\, BE \sin\theta$$

But **BE** is the interplanar spacing, i.e., the spacing between the atomic planes, denoted by d. Hence, we can write the path difference as,

$$\delta = 2d \sin\theta$$

As stated earlier, constructive interference takes place in the direction $\boldsymbol{\theta}$ when $\delta = n\lambda$, where $n = 1,2,3, \ldots$ is called the diffraction order.

The condition for constructive interference is

$$2d \sin\theta = n\lambda$$

The above equation is called the *Bragg's law*, and the angle θ is the glancing angle.

Here, the rays **AB** and **DE** are incident at an angle θ to the Bragg plane. Also, we see that the corresponding scattered rays **BC** and **EF** undergo constructive interference only when they are inclined at the same angle θ to the Bragg plane. Thus, in essence, the constructive interference conditions match the law of reflection in geometrical optics.

5.1.2 X-Ray Diffractometer

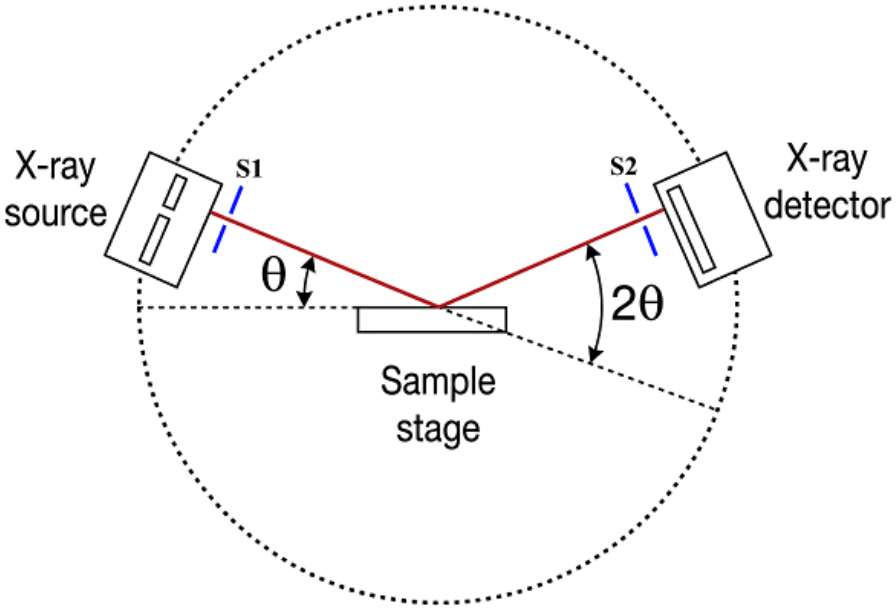


Fig. 5.3: X-ray diffractometer.

Monochromatic X-rays produced from an X-ray source are passed through a slit S_1 to form a fine beam of X-rays (Fig. 5.3). This X-ray beam is directed to the sample placed on the sample stage at an angle θ. The diffracted X-ray is detected using an X-ray detector (such as a semiconductor or scintillation detector). A slit S_2, known as the receiving slit, placed in front of the detector helps in improving the resolution of the system. The detector is kept at an angle of 2θ.

The incident X-rays penetrate the sample, which is composed of many tiny crystallites oriented in all possible directions, in the case of powder samples. When the X-rays hit regular, repeating planes of atoms within the crystal, they are scattered by the electron clouds of the individual atoms. This is a type of coherent elastic scattering (the energy of the scattered X-rays does not change). While scattering occurs in all directions, in most directions the scattered waves are out of phase and undergo destructive interference, cancelling each other out. However, in specific directions, the waves scattered by adjacent atomic planes are perfectly in phase, leading to constructive interference, thus producing an intense diffracted beam.

When the diffracted rays from the sample undergo constructive interference, the intensity of X-rays reaching the X-ray detector increases. The reinforcement occurs only when Bragg's condition is satisfied. One can measure the important crystal parameter, namely the interplanar spacing d, using the relation $2d \sin\theta = n\lambda$, by using the measured value of θ, provided the order (n) and the wavelength (λ) of the incident X-ray are known. The X-ray diffraction pattern is a graph plotted between the angle of the X-ray detector (2θ) and the intensity of X-rays recorded by the X-ray detector. Fig. 5.4 shows a typical XRD pattern recorded from sodium chloride (NaCl) powder sample.

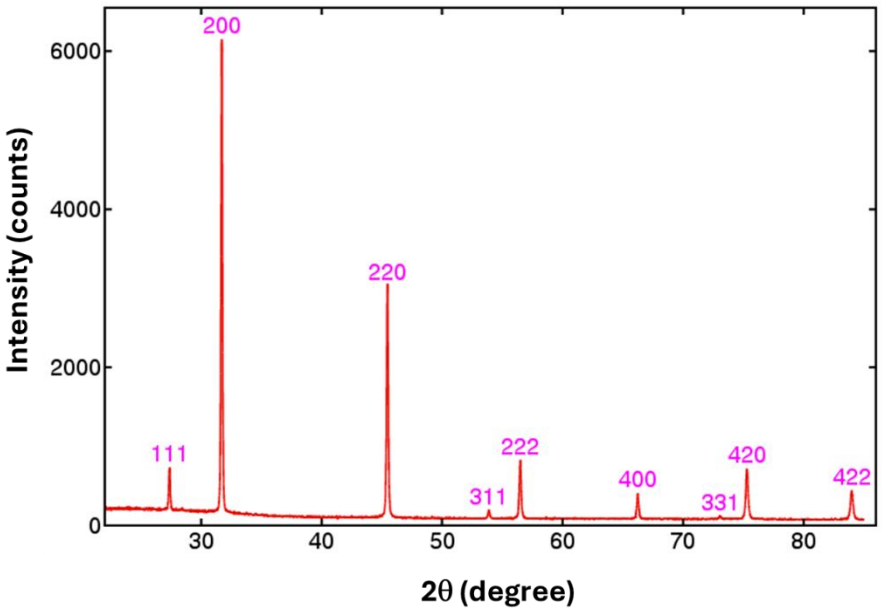


Fig. 5.4: A typical XRD pattern recorded from NaCl powder

5.1.3 Applications of X-Ray Diffraction

Phase identification: X-ray diffraction (XRD) helps us in phase identification of a material since each crystalline material has a unique arrangement of atoms that produces a characteristic diffraction pattern governed by Bragg's law. When X-rays interact with a crystal, they are diffracted at specific angles corresponding to the interplanar spacings of that structure. The resulting pattern—positions (2θ values) and intensities of peaks—acts as a fingerprint of the phase. By comparing an experimental diffraction pattern with standard reference databases, one can identify the phases present in a material.

Determination of crystal structure: Crystal structure refers to the unique ordered arrangement of atoms in a crystalline material, forming symmetric, repeating patterns in 3D space. It is defined by a unit cell—the smallest repeating unit—and classified by symmetry into seven lattice systems (such as cubic, hexagonal, monoclinic etc.). The crystal structure determination by X-ray diffraction relies on the systematic relationship between interplanar spacing and Miller indices. For example, for a cubic crystal, the interplanar spacing, d, is given by,

$$d = \frac{a}{\sqrt{h^2+k^2+l^2}}.$$

where a is the lattice parameter and h, k, l are the miller indices. By measuring the diffraction peak positions (2θ) and using Bragg's law, the corresponding d-spacings can be calculated. Finally, the lattice parameters and the crystal structure can be estimated.

Determination of wavelength: The wavelength of X-rays can be determined using XRD from a crystal of known interplanar spacing by applying Bragg's law. By measuring the diffraction angle θ for a specific set of lattice planes (with known d), the wavelength λ can be directly calculated.

Crystallite size and strain: If the crystallite size is small, peak broadening can be observed in the XRD pattern. The crystallite size can be calculated using the Debye Scherrer formula,

$$D = \frac{K\lambda}{\beta cos\theta}$$

where D is the crystallite size, K is the shape factor (typically ~0.9 for spherical crystals), λ is the X-ray wavelength, β is the full width at half maximum (FWHM, in radians) of the Bragg peak, and θ is the Bragg angle.

When there is a strain in the crystal, the inter planar spacing gets altered. Thus, uniform strain in a crystal leads to the shift in the observed Bragg angle. The uniform lattice strain can be obtained directly from the change in interplanar spacing:

$$\epsilon = \frac{d - d_0}{d_0}$$

where d_0 is the stress-free d spacing and d is the measured spacing.

5.2 SCANNING PROBE MICROSCOPY

Scanning Probe Microscopy (SPM) is a powerful set of techniques for imaging and analyzing surfaces at the nanoscale. SPM was discovered in early 1980s, when it impressed the world by producing the first real-space images of surfaces with an unprecedented resolution of atomic-scale. Presently, SPM is one of the most employed techniques in various disciplines, which include fundamental surface science, routine examinations of the surface roughness, and three-dimensional imaging (from the atoms of silicon to microscale projections on the surface of a living cell). It produces high-definition images of a surface using a physical probe that scans over the sample surface. SPM modes include various techniques for mapping surface properties at the atomic or nanoscale, primarily using Atomic Force Microscopy (AFM) and Scanning Tunnelling Microscopy (STM). AFM measures the forces (like van der Waals) between the sample and the tip while STM measures the tunnelling current between a conductive tip and sample. In AFM, key modes include contact mode (to get the topography), tapping mode (to reduce damage to probe and sample), and non-contact mode (least damage to probe and sample).

The basic components of SPM include the probe and scanner; these are discussed in detail in the following sections.

Probe: A SPM probe is a nanoscale sharp tip mounted on a static probe or a cantilever, designed to physically scan a sample surface to map its properties like topography, roughness, or electrical characteristics as shown in Fig. 5.5 below. As the distance between two objects is decreased to very small values, a variety of interactions occur between them at atomic scale. These interactions form the basis of SPM. STM uses tunnelling current, and AFM uses forces between the probe and the sample to achieve high resolution images of the sample surfaces. The probe is very sensitive towards these interactions and is configured to detect them. In SPM, this probe is brought very close to the sample being imaged. The magnitude of the interaction between probe and sample depends on the distance between them. This gives an SPM the ability to precisely map the surface topography of a specimen.

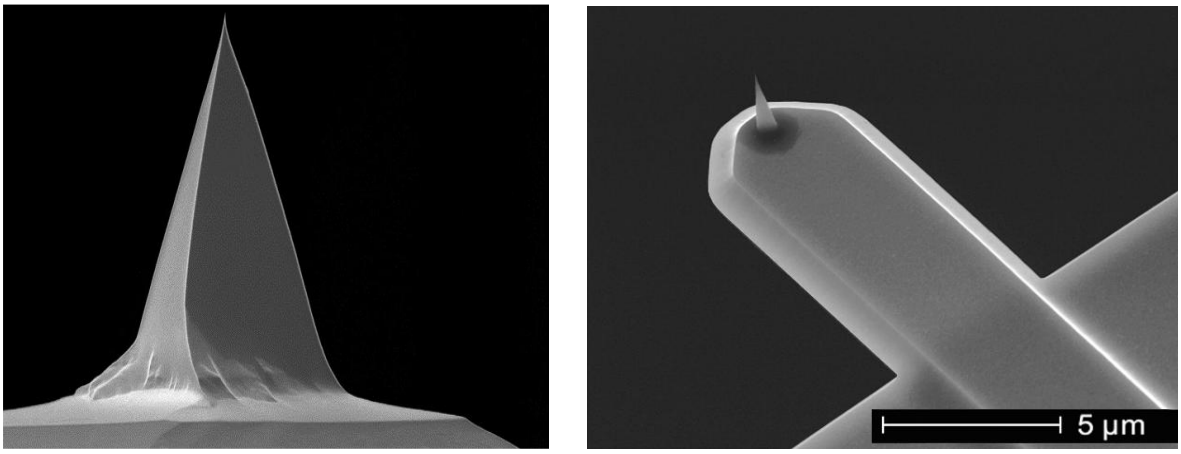


Fig. 5.5: Sharp probes of a scanning probe microscopy system

The type of probe required depends upon the type of SPM being used. The resolution of SPM is determined by the apex of the probe, such that the sharper the probe is, better is the resolution. For many cantilevers-based SPMs (e.g. AFM) the entire cantilever as well as the probe are produced via etching, most commonly from silicon nitride (SiN). The conductive probes for STM are usually constructed from platinum/iridium wire.

Scanner: SPM scanners are made from piezoelectric material, which expands and contracts proportionally to an applied voltage. Whether they elongate or contract depends on the polarity of the voltage applied. Such piezoelectric ceramics are used to provide precise positioning control necessary for the SPM scanner. The application of voltage causes a corresponding change in the geometry of the piezoelectric ceramic. In SPM, the piezoelectric scanner undergoes controlled bending, expansion as well as contraction. Thus, the probe-sample gap and the position of the probe, both can be controlled via this scanner. SPM scans the sample surface in a raster fashion. To create an image, specific values (depending upon the type and

mode of SPM) are recorded while the probe is raster scanned over the sample surface. A map is created from these values, which produces the final image of the sample surface.

5.2.1 Atomic Force Microscopy (AFM)

Fundamentals of AFM Operation: Hooke's Law ($F = -kx$) is fundamental to AFM as it describes the cantilever's deflection (x) as directly proportional to the tip-sample force (F), acting like a tiny spring with a known spring constant (k). This principle allows AFM to map surface topography and measure interaction forces by correlating the bending of the cantilever to the nanometer-scale forces between the tip and sample, enabling high-resolution imaging. Basically, an AFM operation involves the following three steps:

a. Surface Sensing: An AFM scans a sample using a sharp tip mounted on a flexible cantilever. As the tip nears the surface, attractive forces (van der Waals, electrostatic) pull the cantilever downward. Upon contact, repulsive forces dominate, pushing the cantilever upward. This delicate balance enables ultra-precise surface probing at the atomic level.

b. Detection Method: A laser beam reflects off the cantilever onto a position-sensitive photodetector (PSPD). The PSPD is a core component in SPM systems, particularly in AFM, used to detect the minute deflection of the SPM's cantilever. Even nanoscale deflections can alter the laser deflection, allowing the PSPD to track the height variations (topography) and force interactions (such as mechanical, electrical, and magnetic forces).

c. Imaging Mechanism: By scanning the tip across the sample while maintaining constant tip-sample interaction (via feedback control), AFM constructs a 3D topographic map with atomic-level resolution (sub-nanometer precision) and real-time quantitative data (e.g., adhesion, stiffness, conductivity).

Modes of Operation of an AFM

a. Contact Mode: Contact mode is the most basic mode of AFM for measuring topography. In this mode, the cantilever scans while applying a constant force onto the surface of the sample. The cantilever tip is brought close to the sample and eventually contacts the surface. Once engaged, the cantilever bends upwards proportional to the amount of applied imaging force. As the tip passes over higher or lower sample features, the cantilever will bend and deflect the laser beam. The deflected laser spot on the PSPD will move due to the change in contact force. The feedback loop responds by moving the Z scanner to restore the initial cantilever deflection to keep the applied imaging force constant. By tracking the displacement of the Z scanner, one can ultimately determine the surface topography of the sample.

Contact mode typically uses cantilever bending for feedback with a relatively low spring constant to avoid sample damage. As the scanner traces the tip across the sample during contact mode imaging, the repulsive contact forces between tip and sample cause the cantilever to bend to accommodate for changes in topography, as shown in Fig. 5.6(a). For more details, please

refer to the Lennard-Jones Potential curve in Fig. 5.6(b), which represents the distance dependence of tip-sample interactions. When the outermost atoms of tip and sample are gradually brought closer, they start to weakly attract each other. This attractive force increases until the inter-atomic distance is small enough to trigger the Pauli repulsion between electron clouds. The strong repulsion quickly offsets the attractive force as the inter-atomic distance continues to decrease. The inter-atomic forces are balanced when the distance is reduced to a few angstroms, about the length of a chemical bond.

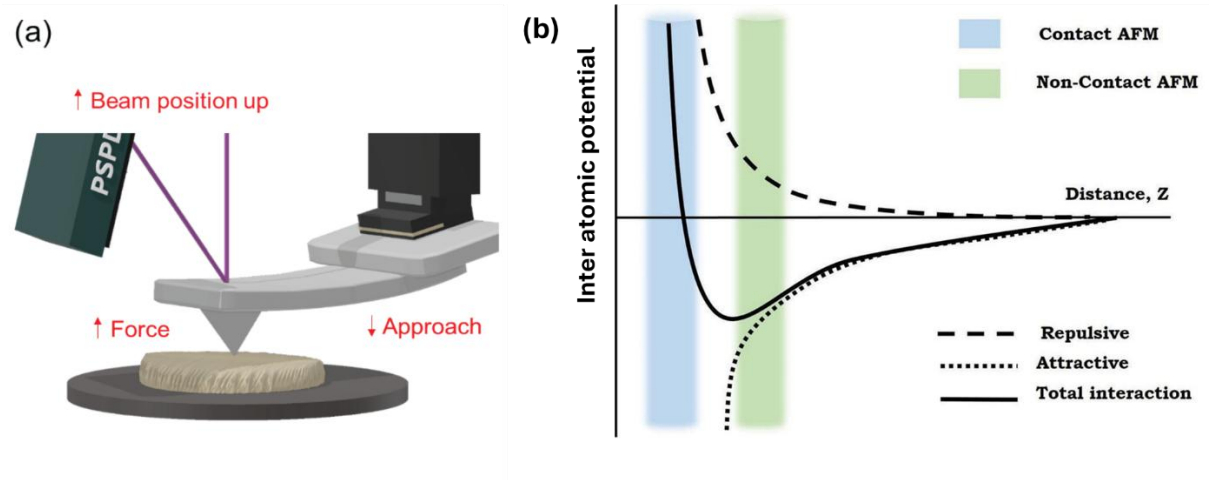


Fig. 5.6: (a) Cantilever bending due to the repulsive contact forces, and (**b**) inter-atomic interaction potential vs. distance.

In Fig. 5. 6(b), the dashed curve represents a purely repulsive interaction, while the dotted curve represents a purely attractive interaction. The solid curve is a combination of long-range attractive and short-range repulsive interaction forces called the Lennard-Jones Potential. At larger distances, the net force is attractive ($-F$), which switches to a net repulsive force ($+F$) when two atoms are brought closer together.

The deflection of the cantilever is made via an optical beam deflection technique as shown in Fig. 5.7. Almost all AFM techniques employ a feedback loop to facilitate keeping a constant interaction between the tip and the sample as the tip scans the surface. The system detects the cantilever motion during the scan by monitoring the vertical deflection. The deflection of the cantilever is kept constant at a target value (called setpoint). The difference between the setpoint and the actual deflection is measured and referred to as the error. This error is scaled by the gain, and this signal is used to control the Z drive of the scanner. Based on the error signal, the feedback changes the Z scanner position to accommodate for the topography change. The AFM topography is then generated based on the motion of the Z scanner.

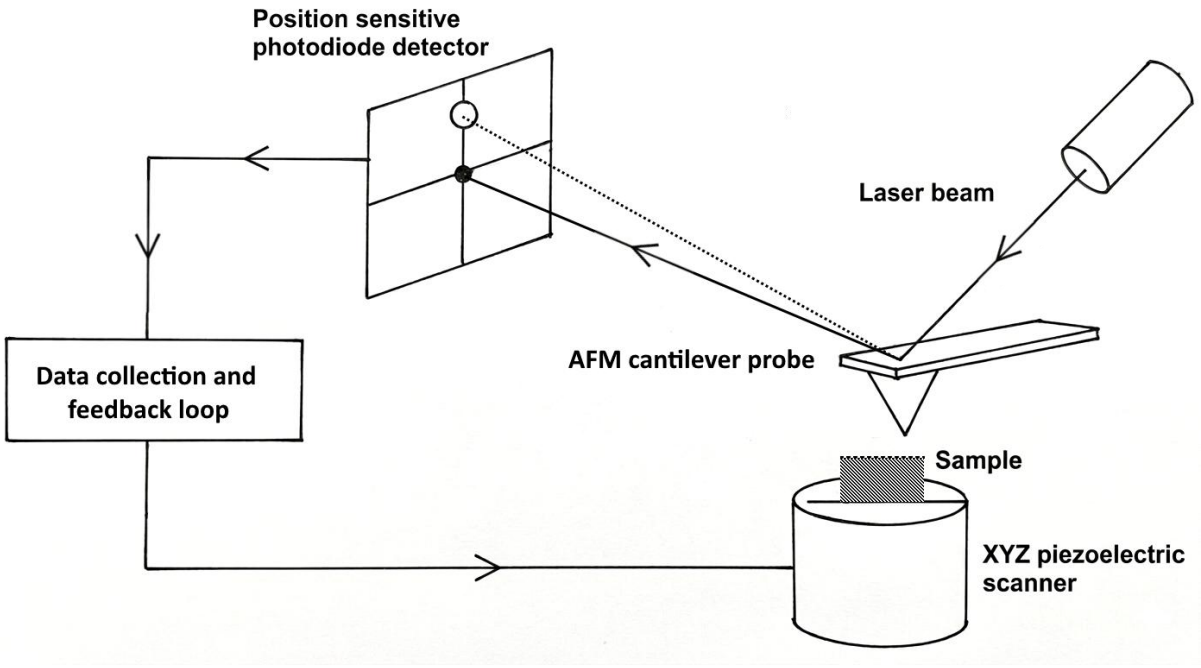


Figure 5.7: Schematic showing the working principle of an AFM system.

b. Tapping Mode: Tapping mode is the most commonly used mode in AFM, where the tip vibrates up and down at high frequency, gently "tapping" the surface thousands of times per second. Tapping mode AFM operates by scanning a tip attached to the end of an oscillating cantilever across the sample surface. The cantilever is oscillated at or near its resonance frequency with an amplitude ranging from 20 nm to 100 nm. The tip lightly "taps" on the sample surface during scanning, contacting the surface at the bottom of each oscillation. Changes in the vibration frequency of the cantilever vibrating near its resonance frequency are measured. The feedback loop maintains a constant oscillation amplitude, and therefore, a constant tip-sample interaction is maintained during imaging. Tapping mode can be operated in ambient and liquid environments. Tapping mode has a number of advantages compared with contact mode, including greatly reduced lateral and normal forces being applied to the sample surface. Therefore, tapping mode is preferred to image samples with structures that are weakly bound to the surface or soft samples.

c. Non-Contact Mode: In contact mode, while scanning, the cantilever applies large shear forces that can damage the samples and/or the tip apex. Particularly, delicate samples such as adsorbed molecules, biological samples, and soft polymers, benefit from non-contact mode for topography imaging. Since non-contact mode operates in the attractive force regime, shear forces are avoided, which significantly increases the tip lifetime. Therefore, non-contact mode is generally recommended for topography imaging.

In this technique, the cantilever oscillates just above the surface as it scans as shown in Fig. 5.8 below. A precise, high-speed feedback loop prevents the cantilever tip from crashing into the surface, keeping the tip sharp and leaving the surface untouched. As the tip approaches the sample surface, the oscillation amplitude of the cantilever decreases. By using the feedback

loop to correct for these amplitude deviations, one can generate an image of the surface topography.

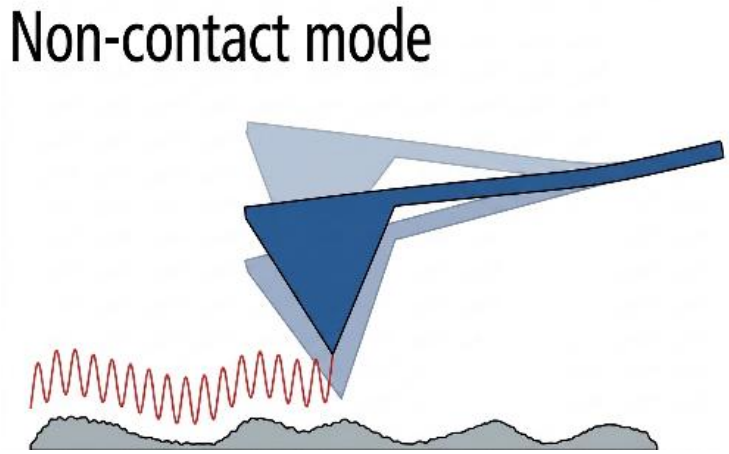


Fig. 5.8: Schematic of the probe oscillation of non-contact AFM.

Contact mode is ideal for high-resolution imaging of hard surfaces. Tapping mode is best for soft/biological samples, reducing lateral damage, while non-contact eliminates surface contamination issues but suffers from lower resolution. The different modes of operation of AFM with different features such as tip-sample separation, cantilever oscillation, feedback, lateral force, sample damage, tip wear, applications and ease of usage in different modes is summarised in below table 1.

Table 1: Comparison of AFM modes

Feature	**Contact Mode**	**Tapping Mode (Intermittent)**	**Non-Contact Mode**
Tip-Sample Separation	< 0.5 nm (Repulsive)	0.5 – 2 nm (Intermittent)	1 – 10 nm (Attractive)
Cantilever Oscillation	Static (no oscillation)	Oscillates at/near resonance	Oscillates at/near resonance
Feedback Parameter	Cantilever deflection	Oscillation Amplitude	Frequency or Amplitude
Lateral Force (Friction)	High	Negligible	None
Sample Damage	High (can damage soft samples)	Low (suitable for soft samples)	Very Low (non-invasive)
Tip Wear	High	Moderate/Low	Negligible
Best Application	Hard, rigid surfaces	Soft/biological samples, polymers	Non-invasive imaging, sticky samples
Ease of Use	High	Moderate (requires tuning)	Low (hard to stabilize)

5.2.2 Scanning Tunnelling Microscopy (STM)

STM is the oldest of the scanning probe microscopes and involves bringing an atomically sharp tip within a couple of nanometres of the sample. When a bias is applied between the tip and sample, electrons will tunnel from one to the other and this current can be used to measure the topography of the sample. The unique aspect of STM is that it can readily be used to acquire images with atomic resolution, which is not generally true for the other SPMs. However, because a current is being used as feedback, only conducting samples can be examined and hence this limits the range of possible materials.

STM relies on "tunnelling current" between the probe and the sample to sense the topography of the sample. A sharp metal tip (in the best case, atomically sharp), is positioned a few atomic diameters above a conducting sample, which is electrically biased with respect to the tip. At a distance under 1 nm, a tunnelling current will flow from sample to tip. In operation, the bias voltages typically range from 10 to 1000 mV while the tunnelling currents vary from 0.2 to 10 nA. The tunnelling current changes exponentially with the tip-sample separation, typically decreasing by a factor of two as the separation is increased 0.2 nm, as shown below:

$$I \propto e^{-2CL}$$

where

$$C \approx \frac{\sqrt{2m\Phi}}{\hbar}$$

Plugging in typical values for m (mass of an electron), L (separation distance between the tip and sample), and Φ (where Φ is the average work function of the tip and the sample), one can calculate the tunnelling current at a finite tip-sample separation distance.

A schematic view of the STM system is shown in Fig. 9. STM relies on a precise scanning technique to produce very high-resolution, three-dimensional images of sample surfaces. The STM scans the sample surface beneath the tip in a raster pattern while sensing and outputting the tunnelling current to the computer control station. The digital signal processor (DSP) in the workstation controls the Z position of the piezo based on the tunnelling current error signal. The STM can operate in both constant height and constant current modes. The DSP always adjusts the height of the tip based on the tunnelling current signal, but if the feedback gains are low the piezo remains at a nearly constant height while tunnelling current data is collected. This is constant height mode. With the gains high, the piezo height changes to keep the tunnelling current nearly constant, and changes in piezo height are used to construct the image. This is constant current mode. The exponential relationship between tip-sample separation and tunnelling current allows the tip height to be controlled very precisely and to follow the surface contours that reflects the surface topography. The main differences between an AFM and an STM are listed in table 2.

Table 2: Differences between AFM and STM

Feature	STM	AFM
Interaction measured	Tunneling current	Interatomic forces (van der Waals, contact)
Sample requirements	Conductive or semiconductive	Any (insulators, semiconductors, conductors)
Resolution	~0.1 nm lateral, ~0.01 nm vertical	~1 nm lateral, ~0.1 nm vertical
Probe	Metal tip, bias voltage applied	Cantilever with sharp tip, deflection sensing
Feedback parameter	Tunneling current	Cantilever deflection (optical or piezo sensor)
Typical applications	Surface atomic structure of metals/semiconductors	Topography of polymers, biomolecules, thin films

Applications of Scanning Probe Microscopy

- Characterizing surface topography, defects, and crystal growth in higher resolution.
- Studying nanomechanical properties (adhesion, friction, wear) in polymers, lubricants and composites materials in nanometer scale.
- Investigating magnetic domain structures (MFM).
- Analyzing corrosion processes and passive oxide layers.
- Inspecting the electronic properties of semiconductor surfaces (Si, GaAs) and integrated circuits (ICs) in nanometer scale.
- Visualizing single biomolecules, viruses, and bacteria. Studying cell membrane stiffness to differentiate healthy vs. sick cells.
- Characterizing drug delivery systems, polymorphs, and drug-polymer interactions.
- Nanolithography: Modifying surfaces and moving atoms to create nanostructures.

5.3 ELECTRON MICROSCOPY

Microscopy is a technique to observe finer details of an object that are not typically visible to the eye. For example, optical microscopes are routinely used for examinations of cells and organisms. However, the use of microscopes to observe small details is limited by the wave nature of light. Owing to the fact that light diffracts significantly around small objects, it becomes impossible to observe details significantly smaller than the wavelength of light. In general, details of the sample smaller than the wavelength of the light used are difficult to resolve. For example, a Radar can detect the size of an aircraft, but not smaller features on it, since the wavelength of most radars is several centimeters or greater. Similarly, visible light cannot resolve individual atoms, since atoms are of the order of a few angstroms in size and visible wavelengths are in the range of a few hundreds of nanometers. Electron microscopes help in achieving sub-nanometer resolution.

5.3.1 Interaction of High-Energy Electrons with a Material

When a high-energy beam of electrons strikes a sample, they don't abruptly a stop at the samples surface. Instead, they penetrate the material and undergo a variety of interactions. These interactions occur within a specific region called the interaction volume which is typically shaped a "teardrop" extending into the sample (see Fig. 5.9). This interaction volume mainly depends on the electron beam energy (higher energy leads to larger interaction volume) and atomic number of the sample (higher atomic numbers reduces interaction volume).

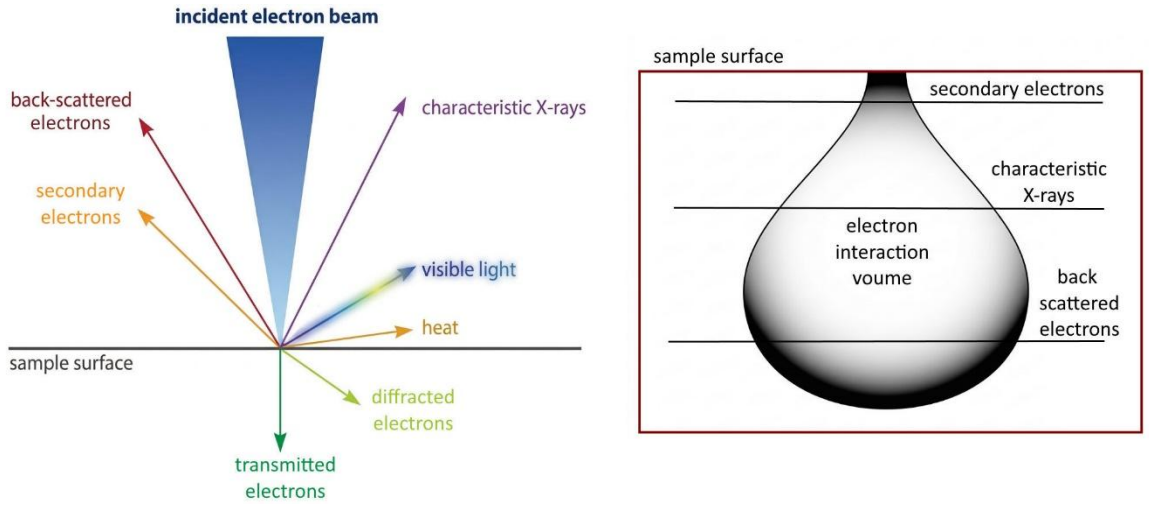


Fig. 5.9: Types of interactions between electrons and a sample and the electron interaction volume inside the sample.

The electrons will be scattered by the material, both in forward and backward directions. The scattering strength depends on the atomic number of the sample material. The scattering can also be elastic or inelastic. Inelastic scattering can lead to the generation of secondary electrons. Inelastic scattering may also cause characteristic X-ray emission from the sample. When the incoming high-energy electron knocks out an inner-shell electron from the material, an electron from a higher-energy outer shell transitions down to fill the inner shell electron vacancy. This transition releases energy in the form of an X-ray, which has a characteristic wavelength as the energy levels of the shells in a material are quantized.

The incident electrons can also get transmitted through the sample, if the sample is thin enough. The electron interaction can also lead to the generation of heat and light. Figure 9 shows various processes occurring when an energetic electron interacts with a sample and the tear drop shaped interaction volume. These interactions of the electron beam with a material can be used to characterize the material. Two of the most commonly used electron-based characterization techniques are transmission electron microscopy and scanning electron microscopy.

5.3.2 Transmission Electron Microscope (TEM)

Transmission Electron Microscope (TEM) is a powerful analytical tool used to visualize the internal structure of samples at the atomic level. Unlike a standard light microscope that uses photons, a TEM uses a high-energy beam of electrons. Because electrons have much shorter wavelengths than visible light, TEMs can achieve resolutions thousands of times higher than optical microscopes, reaching down to less than 0.1 nm. Hence, electron microscopy becomes a very important physical characterization tool at the nano-metric level.

Key Concept: the de Broglie Wavelength

The de Broglie wavelength associated with an electron in a TEM can be written as:

$$\lambda = \frac{h}{\sqrt{(2meV)}}$$

This equation defines the relationship between the electron wavelength, λ, and the accelerating voltage of the electron microscope (V). For example, the de Broglie wavelength of a 10 keV electron will be around 12 pm. This is much smaller than the wavelength of visible light, exemplifying the high-resolution capabilities associated with electron microscopy. One must keep in mind that at high acceleration potentials, the electron attains velocities close to the speed of light and we need to consider effects due to relativity for electron energies more than 100 keV.

Basic Structure of a TEM: A TEM is quite similar to an optical microscope in its basic structure. However, there are some key differences owing to the use of electrons. A TEM is usually housed in a tall, cylindrical vacuum column unlike an optical microscope. The use of a vacuum column is essential as the electrons are easily scattered and absorbed by air molecules. The basic structure of a TEM is given in Fig. 5.10 below.

The electrons are generated by the electron gun situated at the top of the vacuum column. The produced electrons are accelerated by the application of suitable electric fields and are focussed using electromagnetic lenses. The electromagnetic condenser lens focuses the electron beam on to the sample. Electrons transmitted through the thin sample are then magnified by an objective electromagnet. This is further magnified by a projector lens on to a fluorescent viewing screen and recorded by a camera.

The electrons undergo absorption, forward and backward scattering and diffraction when interacting with a sample. The sample specimen preparation is of high importance in TEM measurements. The samples must be very thin so that the electrons can pass through them and be detected on the fluorescent screen. Typically, a sample thickness of <100 nm is preferred.

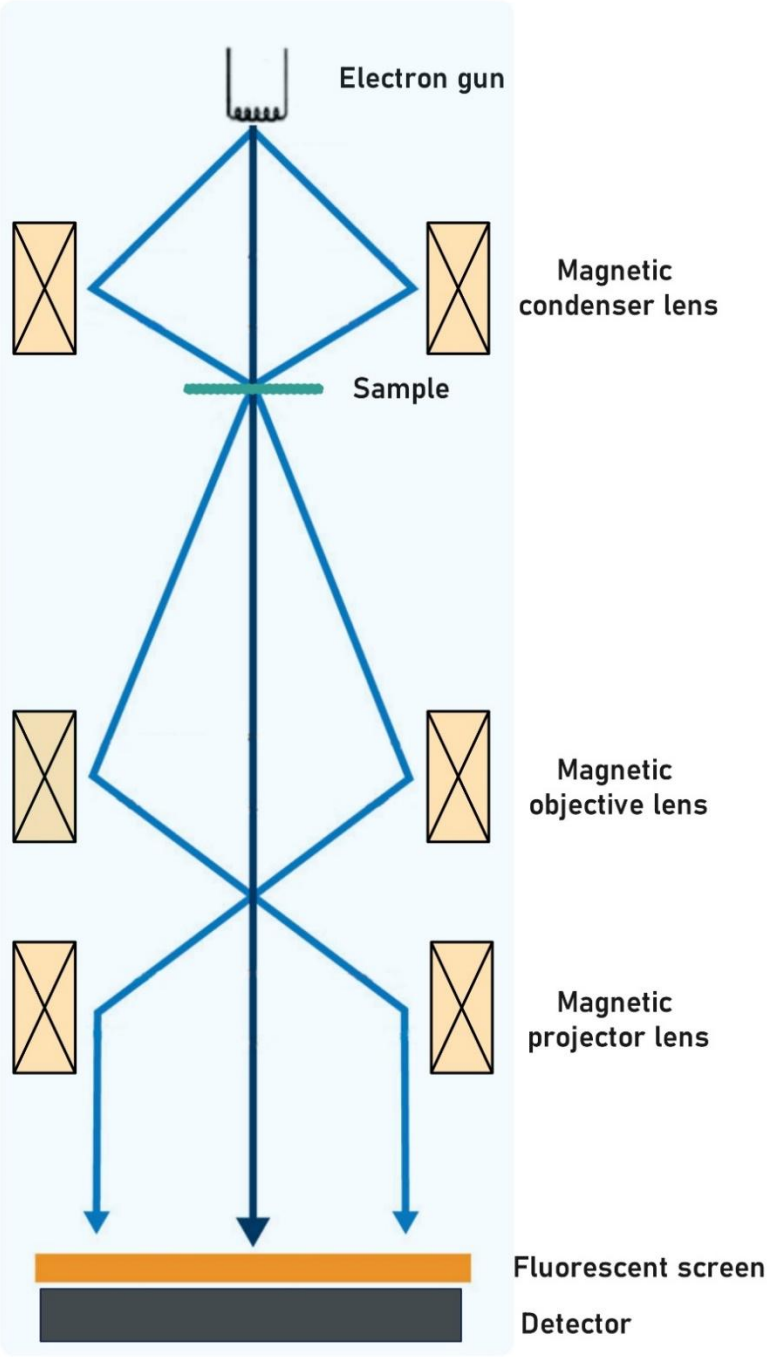


Fig. 5.10: Schematic diagram of transmission electron microscope.

Applications of TEM: TEM can be used to gather information related to the morphology of samples at atomic or molecular level. Its applications span materials science, nanotechnology and life sciences, providing ultra-high-resolution images to study structures from atoms to viruses, analysing elemental composition and morphology for fields such as drug discovery, semiconductors and virology. TEM is widely used to examine nanoparticles, crystal defects, cell organelles, bacteria, and viruses, offering detailed structural and compositional data. The electrons also undergo diffraction similar to X-rays, which can be used to identify the crystal structure of the sample.

5.3.3 Scanning Electron Microscope (SEM)

Similar to the TEM, the Scanning Electron Microscope (SEM) also utilizes electrons for high-resolution imaging. However, unlike the TEM, SEM do not require the sample to be thin for the measurements. Instead of recording the transmitted electrons, SEM utilizes the back scattered electrons and secondary electrons for imaging the specimen.

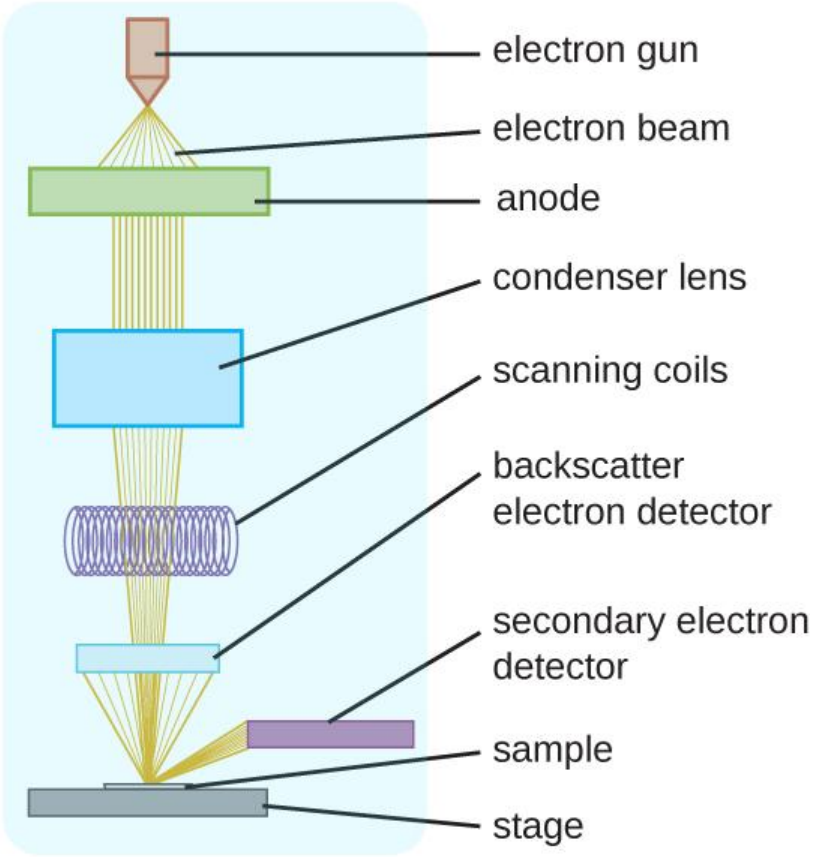


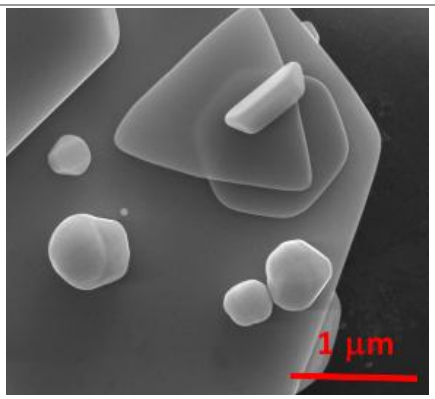
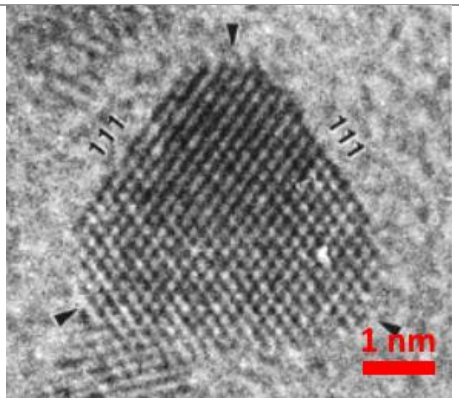
Fig. 5.11: Schematic showing the basic structure of an SEM.

Basic Structure of an SEM: The SEM is also typically housed in a tall vertical vacuum column. The electron gun situated at the top of the column generates a stream of electrons. This electron beam is then accelerated toward the sample by a high-voltage anode, typically ranging from 1 kV to 30 kV. A series of electromagnetic lenses condense and focus the electron beam to a sharp point. Electromagnetic scanning coils are used to deflect the focussed electron beam in a raster pattern across the sample surface. The interaction of the electron beam with the sample atoms results in the generation of various signals. The secondary electrons ejected from the specimen are primarily used to map surface topography using an electron detector placed near to the sample. By raster scanning the focussed beam, the entire sample surface can be mapped. The emitted secondary electrons are converted into electronic signals that are synchronized with the position of the scanning beam to reconstruct the final image on a digital display. In addition to the secondary electrons, the backscattered primary electrons are recorded using a back scattered electron detector. This data is particularly useful for distinguishing between different materials, as heavier elements scatter more electrons and appear brighter in

the resulting image. Though SEM does not require thinning of the samples, non-conductive samples may often require a thin conductive coating to prevent electric charge accumulation. Fig. 5.11 below shows the basic schematic of an SEM.

Applications of SEM: SEM is used in diverse fields such as materials science, biological sciences and semiconductor industry. In materials science, it is used to investigate the surface structure of alloys, microstructures and composite materials. SEM to utilized to view the intricate details of cells, tissues, and microorganisms in biological sciences. It is also used in semiconductor industry for the inspection of integrated circuits, to verify the integrity of features that are on the nanoscale.

Comparison between SEM and TEM: While both instruments utilize electron beams to achieve high-resolution imaging, they serve different purposes. SEM is essentially the preferred instrument for "looking at" an object, giving a thorough view of its surface and three-dimensional structure. On the other hand, TEM is used for "looking through" a specimen, providing insight into the sample's intrinsic atomic structure and crystalline arrangement.

	SEM	TEM
Primary purpose	Surface morphology, topography, and composition.	Internal structure, crystallography, and atomic details.
Image type	3D-like images of the exterior surface.	2D projection images of the internal structure.
Sample thickness	Can accommodate bulk samples of varying sizes.	Requires ultra-thin sections (usually <100 nm).
Resolution	High (down to ~1–5 nm).	Ultra-high (down to <0.1 nm).
Magnification	Typically, up to 500,000x.	Up to 50,000,000x.
Signal source	Secondary and backscattered electrons.	Transmitted and diffracted electrons.
Sample images	 SEM of gold powder	 TEM of platinum nanoparticle

5.4 QUESTIONS

1. Derive Bragg's law.

2. Describe the construction and working of an X-ray diffractometer.
3. Discuss how XRD can be used to identify an unknown crystalline material.
4. Describe how crystallite size can be determined from an XRD spectrum.
5. What is a Scanning Probe Microscope (SPM)? Explain the working principle of Atomic Force Microscopy.
6. With a neat diagram explain the working principle of Scanning Tunneling Microscopy.
7. Explicitly explain the differences between Scanning Tunneling Microscope (STM) and an Atomic Force Microscope (AFM).
8. Write few applications of Scanning Probe Microscopy.
9. Briefly explain the feedback system in an atomic force microscopy.
10. Write few applications of scanning probe microscopy?
11. Explain the working principle of an electron microscope.
12. What are the differences between an SEM and a TEM?
13. Why do a TEM require high vacuum for operation?
14. Why glass lenses cannot be used in TEM & SEM?
15. If you need to identify the internal lattice structure of a gold nanoparticle, which electron microscope would you choose and why?

5.5 PROBLEMS

1. X-rays are diffracted in the first order from a crystal with interplanar spacing 2.82 Å at a glancing angle of 6°. Calculate the wavelength of X-rays.
2. Calculate the energy of the X-rays that produces 1st order Bragg diffraction at 20° when incident on planes separated by a distance of 2 Å.
3. An XRD pattern of a specimen shows a prominent peak at $2\theta = 38.2^{o}$. The full width at half maximum (FWHM) of this peak is measured to be 0.45^{o}. Assuming the X-ray source is Cu $K\alpha$ (wavelength = 0.15406 nm) and the shape factor K = 0.9, calculate the average crystallite size.
4. The tunneling current (I) in an STM is 10 nA at a tip-sample separation (d) of 0.5 nm. If the tip-sample separation increases to 0.6 nm, what is the new current? Decay constant $k = 10\ nm^{-1}$
5. An STM tip is tunneling through a vacuum with a work function (Φ) of 5 eV If the tip-sample distance (L) changes by 0.1 nm (1 A^{0}), what is the approximate percentage change in tunneling current? Assume k = 2.2 $A^{0\ -1}$
6. An AFM cantilever has a spring constant k of 0.5 N/m. During a scan, the cantilever deflects by 5 nm. What is the force?
7. An AFM cantilever has a spring constant k of 0.1 N/m. To obtain a force of 1 nN (10^{-9} N) on the sample on the sample, what is the required deflection?

8. If the electron accelerating voltage is increased from 5 kV to 30 kV, how does it affect the resolution of the SEM?
9. What is the wavelength of electrons of a TEM that uses 100 V accelerating electric field?